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1 Chapter 1: Introduction

Technology has quietly reshaped how people invest. Today, someone in Riyadh can own a fraction of a property in Doha, fund a startup in London, or trade stocks in New York , all from their phone. Yet one sector has remained stubbornly out of reach for the everyday person: agriculture. In Saudi Arabia, date palm farming is not just an industry, it is a cultural identity, a national heritage, and an increasingly strategic pillar of the Kingdom's Vision 2030. With over 37.6 million date palms and a national production of approximately 1.9 million tons in 2024 [1], the scale of this sector is undeniable. However the ability to participate in it financially has always required something most people simply do not have : land, large capital, or the right connections. Conversely, many farmers who own productive land and possess the necessary skills cannot find interested individuals willing to invest in their farms or purchase their future harvests.

The gap between those who want to invest in this thriving sector and those who own the land to make it possible has persisted for years, not because there is no interest on either side, but because no structured, accessible bridge between them has ever existed.

Introducing “Qinwan”,web-based platform that enables simple and transparent investment in date palm farms across Saudi Arabia. The platform bridges the gap between individual investors and farm owners, allowing users to discover and invest in specific land of palm trees, track their growth, and choose flexible returns from their harvest all through a secure and user-friendly digital interface.

At the local level, a technological solution in this domain plays a direct role in supporting farmers and revitalizing agricultural activity across the Kingdom. By increasing visibility of farms and enabling structured investment participation, farmers gain improved access to financial resources, allowing them to sustain operations, enhance productivity, and plan for future agricultural cycles more effectively. This contributes to the continuity and growth of farming activities, particularly in rural regions where agriculture remains a primary source of income.

On a global level, it presents an innovative model for "agricultural fintech" and fractional asset ownership that can be applied to other regions and crops facing similar disconnects between small-scale farmers and dispersed capital. It contributes to knowledge sharing on using technology to enhance sustainability and efficiency in arid-land agriculture, a challenge relevant to many communities worldwide.

This document presents the initial proposal for Qinwan . We begin by detailing the significant real-world problem of inaccessible agricultural investment. Following this, we will elaborate on our proposed technological solution and its core vision. The document will then specify the product's functional objectives for end-users, outline the project's technical and developmental goals, and define the key learning outcomes for the student team. The proposed development timeline is captured in a product roadmap, and the project's Scrum team is formally introduced. All references supporting this chapter are provided in the final section.

1.1 The Problem

In this section, we describe the challenges in agricultural investment, including the difficulties faced by investors and farm owners, and explain the need for a structured digital solution.

Many individuals are interested in agricultural investment but face difficulties entering this field because they do not own farmland or have enough capital to start or manage a farm. At the same time, many farm owners who own productive agricultural land struggle to find interested investors or individuals willing to invest in their farms. These farmers often have the land, expertise, and labor, but lack access to a reliable network of potential investors. As a result, they miss opportunities to expand their operations, improve their farming techniques, or increase their annual production. Agriculture is an important sector for the economy and food security[1][2], yet investment opportunities are often limited to large investors and farm owners, which reduces participation from the wider community and leaves many farmers with underutilized land.

A concrete example of this problem can be seen when a person wants to invest in date palm farming but does not own land and cannot afford to buy or operate a full farm. Even if they wish to invest with a farmer, they may not know which farms are available or how to invest safely. On the other side, a farm owner in Al-Ahsa or Al-Qassim may have 50,000 square meters of productive land with high-quality date palms, but no clear way to reach individuals who would be interested in leasing a portion of that land. As a result, many people lose interest in agricultural investment despite their willingness to participate, and many farmers continue operating below their full potential.

This project focuses on a specific part of this problem: the lack of an easy and organized way for individuals to invest in agriculture without needing land or large capital. It also addresses the absence of a clear digital platform that connects farm owners with investors and presents investment opportunities in a transparent way.

1.2 The Solution

In this section, we propose a solution to the problem described earlier. The proposed solution is a digital platform called Qinwan, which aims to facilitate fractional agricultural investment in date palm farms within Saudi Arabia.

Qinwan brings farm owners and investors together in one platform. Farm owners can register their farms and define specific land areas available for investment, each clearly displaying the total size, the types of date palms planted in that plot, and the expected seasonal yield. Investors can browse available farms through an interactive map of Saudi Arabia or a filterable list view. When selecting a farm, investors can view each available land plot and choose the exact area (in m^2) they wish to lease based on the options provided by the farm owner.

Once an investor selects their desired land area, they can specify the investment duration and choose how they would like to benefit from the harvest at the end of the season: receiving the dates, earning a financial return, or donating the harvest to charity. The investment request is then submitted to the farm owner for review and confirmation.

The platform also allows users to track their investments through a personal dashboard that shows the land area they have leased, the types of date palms included in their plot, and the status of each active investment. In addition, Qinwan provides regular updates from the farms about the condition and progress of the palm trees, including photos and text updates sent directly by the farm owner. The system includes an interactive map that allows users to explore farms across different regions of Saudi Arabia.

This solution helps farm owners reach investors in a more organized and efficient way, while offering investors a transparent, flexible, and reliable agricultural investment experience with clear visibility into exactly which land and trees their money supports.

1.3 The Product

1.3.1 Product Vision

In this section, we define the product vision of Qinwan, clarifying what the platform aims to achieve for investors and farm owners and the value it intends to deliver.

For individual investors and date palm farm owners in Saudi Arabia,

Who seek a transparent, accessible, and impact-driven way to participate in the dates sector and directly support local agriculture,

Qinwan is a web-based agricultural investment platform

That enables users to discover, invest in, and monitor specific date palm trees on verified farms, transforming traditional agriculture into a transparent, trackable, and socially impactful digital asset class through a unified, accessible experience.

Unlike online marketplace platforms such as Haraj and Al-Aswaq Al-Mawsimiyah, which require prohibitive capital or operate through informal agreements that lack oversight and transparency,

Our product directly connects capital to specific assets on a map, provides verifiable farm and tree data, and ensures clear, flexible outcomes for every harvest building a trusted and modern investment ecosystem.[2]

1.3.2 Product Roadmap

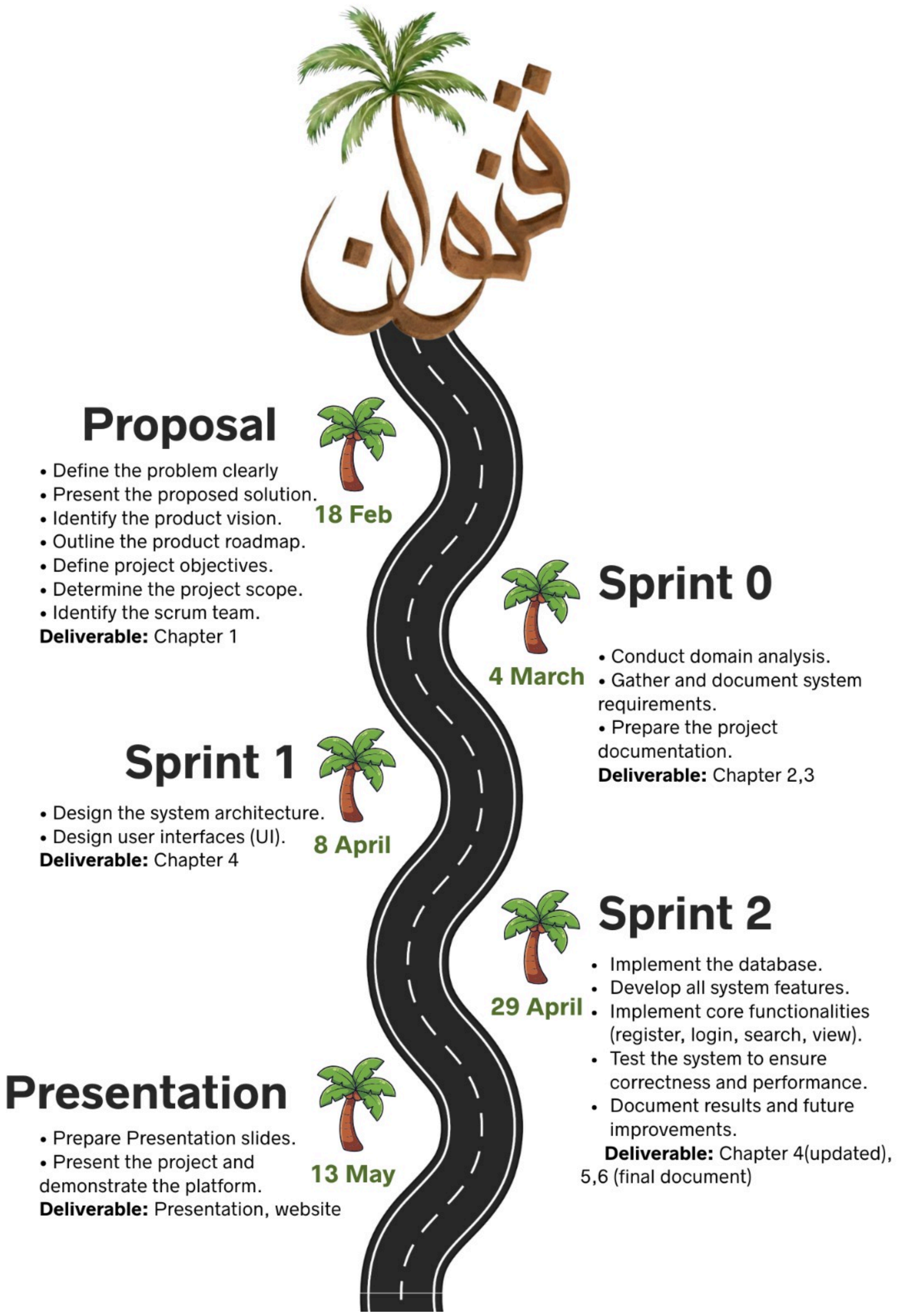

Figure 1 : Product roadmap

1.3.3 Objectives

In this section, we will define the product, project, and learning objectives pertaining to the development of the "Qinwan" web platform. These objectives represent the measurable goals and end-state outcomes we intend to achieve upon successful project completion.

Project Objectives (Customer Focus - Value)):

Farmers:

1. User Registration (Farmer)
The user can create a new account by providing required information (name, email, phone number, password) and selecting the "Farmer" role to access farming features.

2. User Login (Farmer)
The user can log in using their registered email and password to access their farmer profile and dashboard.

3. User Logout (Farmer)
The user can securely log out of their account to end their session.

4. Farm Submission for Approval
The farmer can submit a detailed farm registration form including land size, location, palm tree types, and estimated annual yield for admin review and approval.

5. Edit Farm Details
The farmer can edit their farm information (e.g., description, photos, available land area) after approval.

6. View Investment Requests
The farmer can view a list of incoming investment requests from investors requesting to lease specific land areas within their farm.

7. Accept Investment Requests
The farmer can accept investor requests, confirming the lease agreement for the specified land area.

8. Reject Investment Requests

The farmer can reject or cancel investment requests before or during the review process.

9. Send Periodic Farm Updates

The farmer can send text and photo updates about crop conditions and palm trees to all investors who are currently leasing land on their farm.

10. View Total Leased Land Area

The farmer can view the total land area (in m^2) currently leased by investors on their dashboard.

11. View Number of Active Investors

The farmer can view the total count of active investors currently leasing land on their farm.

12. View Total Expected Yield

The farmer can view the estimated total yield for the current season based on leased land area.

13. View Estimated Profit Distribution

The farmer can view projected profit calculations from current active investments.

For Individual Investors:

1. User Registration (Investor)

The user can create a new account by providing required information (name, email, phone number, password) and selecting the "Investor" role to access investment features.

2. User Login (Investor)

The user can log in using their registered email and password to access their investor profile and dashboard.

3. User Logout (Investor)

The user can securely log out of their account to end their session.

4. Browse Farms via Interactive Map

The investor can discover farms accepting investment through an interactive map of Saudi Arabia.

5. Browse Farms via List View

The investor can discover farms accepting investment through a filterable list view.

6. Filter Farms by Region

The investor can filter available farms based on geographical region.

7. Filter Farms by Farm Size

The investor can filter available farms based on total farm area.

8. Filter Farms by Date Type

The investor can filter available farms based on the type of dates grown.

9. Add Farm to Wishlist

The investor can add farms to a favorites list for quick access later.

10. View Farm Size

The investor can view the total size of a selected farm.

11. View Available Land Area for Investment

The investor can view how much land area is currently open for investment on a specific farm.

12. View Date Palm Types Grown

The investor can view the types of date palms cultivated on the farm.

13. View Number of Current Investors

The investor can view how many investors are currently active on a specific farm.

14. View User Ratings and Reviews

The investor can view ratings and feedback left by other investors about the farm.

15. Select Land Area for Investment

The investor can select the farmer offer of a specific land area (in m^2) within a farm to lease.

16. Add Selected Land Area to Cart

The investor can add their selected land investment item to the cart for review before submission.

17. View Cart Items

The investor can open their cart to view all selected land investment items.

18. Remove Item from Cart

The investor can delete or remove any investment item from their cart before submission.

19. Edit Harvest Return Method in Cart

The investor can edit the preferred harvest return method for any item while it is still in the cart.

20. Select Home Delivery as Return Method

The investor can choose to receive physical dates at their address after harvest.

21. Select Sell Harvest as Return Method

The investor can choose to have their share of the harvest sold and receive the financial profit.

22. Select Donate Harvest as Return Method

The investor can choose to donate their share of the harvest to charity.

23. Specify Investment Duration in Cart

The investor can set the lease duration (e.g., 6 months, 12 months) for each item while in the cart.

24. Edit Investment Duration in Cart

The investor can modify the lease duration for any item while it is still in the cart.

25. Submit Investment Request

The investor submits all items from their cart as a single investment request.

26. View Pending Investment Requests

The investor can view all investment requests currently waiting for farmer approval.

27. View Accepted Investment Requests

The investor can view all investment requests that have been confirmed by the farmer.

28. View Cancelled Investment Requests

The investor can view all investment requests that were rejected or cancelled.

29. Cancel Investment Request (Pre-Approval)

The investor can cancel or delete their investment request only while its status is Pending.

30. View Leased Land Area in Dashboard

The investor can view the total land area they have leased and its location within the farm.

31. View Farm Name and Farmer Contact

The investor can view the farm name and farmer contact information in their dashboard.

32. View Date Palm Types in Leased Area

The investor can view the types of date palms included in their leased land area.

33. View Expected Harvest Yield

The investor can view the estimated harvest yield from their leased land area.

34. View Profit and Loss Summary

The investor can view a summary of financial returns from their investments.

35. View Farmer Update Feed

The investor can view a dedicated feed of text and photo updates received from the farmer about their leased land.

36. Request Edit Return Method (Post-Acceptance)

The investor can request to change their chosen return method after the request is accepted, subject to farmer approval.

For Administrators:

1. View All Registered Users

The admin can view a complete list of all registered users on the platform.

2. View All Registered Farms

The admin can view a complete list of all farms registered on the platform.

3. Review Farm Submissions

The admin can review detailed farm registration forms submitted by farmers.

4. Approve Farm Registrations

The admin can approve farm submissions, making them visible to investors.

5. Reject Farm Registrations

The admin can reject farm submissions that do not meet platform requirements.

6. Edit Farm Information

The admin can edit or correct farm details when necessary.

7. Remove Farm Listings

The admin can remove farms from the platform if violations occur.

8. View All Investment Transactions

The admin can view a complete record of all investment transactions across the platform.

9. View User Activity Logs

The admin can view activity logs for both farmers and investors.

10. Edit User Content

The admin can edit or remove inappropriate content including farm descriptions and user reviews.

11. Receive User Complaints

The admin can receive and review complaints reported by users.

12. View Total Registered Users Count

The admin can view the total number of users registered on the platform.

13. View Total Active Farms Count

The admin can view the total number of approved and active farms.

14. View Total Invested Land Area

The admin can view the total land area currently under investment across all farms.

15. View Total Transaction Volume

The admin can view the total value of all investment transactions processed.

Project Objectives (solution focus–plan):

These objectives describe the stages required to complete the Qinwan platform:

1. Conduct interviews and questionnaires with potential users (investors and farmers).
2. Collect and analyze data related to date palm farming and agricultural investment (domain analysis).
3. Identify user requirements and apply requirements elicitation techniques.
4. Design user interfaces for the Qinwan platform.
5. Design a database for users, farms, palm trees, and investments.
6. Develop and implement the platform and its core features.
7. Integrate interactive map features to display farm locations.
8. Test the platform to ensure functionality and usability.
9. Deploy the platform and prepare project documentation.

Learning Objectives (student focus):

The following objectives outline what the team will learn from completing this project:

- Learn a new programming language, e.g., XML.
- Use new tools, such as GitHub and Jira.
- Learn and implement the Agile development methodology through sprint-based project work.
- Learn how web applications connect to databases to store and retrieve user information.
- Practice software documentation throughout the project lifecycle.

By achieving these objectives, our Qinwan platform will contribute to supporting digital transformation in the agricultural investment sector while helping our team develop practical skills in web-based application development.

In the next section, we will discuss the scope of Qinwan and explore the specific functionalities and features it will provide to users.

1.3.4 Scope

In this section, we define the scope of Qinwan by clarifying what features and functionalities are included in the project and what falls outside its boundaries.

Qinwan is a web-based digital agricultural investment platform designed for individual investors and date palm farm owners in Saudi Arabia aged 18 years and older, with basic to average technical skills. The platform is built as a fully responsive web application, adapting seamlessly to different screen sizes and devices , delivering an optimized experience on desktop, tablet, and mobile through modern web browsers (Chrome, Safari, Edge). The user interface and all platform content are in Arabic only. The system supports three defined user roles: Investor, Farmer, and Administrator ,each with distinct access rights, and does not support guest browsing without registration. Core functionality includes account creation, farm browsing and filtering, fractional land-area investment, investment tracking via a personal dashboard, harvest yield management (receive, sell, or donate), and farm update feeds. The platform does not include real-time IoT farm monitoring, external payment gateway integration, advanced financial analytics, legal contract enforcement, government system integration, or automated smart contract execution. Any features beyond those defined in the objectives are considered future enhancements and fall outside the scope of this project.

1.4 The Scrum Team

Scrum Team	
Product Owner	Noora Aluqaili.
Developers	Noora Aluqaili , Jana Alothman , Haya Alhajri , Noura Altuwiam.
Scrum Master	I. Ghaida ALFayez.
Stakeholders	I. Ghaida ALFayez.

Table 1 : Scrum Team

2 Chapter 2: Domain Analysis

2.1 Literature Review

Competitor Haraj [3]:

Brief Description: Haraj is a popular Saudi online marketplace that allows users to buy and sell various products, including land and farms. Some users post agricultural lands or farms for sale or lease.

Strengths:

1- Large user base across Saudi Arabia.

2- Easy posting and browsing of listings.

3- Covers many categories including farms.

Limitations:

1- Not specialized in agricultural investment.

2- No fractional investment option (users must buy full property).

4- No structured system connecting investors with farm owners.

Competitor Atyab Al-Tamor[4] :

Brief Description : Atyab Al-Tamor is an online store specialized in selling dates and related products in Saudi Arabia. The platform provides various types of packaged dates for individual customers and gift orders.

Strengths:

1. Specialized in date products.
2. Professional product presentation and packaging.
3. Online ordering and delivery services.

Limitations:

1. Focuses only on selling final date products.
2. Does not provide palm tree investment opportunities.
3. No transparency regarding farm-level ownership tracking.

Features	Haraj	Atyab Al-Tamor	Qinwan
Palm tree investment	✗	✗	✓
Agricultural land reservation	✓	✗	✓
Investment tracking system	✗	✗	✓
Structured & transparent system	✗	✗	✓

Table 2 : Competing Platforms and Their Features Compared with Qinwan

Based on the comparison table above, it is clear that existing platforms such as Haraj and Atyab Al-Tamor mainly focus on agricultural listings or product sales, without providing a structured investment framework. While some platforms may support land reservation, they do not offer palm tree–level investment or a dedicated tracking system.

In contrast, Qinwan integrates palm tree investment, agricultural land reservation, and a built-in investment tracking system within a structured digital environment. The platform uniquely connects investors to specific palm trees and enables organized monitoring of their investment lifecycle. This combination of transparency, structure, and asset-level visibility clearly differentiates Qinwan from existing competitors and positions it as a comprehensive agricultural investment solution.

3 Chapter 3: Requirements Engineering

3.1 Requirements Elicitation and Analysis

To identify the system requirements for Qinwan, several requirement elicitation methods were used. First, online questionnaires were distributed to two target groups: potential investors and date palm farm owners. The questionnaires aimed to understand user expectations, preferred features, and challenges in agricultural investment.

In addition, informal interviews were conducted with farm owners to better understand their needs regarding land leasing, investment management, and transparency requirements. Market research was also performed by analyzing similar platforms such as Haraj and Atyab Al-Tamor to identify existing gaps and limitations.

Furthermore, official reports related to date production and agricultural development in Saudi Arabia were reviewed to ensure that the proposed system aligns with market needs and local agricultural practices.

The collected data was analyzed and translated into functional and non-functional requirements that define the core features of the Qinwan platform.

3.1.1 Interviews

Participants' information

Name	Age	Nationality	gender
1. Mohammed	55 Years	Saudi	Male
2. Ibrahim	50 Years	Saudi	Male
3. Abdullah	47 Years	Saudi	Male
4. Saad	60 Yeas	Saudi	Male

Table 3 : Interview participants' information

Interviews results and findings

Interviews were conducted with two investors and two farmers to understand their needs and expectations for a fractional agricultural investment platform.

From the investors' side, experience level affected their expectations. As shown in *Appendix A (Investor 1, Q1–Q3)*, Investor 1 has previous experience in agriculture and knows the importance of regular follow-up and paying attention to environmental factors such as water quality and soil fertility. In contrast, Investor 2, as indicated in *Appendix A (Investor 2, Q1–Q2)*, has no previous experience but is interested in investing without owning land. This shows that the platform can attract both experienced and new investors.

Both investors said that clear information is very important before making any investment decision. As shown in *Appendix A (Investor 1, Q3)*, Investor 1 focused more on the farm's environmental conditions, while *Appendix A (Investor 2, Q3)* indicates that Investor 2 focused on profit percentage and operational details. Both wanted the platform to be simple and easy to use, as reflected *in Appendix A (Investor 1, Q4; Investor 2, Q4)*. They also highlighted the importance of direct communication with farmers and regular visual updates to help build trust, as shown in *Appendix A (Investor 1, Q5; Investor 2, Q5).*

From the farmers' side, both participants showed strong agricultural experience, as shown in *Appendix A (Farmer 1, Q1; Farmer 2, Q1)*, and were open to external investment to reduce financial pressure and increase production, as indicated in *Appendix A (Farmer 1, Q2; Farmer 2, Q2)*. They are willing to share detailed information about their farms, such as expected production, water sources, infrastructure, and operating costs, as shown *in Appendix A (Farmer 1, Q3; Farmer 2, Q3)*. However, they also emphasized that the platform should be easy to use, as reflected in *Appendix A*

(Farmer 1, Q4; Farmer 2, Q4). One farmer mentioned that high commission fees might make him stop using the platform, as shown in *Appendix A (Farmer 2, Q4),* which shows that fair pricing is important.

Overall, the interviews show that investors and farmers have similar expectations. The success of the platform depends on clear information, easy use, fair pricing, and regular communication that builds trust between both sides.

3.1.2 Questionnaire

Executive Summary

A total of 81 participants completed two separate questionnaires for the Qinwan platform between February 27–28, 2026. The investor survey (n=45) and farmer survey (n=36) together reveal a consistent and actionable picture of user needs.

The majority of investors (62%) are aged 46 and above, indicating a mature, financially capable audience who expect a simple and trustworthy experience. When it comes to browsing farms, 64% want both a map view and a list view available simultaneously. Farm photos and short video updates are the most demanded content types, with photo importance scoring an average of 4.3 out of 5. On the farmer side, 69% have never leased their land before but are actively interested ,meaning Qinwan is entering a largely untapped market. Both user groups independently chose WhatsApp as their preferred notification channel (73% of investors, 75% of farmers), and both want visual content in farm updates (photos and short videos). The single most repeated insight across both surveys: users want transparency, verification, and trust between investors and farm owners. Features such as investor ratings, farm activity feeds, and a clear dashboard are not nice to haves they are core expectations.

3.1.2.1 Part 1 — Investor Survey (n = 45)

Key Findings at a Glance

Map + List dual view, palm type and farm rating filters, photo/video upload for updates, WhatsApp notifications, harvest sale & profit transfer system, investor rating system, and automated response reminders for farmers.

Question	Key Finding
Age group	**46 and above — 62%**
Preferred farm browsing	**Both Map + List — 64%**
Top filter criteria	**Farm rating / Palm type — 29 participants**
Photo importance (Likert)	**Average 4.3 / 5 — 60% gave full score of 5**
Preferred update type	**Short video clips — 71%**
Preferred notification channel	**WhatsApp — 73%**
Preferred harvest method	**Sell harvest & receive profits — 67%**

Table 4 : Summary of Key Findings from the Investor Survey

Q1 — What is your age group?

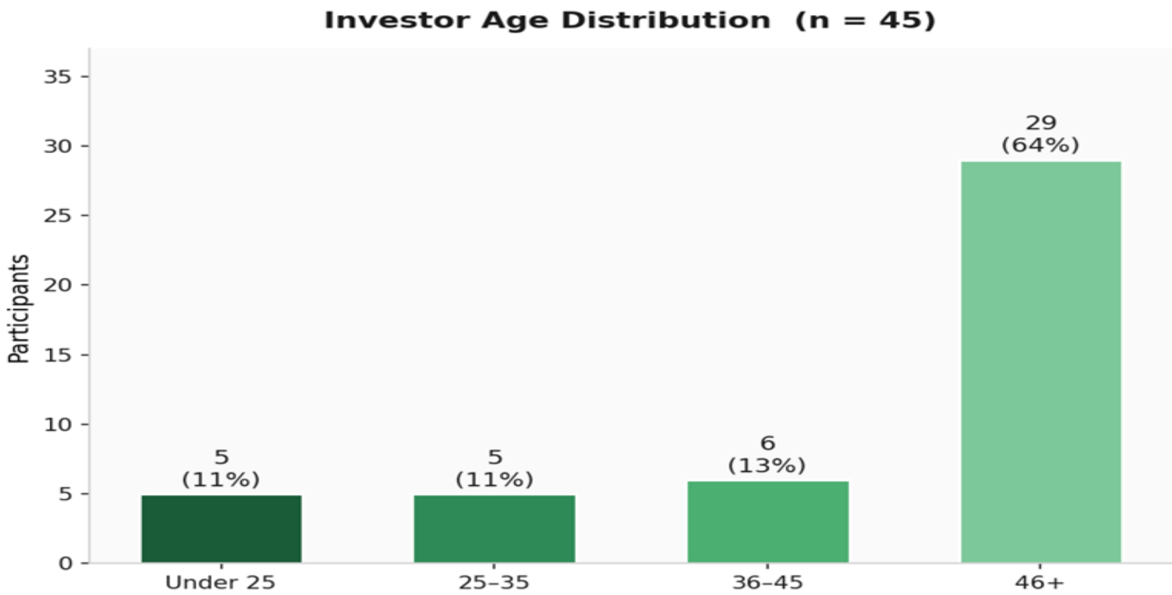

Figure 2 : Questionnaire Results (Q1 | Investor)

The dominant age group is 46+ (62%), followed by 36–45 (16%). This means the platform's primary users are experienced, financially established individuals. The UI must be clean and straightforward, avoiding overly technical language or complex navigation patterns.

Q2 — How do you prefer to browse farms?

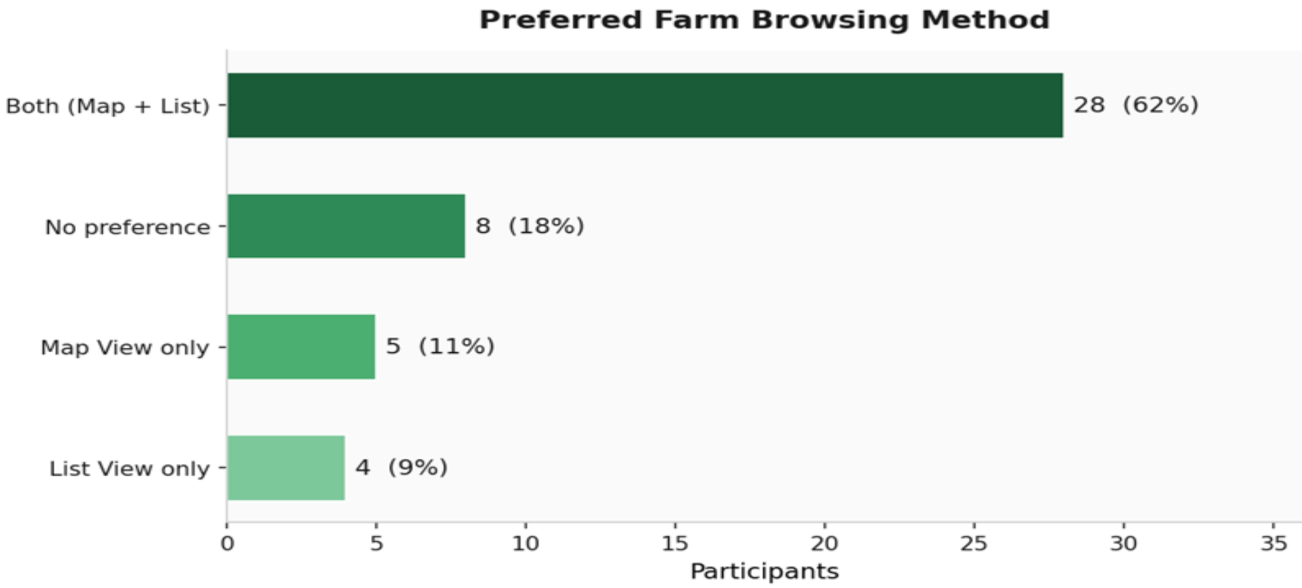

Figure 3 : Questionnaire Results (Q2 | Investor)

64% want both views available (Map + List), and only 9% prefer Map alone or List alone. Offering just one view would alienate most users. Both views must be built and accessible from the same screen.

Q3 — What are your most important filter criteria? (multiple choice)

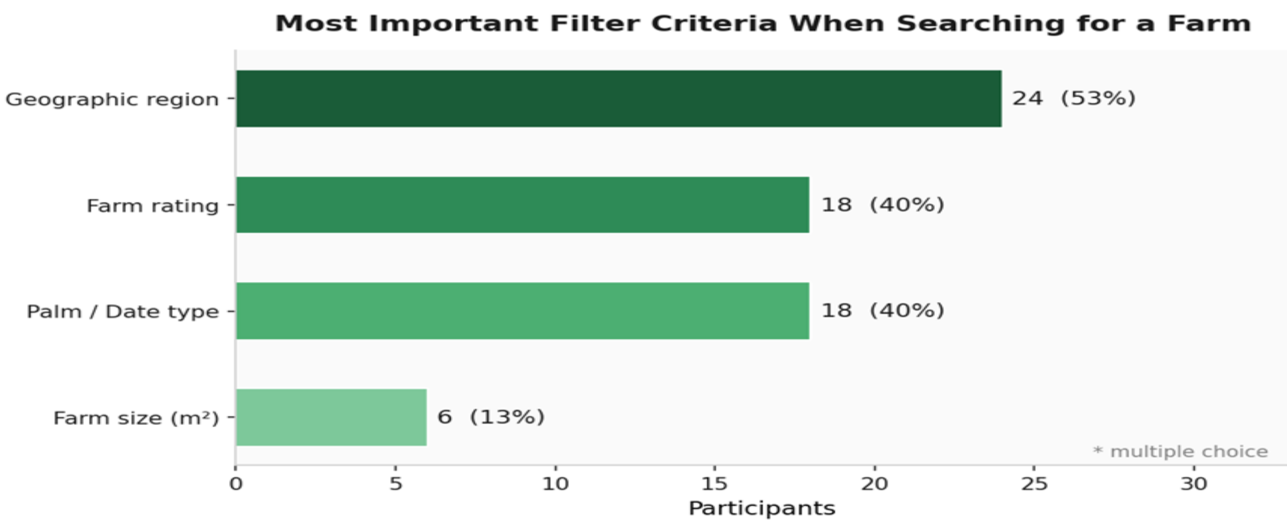

Figure 4 : Questionnaire Results (Q3 | Investor)

Geographic region (53%) leads, closely followed by palm/date type and farm rating (both 40%). Farm size scored notably lower at 13%, suggesting investors care more about what is grown and where, rather than raw acreage. The filter by palm type must be a prominent, first-class feature in the UI.

Q4 — How important are high-quality farm photos before investing? (1–5 scale)

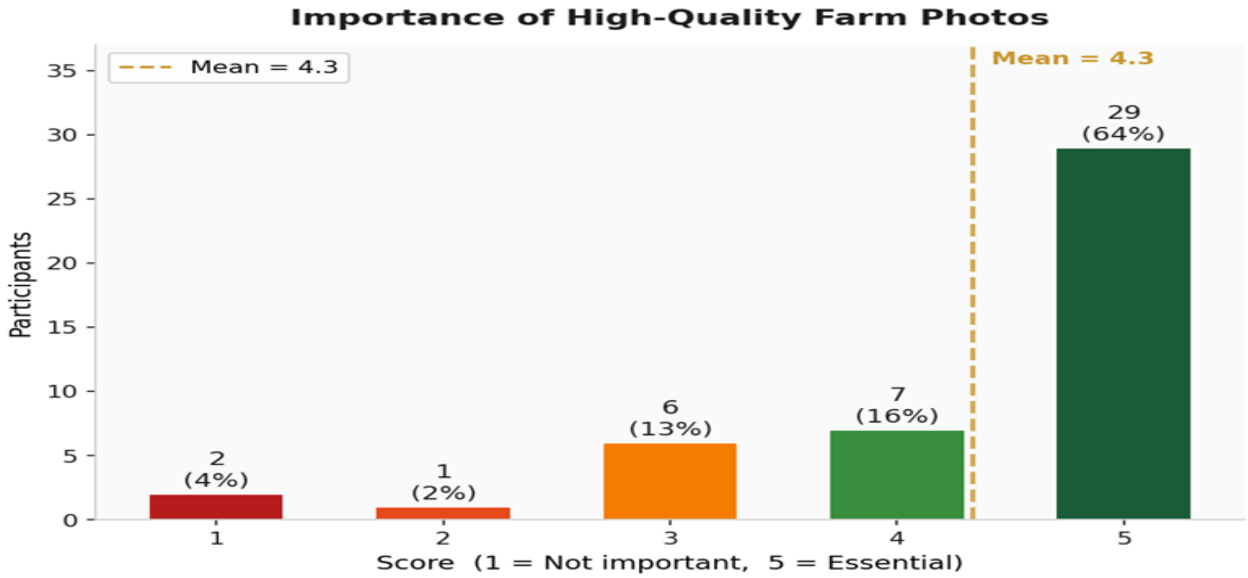

Figure 5 : Questionnaire Results (Q4 | Investor)

Average score: 4.3/5. A striking 60% gave the maximum score of 5, and only 6% rated this below 3. Photos are not cosmetic .they are a critical trust signal that directly influences investment decisions. Farmers must be required (or strongly encouraged) to upload quality photos when registering their farm.

Q5 — What type of farm updates do you want to receive? (multiple choice)

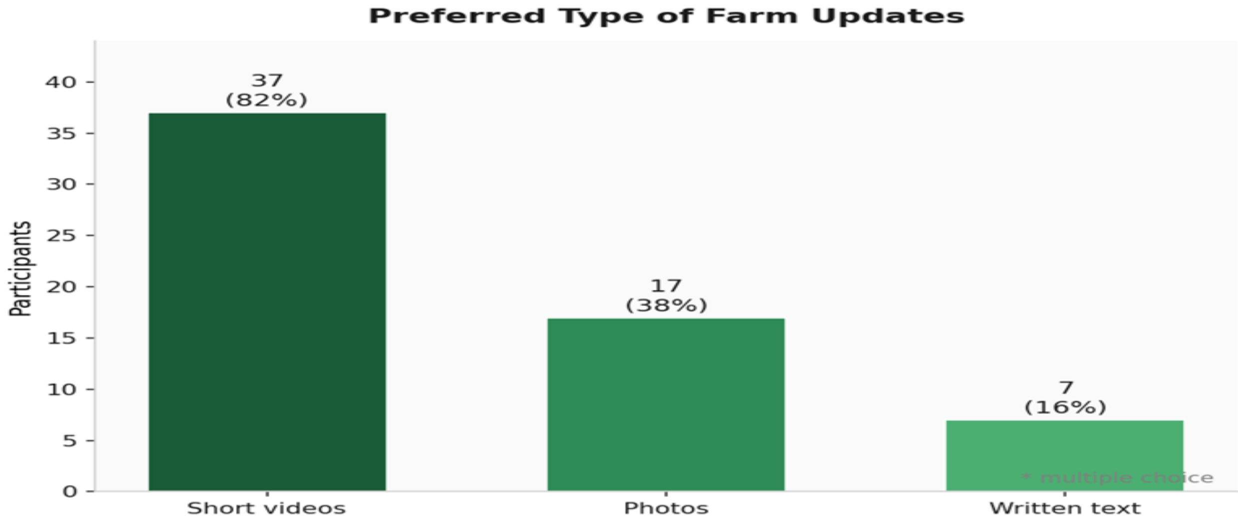

Figure 6 : Questionnaire Results (Q5 | Investor)

Short video clips dominate at 71%, followed by photos (38%) and text (27%). This directly aligns with farmer preferences in Part 2. The update feature must support video uploads, not just text or images. Both sides of the platform independently arrived at the same conclusion.

Q6 — How do you prefer to receive platform notifications?

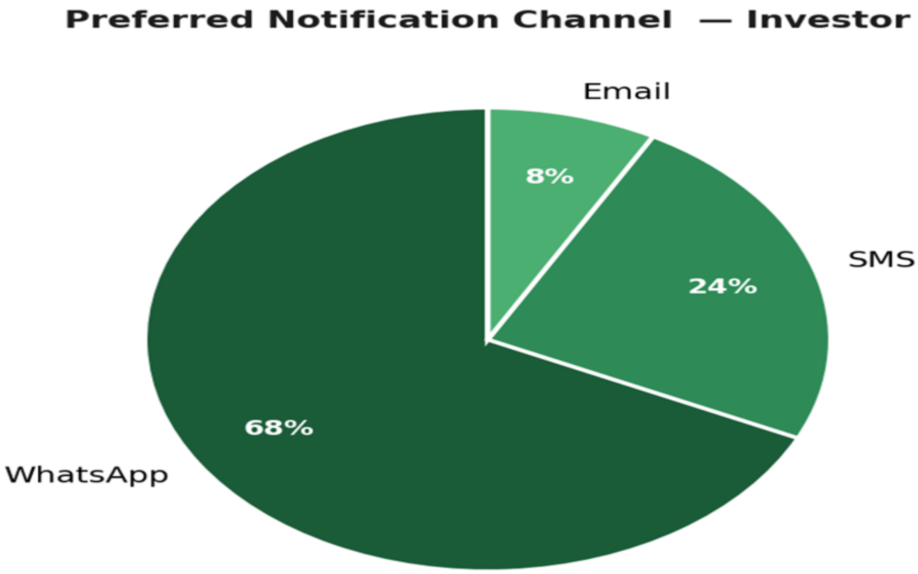

Figure 7 : Questionnaire Results (Q6 | Investor)

WhatsApp accounts for 73% of preferences. SMS and email are distant second choices. Integrating with WhatsApp Business API should be treated as a high-priority infrastructure requirement, not an optional feature.

Q7 — How do you prefer to benefit from the harvest?

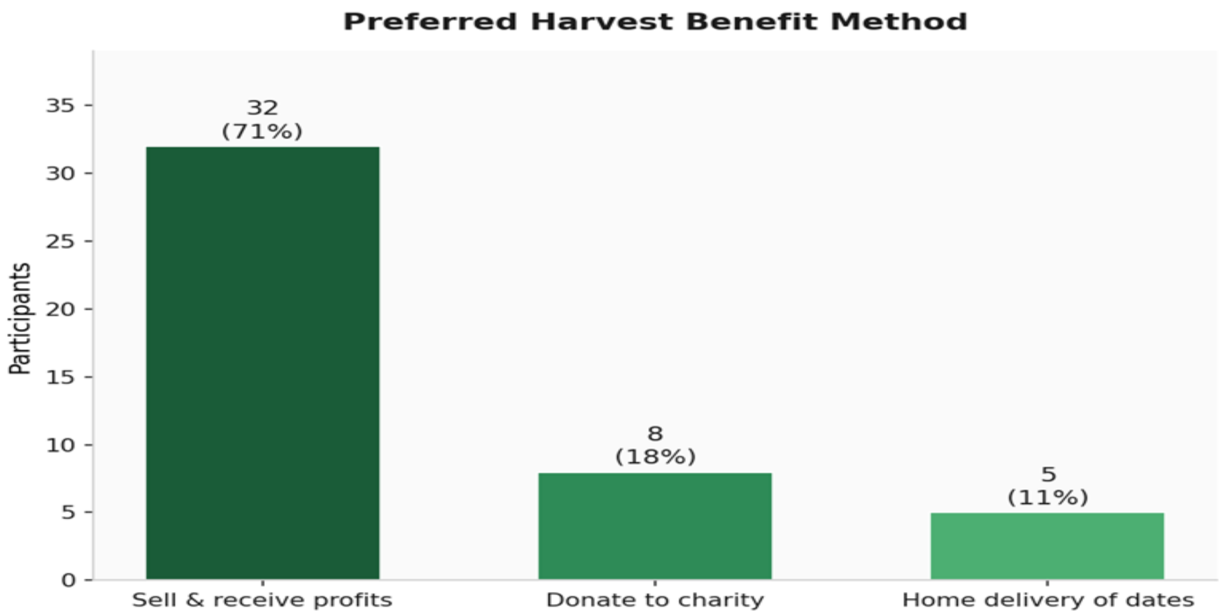

Figure 8 : Questionnaire Results (Q7 | Investor)

67% want to sell the harvest and receive cash profits . the classic return on investment model. Only 18% want physical date delivery, and 13% prefer to donate. This confirms the platform must build a robust harvest sale and profit transfer system as a primary use case.

Open-Ended Question — Investor Suggestions

#	Response
1	Would love a simple investment dashboard showing farm status and production, with photo/video reports for transparency, plus a direct messaging feature inside the platform.
2	Suggestion: add 'water availability' as a farm filter criterion.
3	Want to see estimated profit and loss breakdowns before committing to an investment.
4	Focus on ensuring qualified labor on farms.
5	Option to split the harvest between selling, donating, and home delivery would be valuable.

6	Farm detail pages should include road type (paved/dirt), groundwater availability, and soil type.
7	Would like to see harvest statistics and expected crop count estimates.
8	Groundwater depth and abundance should be a filter/detail field for farms.

Table 5 : Investor Suggestions

The most repeated theme in open responses: environmental farm data (water, soil, road access) is a missing but desired filter. This is not currently planned and represents a potential future feature or an enhancement to the farm registration form.

3.1.2.2 Part 2 — Farmer Survey (n = 36)

Key Findings at a Glance

Question	Key Finding
Prior leasing experience	**69% interested but never leased before**
Most wanted investor info	**Full name & ID + Area & Duration — highest**
Preferred response window	**One full week — 44%**
Update content type	**Photos — 69%**
Update frequency	**Monthly — 33%**
Preferred notification channel	**WhatsApp — 75%**
Most wanted dashboard info	**Estimated profits + investor name list — highest**

Table 6 : Summary of Key Findings from the Farmer Survey

Q1 — Have you previously leased part of your farm?

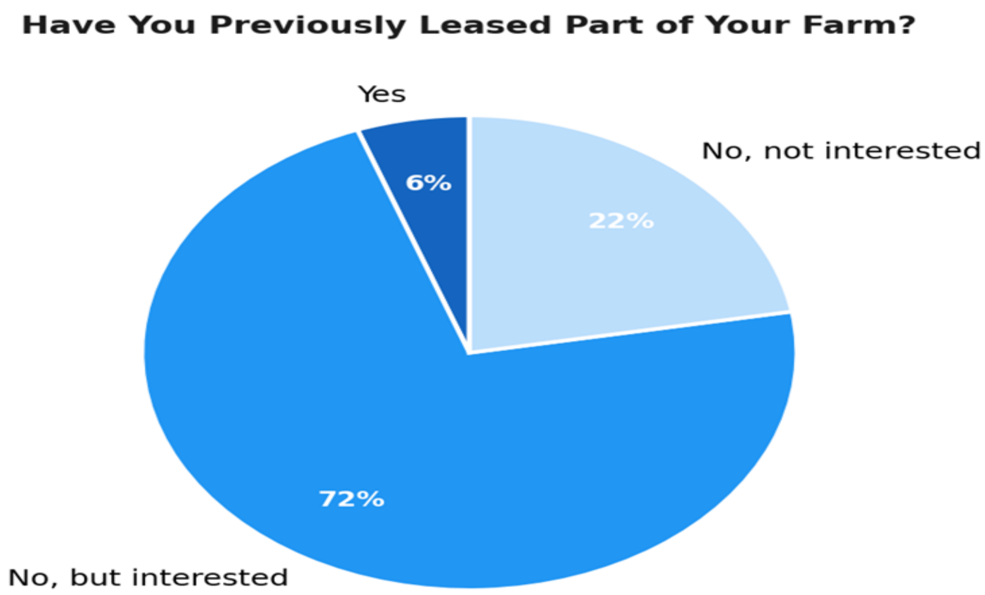

Figure 9 : Questionnaire Results (Q1 | Farmer)

69% are interested but have never leased before ,Qinwan is entering an untapped market. These first-time users will need an onboarding flow that clearly explains the leasing process and builds confidence in the platform from day one.

Q2 — What investor information do you need when receiving a request? (multiple choice)

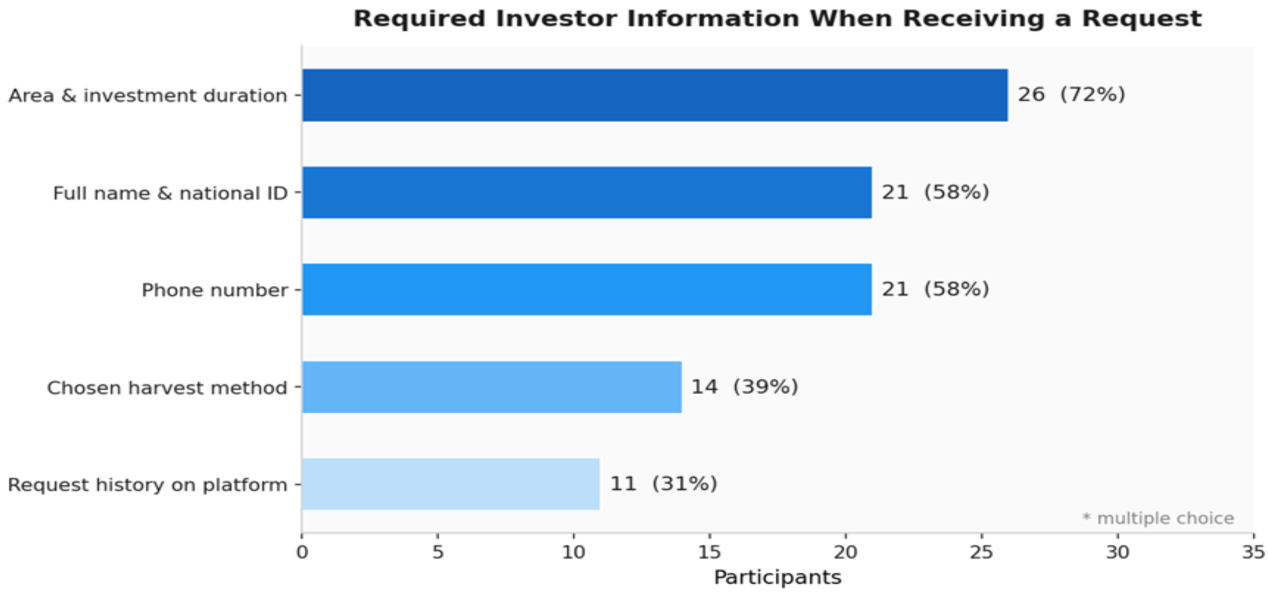

Figure 10 : Questionnaire Results (Q2 | Farmer)

Full area & duration (72%) leads, followed by full name & ID and phone number (both 58%). Farmers want identity verification before they approve any investment. The investor profile page must surface this information clearly and upfront.

Q3 — How long do you need to respond to an investment request?

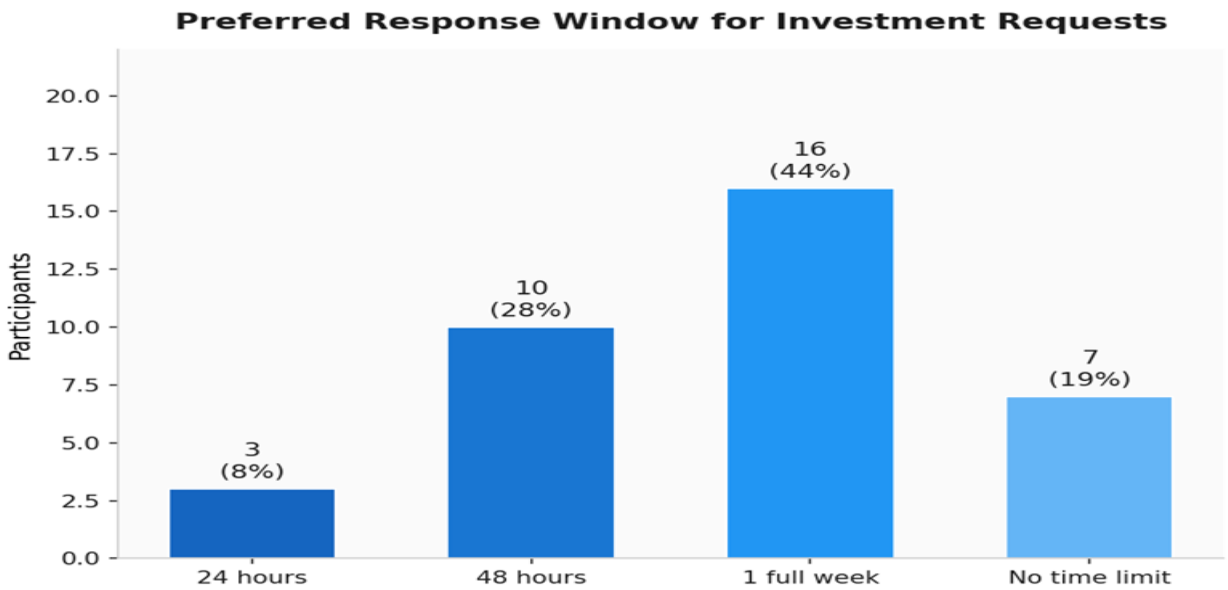

Figure 11 : Questionnaire Results (Q3 | Farmer)

44% need a full week to respond. The system must include automated reminders sent to the farmer as the deadline approaches , otherwise requests will expire silently and investors will be left waiting with no feedback.

Q4 — What content will you share in periodic updates? (multiple choice)

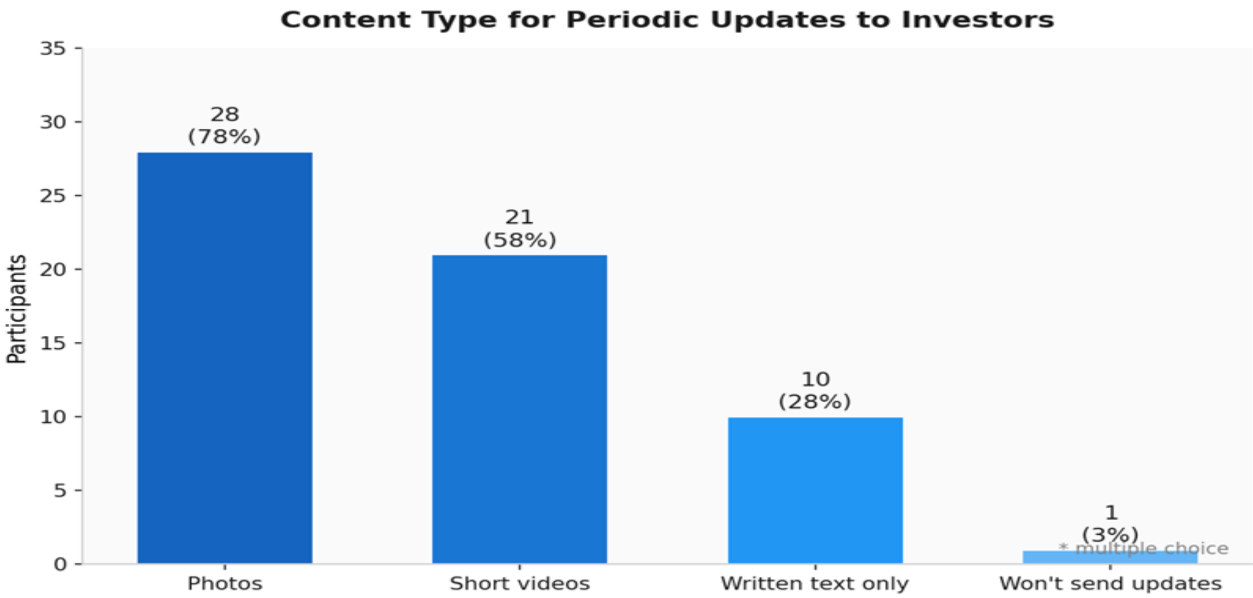

Figure 12 : Questionnaire Results (Q4 | Farmer)

Photos lead at 69%, followed by video (58%) and text+photos (42%). This is consistent with investor preferences , both sides want visual content. The update posting feature must make it easy to attach photos and videos directly from a mobile device.

Q5 — How often will you send updates to investors?

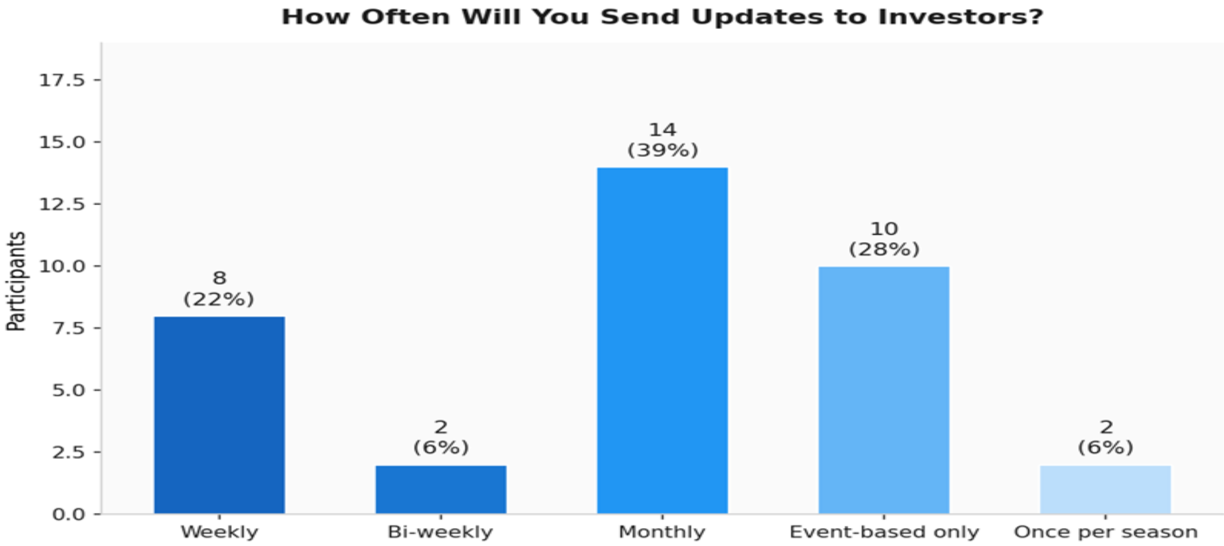

Figure 13 : Questionnaire Results (Q5 | Farmer)

Monthly (33%) and event-driven updates (25%) are the top two answers combined account for 58% of farmers. The platform should allow farmers to schedule recurring updates and trigger event-based ones (e.g., 'harvest started'). A gentle monthly reminder to post an update would improve investor experience significantly.

Q6 — How do you prefer to receive platform notifications?

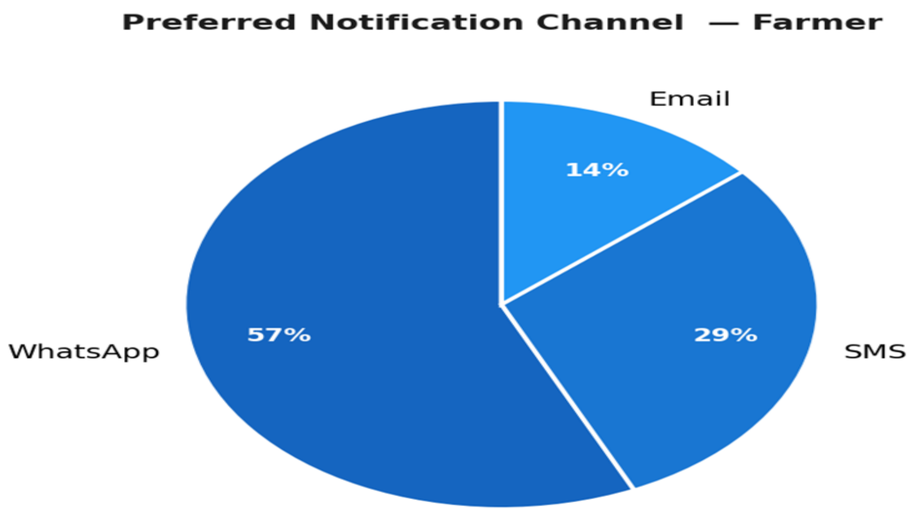

Figure 14 : Questionnaire Results (Q6 | Farmer)

WhatsApp is the top choice at 75% , matching investor preference at 73%. This cross-survey alignment makes WhatsApp integration the single highest-priority notification infrastructure decision for the entire platform.

Q7 — What information do you want in your farm dashboard? (multiple choice)

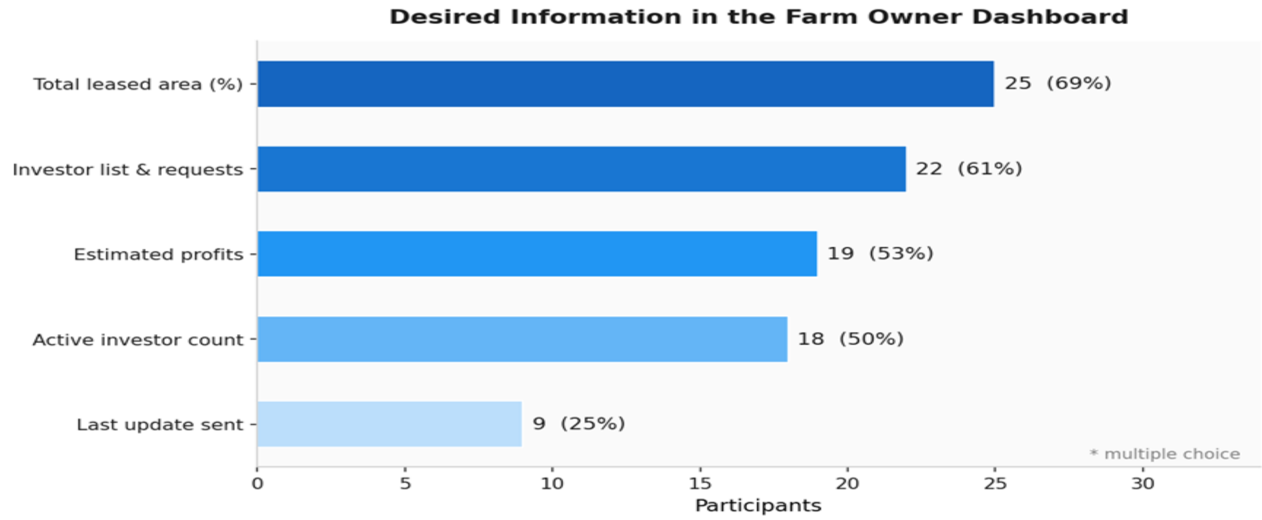

Figure 15 : Questionnaire Results (Q7 | Farmer)

Farmers want a financial + operational overview: estimated profits, list of investors and their requests, and the number of active investors are the top choices. The dashboard must act as both a management tool and a business summary, not just a status page.

Open-Ended Question — Farmer Challenges

#	Response
1	Hardest challenge: finding trustworthy investors who follow through. Qinwan can help by providing a vetted investor base, a smart tracking dashboard, and periodic updates — increasing transparency for both sides.
2	Finding serious tenants who pay on time and treat the farm with care is the main challenge.
3	Farm ownership documentation (shared deeds, old paper titles) may complicate registration on the platform.
4	There is no dedicated platform for farm leasing in Saudi Arabia — Qinwan fills this gap directly.
5	Marketing the farm and finding investors is the main challenge. Confident that Qinwan can solve this.

6	Collecting rent from tenants is a recurring obstacle — the platform's payment system should address this.
7	Suggest expanding investment types: buying dates from palm trees, full farm leasing, seasonal harvesting, etc.
8	Finding an investor who pays, respects the farm, and returns it in the same condition is the hardest part — an investor rating system would directly solve this.

Table 7 : Farmer Challenges

The most frequently mentioned challenge across farmer responses: finding a trustworthy, serious investor. An investor rating and review system, combined with a verified identity requirement, would directly address this and increase farmer confidence in using the platform.

Summary of Requirements Elicitation Findings

The system requirements were identified using surveys and interviews with potential investors and farm owners. The survey results highlighted key user needs such as browsing farms using map and list views, receiving visual updates through photos and videos, and using WhatsApp for notifications.

The interviews provided deeper insights, showing that investors require clear farm information such as water availability, soil quality, and expected profit, while farm owners emphasized the importance of simple platform usage and sharing farm updates with investors.

Confirmed Features by Priority

Priority	Feature	Evidence
High	**Map View + List View (dual browsing)**	64% of investors — both views required
High	**Filter by palm/date type & farm rating**	Top two filter criteria across investors
High	**High-quality photo & video upload for farms**	4.3/5 avg importance + 71% want video updates; interviews confirmed preference for visual farm monitoring through photos, videos, or farm visits.
High	**WhatsApp notifications**	73% investors, 75% farmers prefer WhatsApp; interviews highlighted the need for direct communication between investors and farmers.

High	**Harvest sale & profit transfer system**	67% of investors prefer sell & receive profits
Medium	**Investor rating & review system**	Repeated in 4+ open-ended farmer responses
Medium	**Automated farmer reminder before request expires**	44% need a full week — silent expiry is a risk
Medium	**Environmental farm filters (water, soil, road)**	Mentioned by 4+ investors in open responses; interviews confirmed the importance of environmental information such as water availability, soil fertility, and palm type before investing.
Low	**Harvest split option (sell + donate combined)**	Suggested by 1 investor as a nice-to-have

Table 8 : Confirmed Features by Priority

3.2 System Users

This section describes the detailed characteristics of the main users of the platform. The system includes two main types of users: individual investors and farm owners who participate in agricultural investment through the platform.

1. Individual Investors

- Mainly **Saudi citizens or residents** interested in local agricultural investment.
- May have **different educational levels**.
- May **have previous experience in agriculture or may be new to the sector**, as some investors are interested in passive investment without managing the farm directly.
- Have **basic to intermediate technical knowledge** in using websites and mobile applications.
- Usually use **smartphones or personal computers** to access the platform.
- Usually interested **in agricultural investment opportunities** and following farm performance through the platform

2. Farm Owners

- Typically adults aged between **30 and 65 years**.
- Mainly **Saudi farm owners** who manage palm farms.
- May have **different educational levels**.
- Have **strong experience in agriculture and palm farming**, including farm management and production knowledge.
- Have **basic to intermediate technical knowledge** in using websites and mobile applications.
- Usually use **smartphones** to access the platform.
- Aim to **promote their farms, attract reliable investors, increase production, and generate additional revenue** through the platform.

3.3 Use Case Diagram

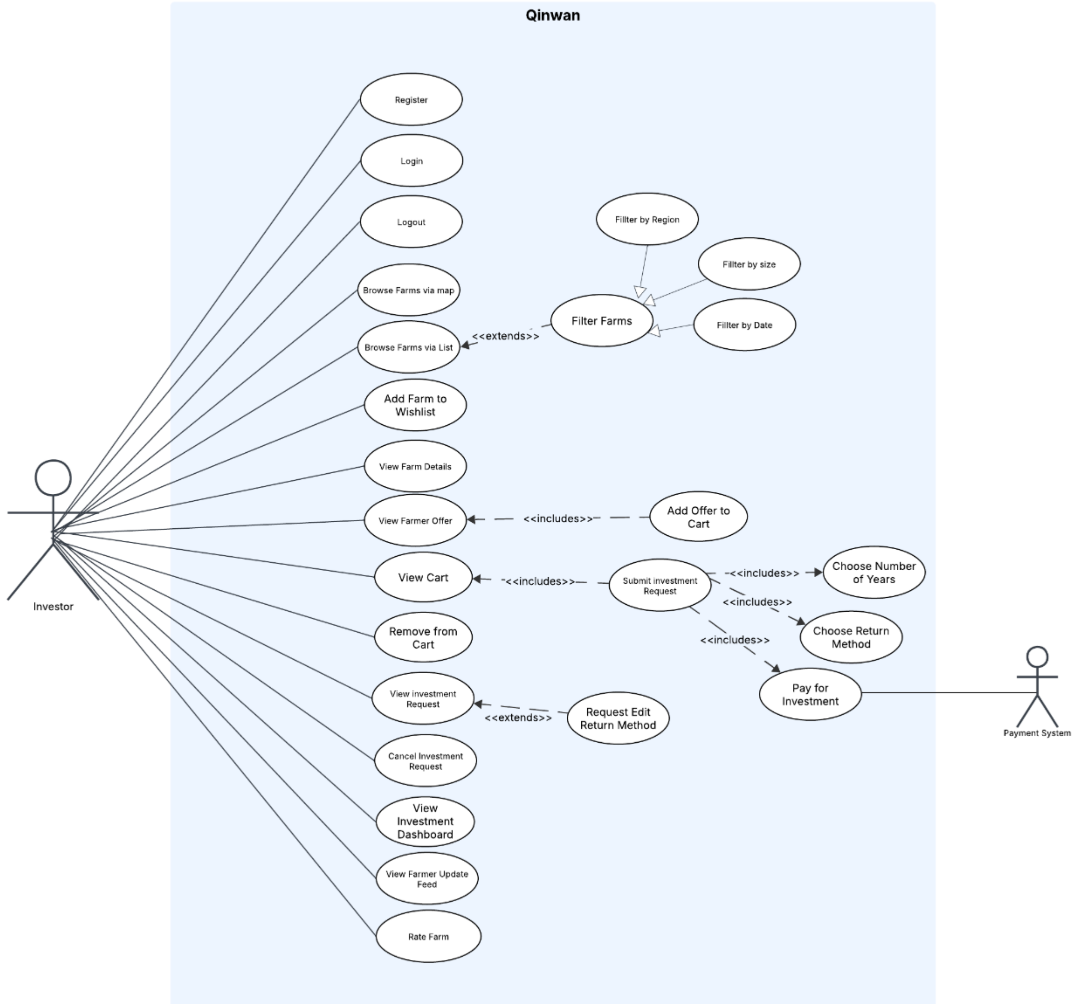

Figure 16 : Use Case Diagram -1

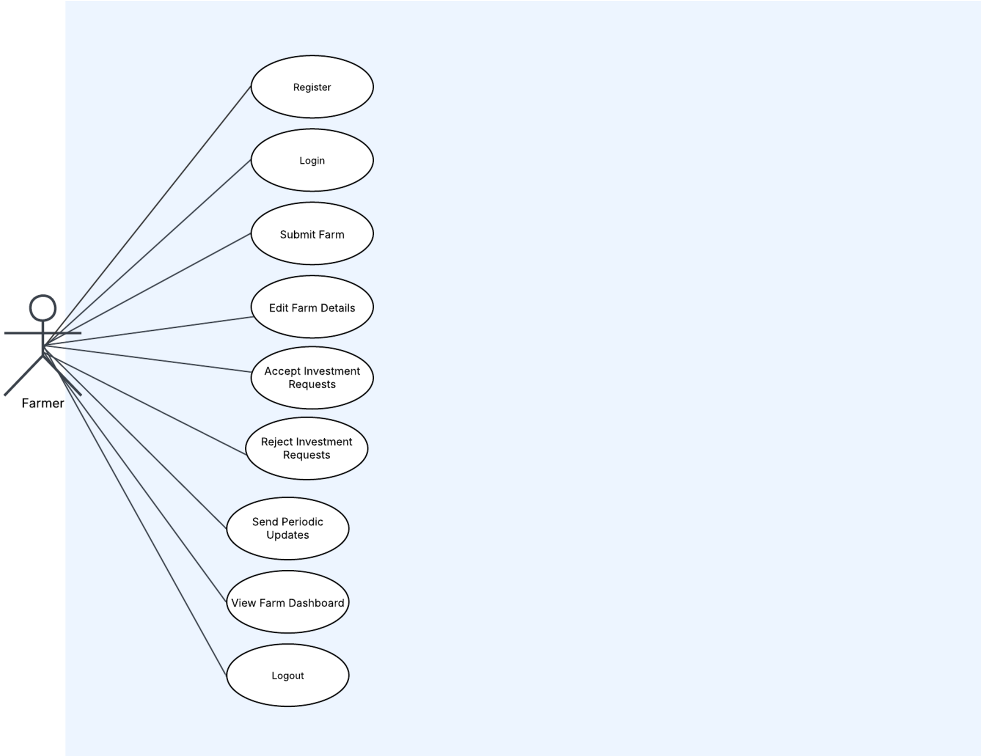

Figure 17 : Use Case Diagram -2

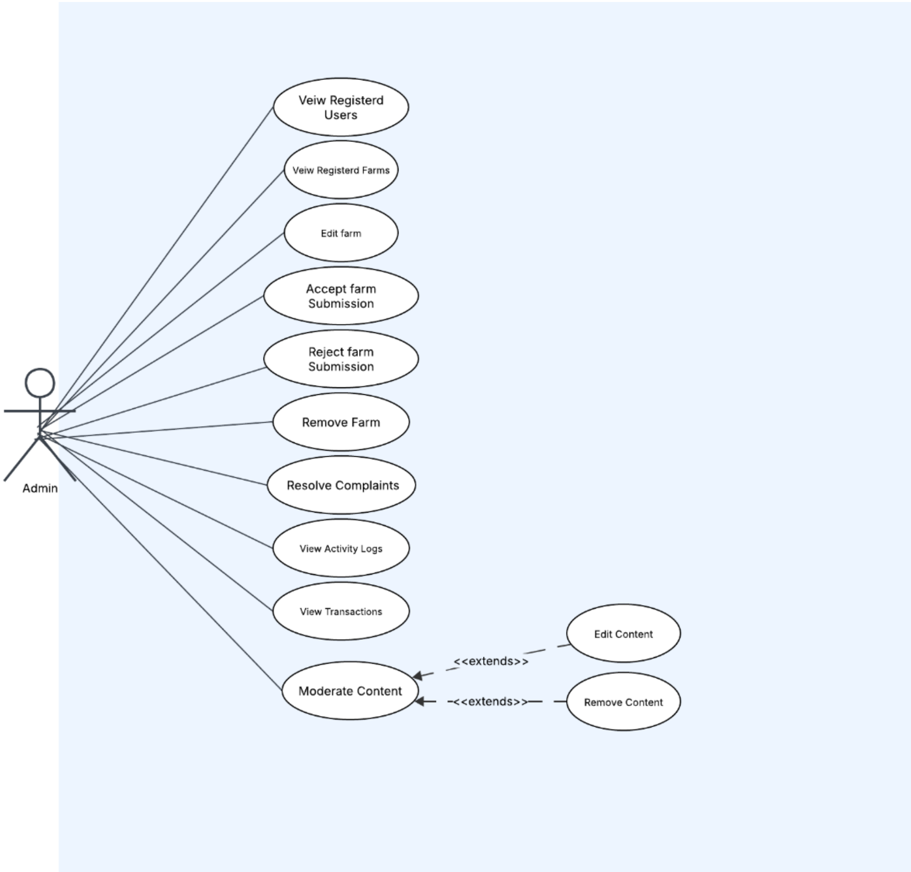

Figure 18 : Use Case Diagram –3

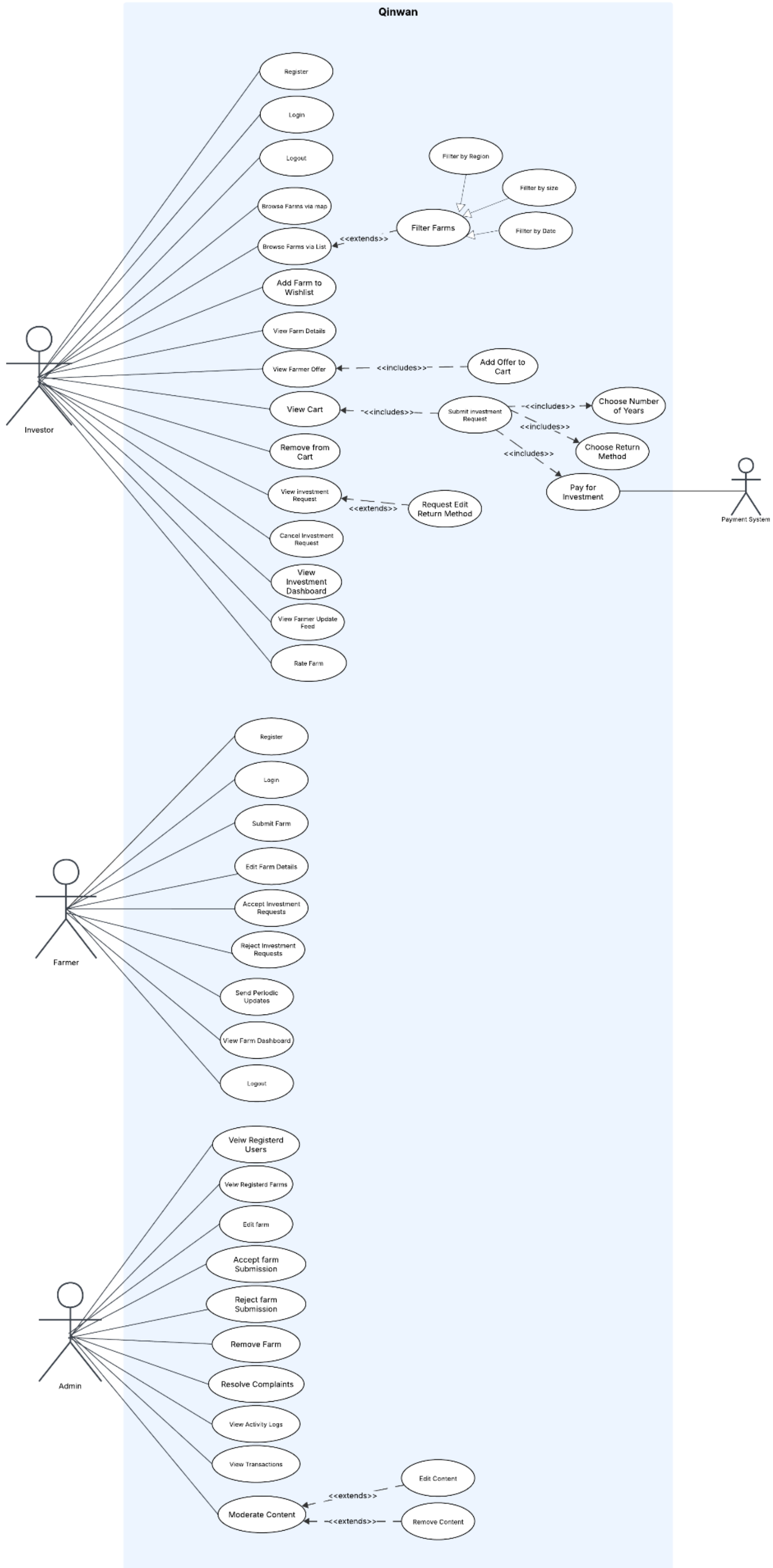

Figure 19 : Use Case Diagram - full diagram

3.4 Product Backlog

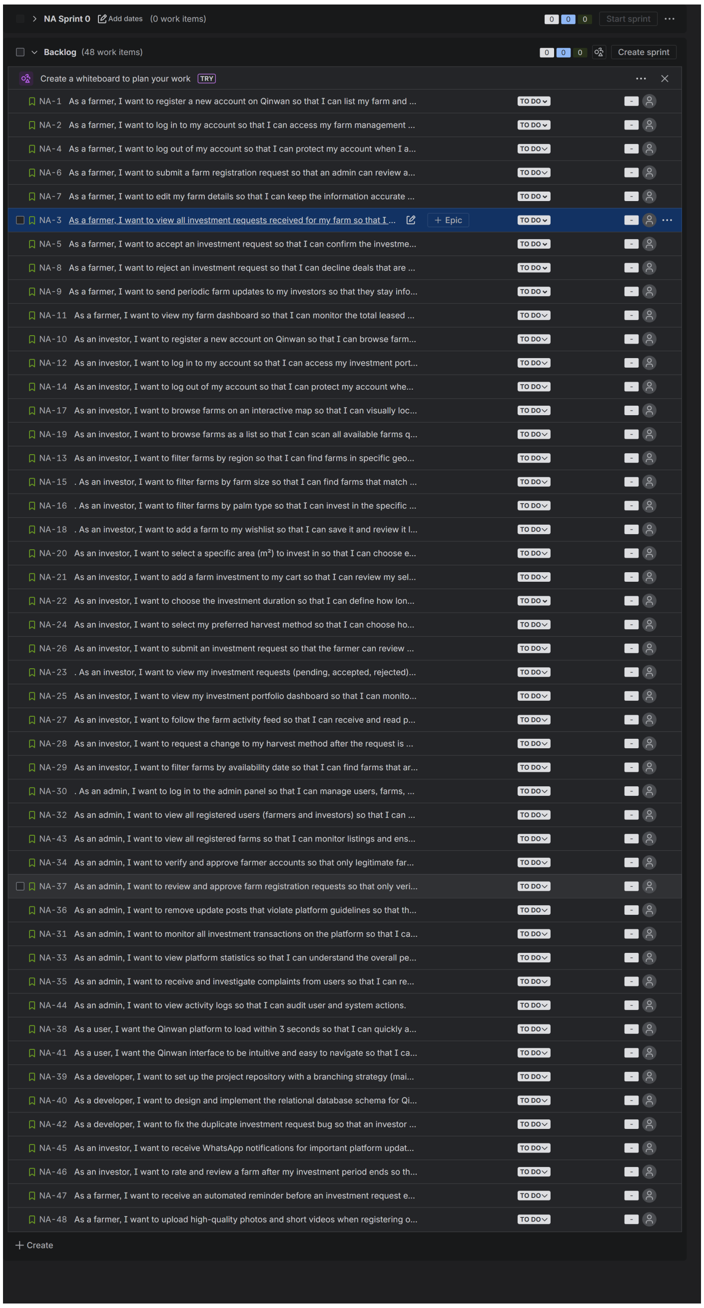

Figure 20 : Product Backlog

PBI (User Story)	Size (Story Points)	Type	Acceptance Criteria
Farmer			
US-01. As a farmer, I want to register a new account on Qinwan so that I can list my farm and receive investment requests.	**5**	**Feature**	• The registration form must include: full name, national ID, phone number, email, and password. • The username must be 4–20 characters (letters, numbers, underscores, or periods only). • Email must follow the format xxxx@xxx.xxx; duplicates are rejected with a clear error message. • Password must contain at least 8 characters, one uppercase, one lowercase, one number, and one special character. • Passwords must match in the confirm-password field before submission. • The system must hash and salt passwords before storing them in the database. • A confirmation email is sent to the farmer after successful registration. • After registration, the farmer is automatically logged in and redirected to their dashboard.
US-02. As a farmer, I want to log in to my account so that I can access my farm management dashboard securely.	**3**	**Feature**	• Login requires a valid registered email and correct password. • If credentials are invalid, the system displays: 'Invalid email or password. Please try again.' • After 5 consecutive failed attempts, the account is temporarily locked for 15 minutes. • Upon successful login, the farmer is redirected to their personal dashboard. • The system must use HTTPS for all login data transmission.
US-03. As a farmer, I want to log out of my account so that I can protect my account when I am done.	**1**	**Feature**	• A 'Log Out' button must be visible and accessible from any page within the dashboard.

			• The farmer is redirected to the login page after logout. • Accessing any protected page after logout must redirect the user to the login page.
US-04. As a farmer, I want to submit a farm registration request so that an admin can review and approve my farm to appear on the platform.	**8**	**Feature**	• The farm submission form must include: farm name, location (city, region), total area (m^2), type of palm, number of palm trees, expected yield (kg/season), and photos. • The farmer must be logged in before submitting a farm registration request. • All required fields must be validated before submission; missing fields display specific error messages. • After submission, the farm status is set to 'Pending Review'. • The farmer receives an notification confirming that the request has been submitted. • The system prevents duplicate farm registrations with the same name and location.
US-05. As a farmer, I want to edit my farm details so that I can keep the information accurate and up to date.	**5**	**Feature**	• The farmer can edit: farm name, total area, expected yield, palm type, photos, and description. • The farmer must be authenticated and own the farm to make any edits. • Edited farms that change core details (area, location) must be re-reviewed and re-approved by the admin. • Changes are saved with a timestamp and logged for audit purposes. • The farmer receives a confirmation notification after successful edits.
US-06. As a farmer, I want to view all investment requests received for my farm so that I can make informed acceptance or rejection decisions.	**5**	**Feature**	• The farmer's dashboard must show a list of all investment requests with: investor name, requested area (m^2), selected harvest method, and requested start date. • Requests must be filterable by status: Pending, Accepted, Rejected. • Each request must display the investor's verified profile badge if available.

			• The list must update in real-time or refresh upon page reload.
US-07. As a farmer, I want to accept an investment request so that I can confirm the investment deal with the investor.	**5**	**Feature**	• The farmer can only accept a request if the requested area does not exceed the remaining available area on the farm. • Upon acceptance, the system updates the farm's available area and marks the request as 'Accepted'. • The investor receives an immediate notification about the acceptance. • The system logs the acceptance with a timestamp. • The farmer cannot accept the same area from two different investors simultaneously.
US-08. As a farmer, I want to reject an investment request so that I can decline deals that are not suitable.	**3**	**Feature**	• The farmer can reject any pending request with an optional reason field. • Upon rejection, the request is marked as 'Rejected' and the investor is notified immediately. • Rejected requests are moved to a separate 'Rejected' section in the dashboard. • The farmer cannot reverse a rejection without creating a new manual arrangement.
US-09. As a farmer, I want to send periodic farm updates to my investors so that they stay informed about the farm's progress.	**5**	**Feature**	• The farmer can publish text and image updates from the farm management dashboard. • Updates are only delivered to investors who have an accepted investment in that specific farm. • Each update is timestamped and appears in the investor's activity feed. • The farmer can only send updates if they have at least one active investor. • The admin can remove any update that violates platform guidelines.
US-10. As a farmer, I want to view my farm dashboard so that I can monitor the total leased area, number of investors, and expected yield.	**5**	**Feature**	• The dashboard must display: total farm area (m^2), total leased area (m^2), remaining available area (m^2), number of active investors, expected yield (kg/season), and estimated revenue.

			• Data must refresh automatically or upon page reload. • All numeric values must be accurate and consistent with the accepted investment records in the system.
US-11 . As a farmer, I want to receive an automated reminder before an investment request expires so that I do not miss responding to pending requests.	**3**	**Feature**	•The system sends a reminder notification to the farmer 24 hours before a pending investment request expires. •A second reminder is sent 2 hours before expiry if the farmer has not yet responded. •Reminders are delivered via WhatsApp and SMS. •If the farmer does not respond before the deadline, the request is automatically marked as 'Expired' and the investor is notified. •Expired requests are moved to a separate 'Expired' tab in the farmer's dashboard.
Investor			
US-01. As an investor, I want to register a new account on Qinwan so that I can browse farms and submit investment requests.	**5**	**Feature**	• The registration form must include: full name, phone number, email, and password. • Email format must be validated (xxxx@xxx.xxx); duplicate emails are rejected with a clear error message. • Password must contain at least 8 characters, one uppercase, one lowercase, one number, and one special character. • Passwords must match in the confirm-password field. • The system hashes and salts all passwords before storage. • A confirmation email is sent to the investor after registration. • After registration, the investor is automatically logged in and redirected to the home/browse page.
US-02. As an investor, I want to log in to my account so that I can access my investment portfolio and browse farms.	**3**	**Feature**	• Login requires a valid registered email and correct password. • If credentials are invalid, the system displays a clear error message.

			• After 5 consecutive failed attempts the account is locked for 15 minutes. • Upon successful login, the investor is redirected to their personal dashboard.
US-03. As an investor, I want to log out of my account so that I can protect my account when I finish.	**1**	**Feature**	• A 'Log Out' button must be visible and accessible from any page of the investor interface. • Upon logout, the session is terminated immediately and all session tokens are invalidated. • The investor is redirected to the login page.
US-04. As an investor, I want to browse farms on an interactive map so that I can visually locate farms across Saudi Arabia and explore their details.	**8**	**Feature**	• The map must display all admin-approved farms as interactive pins. • Clicking a farm pin must display a summary card: farm name, region, total area, available area, palm type, and expected yield. • The map must be integrated with a national mapping system . • The map must load within 3 seconds for 90% of users. • If no farms exist in a region, no pins are shown and no error is thrown.
US-05. As an investor, I want to browse farms as a list so that I can scan all available farms quickly without using the map.	**5**	**Feature**	• The list view must show all approved farms with: name, region, available area, palm type, and expected yield. • Each list item must be clickable and redirect to the full farm detail page. • The list must reflect real-time availability of farm areas.
US-06. As an investor, I want to filter farms by region so that I can find farms in specific geographic areas of interest.	**3**	**Feature**	• The region filter must list all available Saudi regions/cities where farms exist. • Selecting a region updates both the map pins and the list view dynamically. • If no farms match the selected region, the system displays: 'No farms found in this region.' • The filter can be cleared with a single 'Clear Filters' button.

US-07. As an investor, I want to filter farms by farm size so that I can find farms that match my investment capacity.	**3**	**Feature**	• The size filter must offer predefined ranges (e.g., 0–500 m^2, 500–2000 m^2, 2000+ m^2). • Applying the filter updates the results list and map dynamically. • If no farms match the size filter, the system displays: 'No farms match your selected size range.' • The filter can be cleared with a single 'Clear Filters' button.
US-08. As an investor, I want to filter farms by palm type so that I can invest in the specific variety of dates I prefer.	**3**	**Feature**	• The palm type filter must list all available date varieties registered in the system. • Applying the filter updates the results list and map dynamically. • Multiple palm types can be selected simultaneously. • If no farms match the selected type, the system displays: 'No farms match your selected palm type.'
US-09. As an investor, I want to add a farm to my wishlist so that I can save it and review it later before making an investment decision.	**3**	**Feature**	• A 'Add to Wishlist' button must be visible on every farm card and farm detail page. • Clicking the button adds the farm to the investor's personal wishlist section. • Wishlist items persist across sessions (saved to the investor's account). • The investor can remove items from the wishlist at any time. • The wishlist is private and not visible to other users.
US-10. As an investor, I want to select a specific area (m^2) to invest in so that I can choose exactly how much land I wish to lease.	**5**	**Feature**	• The area selection input must only accept numeric values greater than zero. • The selected area must not exceed the farm's remaining available area; the system displays an error if exceeded. • The system must display the cost preview based on the selected area before the investor proceeds. • The investor cannot proceed to the cart without specifying a valid area.

US-11. As an investor, I want to add a farm investment to my cart so that I can review my selections before submitting a request.	**5**	**Feature**	• The cart must display: farm name, selected area (m^2), investment duration, selected harvest method, and estimated cost. • The investor can add multiple farms to the cart in a single session. • Items in the cart are preserved for up to 24 hours. • The investor can delete any item from the cart at any time.
US-12. As an investor, I want to choose the investment duration so that I can define how long I wish to lease the farm area.	**3**	**Feature**	• The investor must select a duration from predefined options (e.g., 1 season, 1 year, 2 years). • The selected duration must be shown clearly in the cart and on the submitted request. • The investor cannot submit an investment request without selecting a valid duration.
US-13. As an investor, I want to select my preferred harvest method so that I can choose how I will benefit from the farm's produce.	**5**	**Feature**	• The investor must choose one harvest method from: (1) receive dates delivered to their home, (2) sell the harvest and receive profits, or (3) donate the harvest to a charity. • The selected method must be displayed in the cart and on the investment request. • If the investor selects home delivery, a delivery address field becomes mandatory. • The investor cannot submit a request without selecting a harvest method.
US-14. As an investor, I want to submit an investment request so that the farmer can review and accept my proposed investment.	**8**	**Feature**	• The investor must be logged in with a verified account to submit a request. • The cart must contain at least one valid item before submission is allowed. • Upon submission, the request is sent to the corresponding farmer and marked as 'Pending'. • The investor receives an on-platform and email notification confirming request submission.

			• The investor can view the submitted request in the 'Pending Requests' section of their dashboard. • If the cart is empty, the system shows: 'Please select a farm area before submitting a request.'
US-15. As an investor, I want to view my investment requests (pending, accepted, rejected) so that I can track the status of all my submissions.	**5**	**Feature**	• The requests section must display three tabs: Pending, Accepted, Rejected. • Each request must show: farm name, requested area, duration, harvest method, submission date, and current status. • The investor can cancel a pending request at any time before the farmer responds. • Accepted and rejected requests are read-only and cannot be cancelled.
US-16. As an investor, I want to view my investment portfolio dashboard so that I can monitor all my active investments at a glance.	**8**	**Feature**	• The dashboard must display for each active investment: farm name, owner's contact, location on mini-map, leased area (m^2), palm type in the region, expected yield, and estimated profit/loss summary. • The dashboard must show a total summary: total m^2 leased, total estimated yield, total estimated profit. • Data must be accurate and reflect the accepted investment records in real-time.
US-17. As an investor, I want to follow the farm activity feed so that I can receive and read periodic updates sent by the farmers I have invested in.	**5**	**Feature**	• The activity feed must display updates only from farms where the investor has an accepted investment. • Each update must show: farm name, update text, attached image (if any), and timestamp. • New updates must appear at the top of the feed. • The investor receives a push notification when a new update is posted by a farmer they are invested with.
US-18. As an investor, I want to request a change to my harvest method after the request is accepted so that I can adjust how I receive my produce.	**5**	**Feature**	• This option is only available for investments that have already been accepted by the farmer. • The investor can submit a change request; the farmer must approve or reject the change.

			• The investor is notified of the farmer's decision via the platform and email. • Only one pending harvest-method change request is allowed per investment at a time.
US-19. As an investor, I want to receive WhatsApp notifications for important platform updates so that I can stay informed .	**3**	**Feature**	•The investor can enable WhatsApp notifications from their profile settings. •Notifications are sent for the following events: investment request accepted or rejected, new farm update posted, harvest method change request approved or rejected. •The investor must provide and verify a valid WhatsApp number during registration or from settings. •If WhatsApp delivery fails, the system falls back to SMS. •The investor can disable WhatsApp notifications at any time from their settings.
US-20. As an investor, I want to rate and review a farm after my investment period ends so that I can share my experience with other investors.	**3**	**Feature**	•The rating option only appears after the investment period for that farm has officially ended. •The investor can submit a rating from 1 to 5 stars and an optional written review. •Each investor can only submit one rating per investment. •The rating is publicly visible on the farm's detail page. •The investor cannot edit or delete their rating after submission. •The farm's overall rating displayed on its card and detail page is calculated as the average of all submitted ratings.
Admin			
US-01. As an admin, I want to view all registered users	**3**	**Feature**	• The admin panel must display a list of all users with: name, role

(farmers and investors) so that I can monitor platform membership.			(farmer/investor), registration date, and status (active/suspended). • The list must be searchable by name and filterable by role or status. • The admin can click on any user to view their full profile and activity.
US-02. As an admin, I want to verify and approve farmer accounts so that only legitimate farm owners operate on the platform.	**8**	**Feature**	• The admin reviews the farmer's submitted national ID and farm ownership documents. • The admin can approve or reject a farmer account with a mandatory note. • Upon approval, the farmer receives a notification and can begin listing farms. • Upon rejection, the farmer receives a notification with the stated reason. • Approved farmers are marked with a verification badge on the platform.
US-03. As an admin, I want to review and approve farm registration requests so that only verified and accurate farms appear on the platform.	**8**	**Feature**	• The admin sees all pending farm registration requests with full details (name, location, area, photos, palm type). • The admin can approve the farm, which makes it visible to investors on the map and list. • The admin can reject the farm with a mandatory rejection reason sent to the farmer. • The admin can request additional information from the farmer before making a decision. • Approval and rejection actions are logged with a timestamp and admin ID.
US-04. As an admin, I want to edit or remove farm listings that violate platform guidelines so that the platform maintains quality and trust.	**5**	**Feature**	• The admin can edit any farm field (name, description, area, photos) at any time. • The admin can deactivate or permanently delete a farm with a mandatory reason. • Affected farmers are notified immediately upon any admin edit or deletion. • All admin edits and deletions are logged in the audit trail with a timestamp.

US-05. As an admin, I want to monitor all investment transactions on the platform so that I can ensure financial integrity and resolve issues.	**8**	**Feature**	• The admin panel must display all transactions with: investor name, farmer name, farm name, area, duration, harvest method, status, and date. • Transactions are filterable by status (pending, accepted, rejected, completed) and date range. • The admin can view the full detail of any individual transaction. • The admin receives an alert for any transactions flagged as suspicious.
US-06. As an admin, I want to view platform statistics so that I can understand the overall performance and growth of Qinwan.	**5**	**Feature**	• The admin dashboard must display: total registered users, total active farms, total leased area (m^2), total investment transaction volume, and new registrations per month. • Statistics must update in real-time or at most with a 5-minute delay. • The admin can export statistics as an Excel or PDF report.
US-07. As an admin, I want to receive and investigate complaints from users so that I can resolve disputes fairly and maintain platform trust.	**8**	**Feature**	• Users (farmers and investors) can submit a complaint via a dedicated 'Report a Problem' form. • The admin sees all complaints in a dedicated complaints queue with: submitter, subject, description, and date. • The admin can change the complaint status to: Under Investigation, Resolved, or Dismissed. • Resolved complaints are archived and not deleted.
US-08. As an admin, I want to remove update posts that violate platform guidelines so that the community content stays appropriate and respectful.	**3**	**Feature**	• The admin can view all farmer-posted updates across all farms. • The admin can delete any update with a mandatory reason. • The farmer whose update was deleted is notified with the reason.
US-09.As an admin, I want to view all registered farms so that I can monitor listings and ensure platform compliance.	**3**	**Feature**	•The admin panel displays a complete list of all registered farms.

			•Each farm record shows: farm name, owner name, location, area (m^2), palm type, and status. •The list is searchable by farm name or owner name. •The list is filterable by status (pending, approved, rejected, deactivated). •The admin can click on any farm to view full details.
US-10. As an admin, I want to view activity logs so that I can audit user and system actions.	**5**	**Feature**	•The admin can access a dedicated "Activity Logs" page. •The system displays logs including: user ID, action type, timestamp, and affected entity •Logs include actions such as: account approval, farm approval, edits, deletions, and transaction updates. •The logs are searchable by user name or action type. •The logs are filterable by date range. •Logs cannot be edited or deleted by any user. •The system records all actions in real-time.
US-11. As an admin, I want to edit user-generated content so that I can correct minor violations without deleting posts.	**5**	**Feature**	•The admin can access a dedicated "Activity Logs" page. •The system displays logs including: user ID, action type, timestamp, and affected entity. •Logs include actions such as: account approval, farm approval, edits, deletions, and transaction updates. •The logs are searchable by user name or action type. •The logs are filterable by date range. •Logs cannot be edited or deleted by any user. •The system records all actions in real-time.
Non-Functional Requirements			

US-01. As a user, I want the Qinwan platform to load within 3 seconds so that I can quickly access my data without frustration.	**5**	**Feature**	• The homepage and main browse page must load in under 3 seconds for 90% of users . • API response time must not exceed 1 second. • The interactive farm map must fully render within 3 seconds after page load.
US-02. As a user, I want the Qinwan interface to be intuitive and easy to navigate so that I can complete key tasks without confusion.	**5**	**Feature**	• A new user must be able to register and submit their first investment request within 5 minutes. • Key tasks (browse farms, submit request, view dashboard) must be completable in no more than 4 clicks from the home screen. • Error messages must be in clear Arabic and English, describing the problem and actionable steps to fix it.
Developer Stories			
US-01. As a developer, I want to set up the project repository with a branching strategy (main / develop / feature branches) so that the team can collaborate without overwriting each other's work.	**3**	**Technical Work**	• A Git repository is created and shared with all team members. • Branching conventions are documented: main (production), develop (staging), feature/xxx (per story). • Pull requests require at least one code review approval before merging into develop. • No direct commits are allowed to the main branch. • A README file documents the project setup steps and branching rules.
US-02. As a developer, I want to design and implement the relational database schema for Qinwan so that all entities (users, farms, investments, transactions) are stored consistently and efficiently.	**8**	**Technical Work**	• The database schema includes tables for: Users, Farmers, Investors, Admins, Farms, InvestmentRequests, Transactions, FarmUpdates, Complaints, and Wishlist. • All foreign key relationships are correctly defined and enforced at the database level. • All text fields storing sensitive data (passwords, IDs) use appropriate encrypted or hashed column types. • An Entity-Relationship Diagram (ERD) is documented.

Defect Storie			
US-01. As a developer, I want to fix the duplicate investment request bug so that an investor cannot submit more than one pending request for the same farm.	**3**	**Defect**	• BUG: An investor can currently submit multiple pending requests for the same farm simultaneously, causing inconsistency in farm availability tracking. • FIX: Before allowing a new investment request submission, the system must check if the investor already has a 'Pending' or 'Accepted' request for the same farm. • If a duplicate is detected, the system displays: 'You already have an active request for this farm. Please wait for the farmer's response before submitting a new one.' • Verified by test: attempting to submit a second request for the same farm while the first is pending must be blocked.

Table 9 : Product Backlog

4 Chapter 4: System Design

4.1 System Architecture

The Qinwan platform adopts a Client–Server Architecture, where the system is divided into two main components: the client side and the server side (Figure 4.1)

The client side represents the user interface accessed through a web browser. It enables different types of users, including investors, farmers, and administrators, to interact with the platform. Through this interface, users can register, log in, browse farms, submit investment requests, and manage their activities efficiently.

The server side is responsible for handling the application logic and processing all user requests. It manages authentication, farm data, investment requests, complaints, and controls access to system functionalities. The server ensures that all operations follow the system's business rules and validations.

The system uses a MySQL database to store all platform data, including users, farms, investments, updates, and complaints. The database communicates only with the server to ensure secure data handling and maintain data integrity.

Furthermore, the system follows a layered structure (MVC-inspired design) within the server side, separating the presentation, control, and data layers. This enhances modularity and simplifies maintenance and future development.

Therefore, the Client–Server Architecture was chosen because it separates the user interface from the system logic and data management, making the platform more maintainable, secure, and scalable.

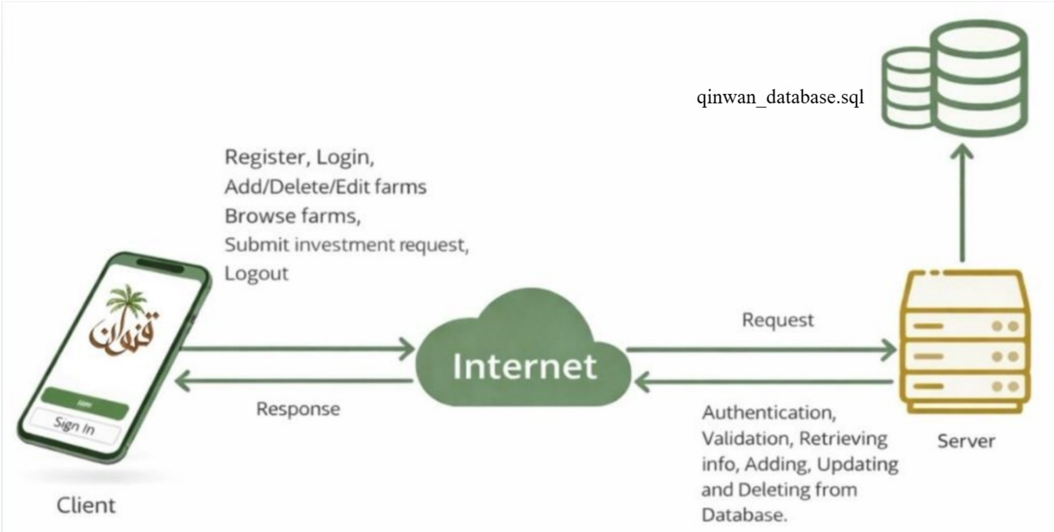

Figure 21: System Architecture of Qinwan

4.2 Class Diagram

In this section, we present the Class Diagram of the "Qiwan" application. The diagram shows the main system classes along with their attributes, methods, and the relationships between them. It provides a high-level view of how the system is structured and how different components interact to support the application's functionality within the object-oriented design.

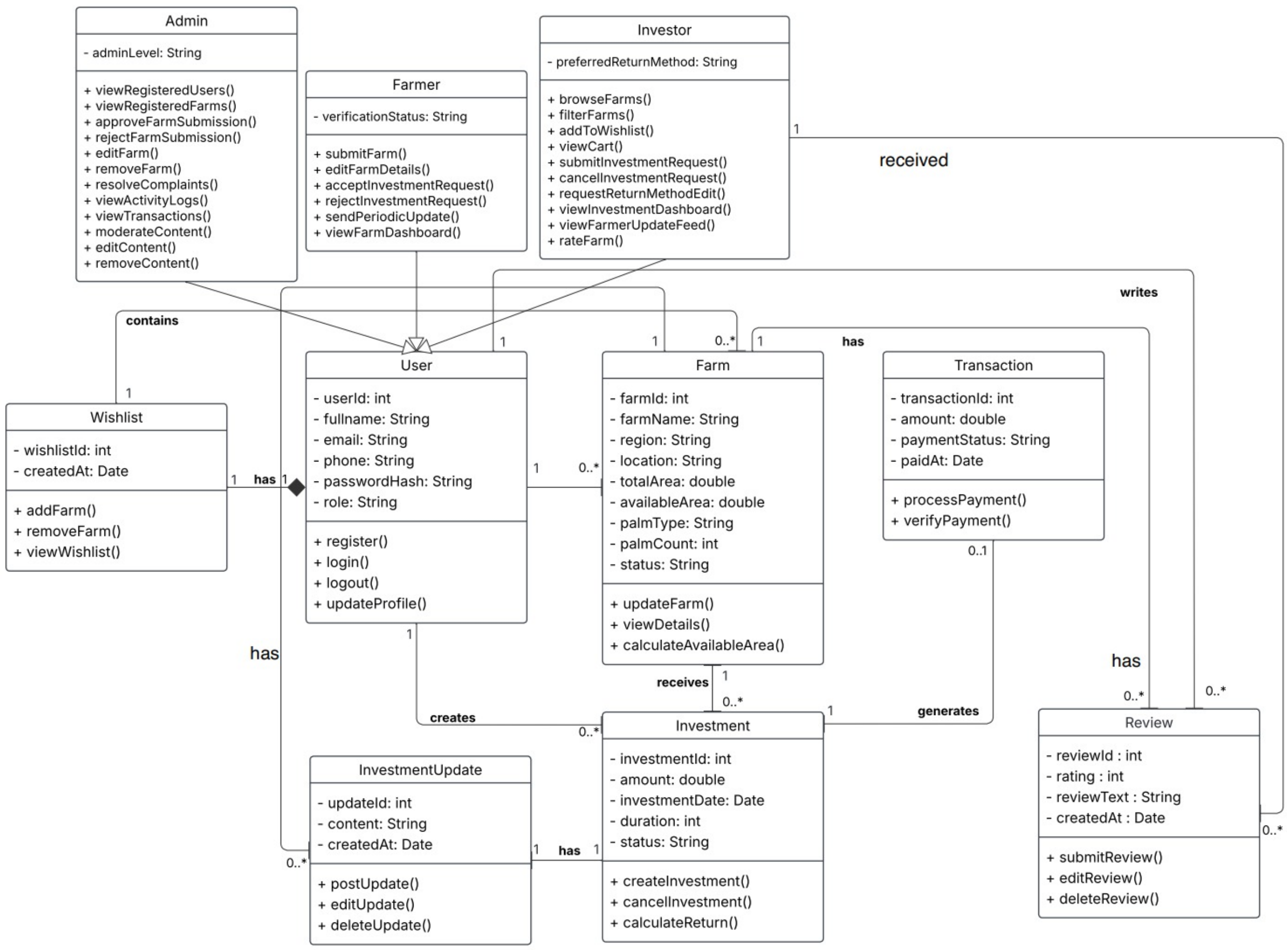

Figure 22: Class Diagram of Qinwan

4.3 Data Design

4.1.1 ER Diagram

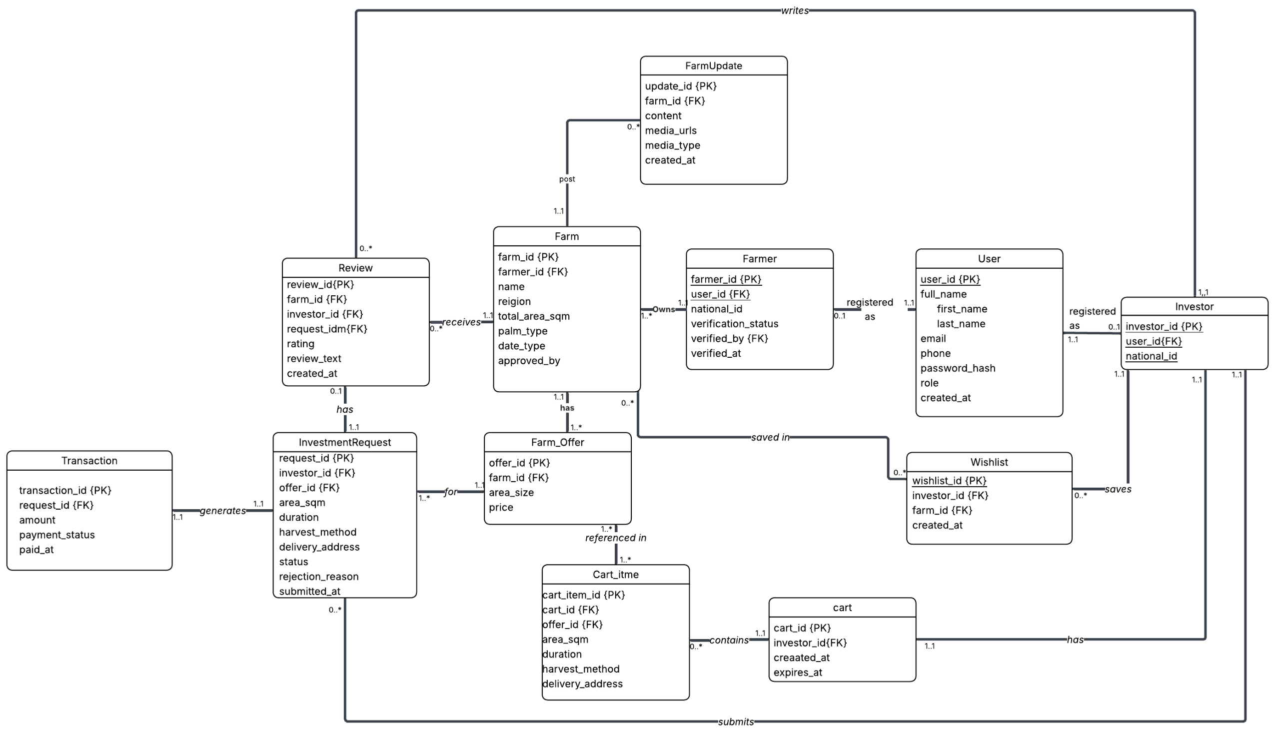

Figure 23: ER Diagram of Qinwan

4.1.2 Relational Schema

The following schema defines all tables in the Qinwan MySQL database, including primary keys, foreign key references, and unique constraints.

-- USER TABLE

User (user_id, first_name, last_name, email, phone, password_hash, role, created_at)

Primary Key: user_id

Unique: email

-- FARMER TABLE

Farmer (farmer_id, user_id, national_id, verification_status, verified_by, verified_at)

Primary Key: farmer_id

Foreign Key: user_id references User(user_id)

Foreign Key: verified_by references User(user_id)

-- INVESTOR TABLE

Investor (investor_id, user_id, national_id)

Primary Key: investor_id

Foreign Key: user_id references User(user_id)

-- FARM TABLE

Farm (farm_id, farmer_id, name, region, total_area_sqm, palm_type, date_type, approved_by)

Primary Key: farm_id

Foreign Key: farmer_id references Farmer(farmer_id)

Foreign Key: approved_by references User(user_id)

-- FARM_OFFER TABLE

Farm_Offer (offer_id, farm_id, area_size, price)

Primary Key: offer_id

Foreign Key: farm_id references Farm(farm_id)

-- FARMUPDATE TABLE

FarmUpdate (update_id, farm_id, content, media_urls, media_type, created_at)

Primary Key: update_id

Foreign Key: farm_id references Farm(farm_id)

-- CART TABLE

Cart (cart_id, investor_id, created_at, expires_at)

Primary Key: cart_id

Foreign Key: investor_id references Investor(investor_id)

-- CART_ITEM TABLE

Cart_Item (cart_item_id, cart_id, offer_id, area_sqm, duration, harvest_method, delivery_address)

Primary Key: cart_item_id

Foreign Key: cart_id references Cart(cart_id)

Foreign Key: offer_id references Farm_Offer(offer_id)

-- INVESTMENTREQUEST TABLE

InvestmentRequest (request_id, investor_id, offer_id, area_sqm, duration, harvest_method, delivery_address, status, rejection_reason, submitted_at)

Primary Key: request_id

Foreign Key: investor_id references Investor(investor_id)

Foreign Key: offer_id references Farm_Offer(offer_id)

-- TRANSACTION TABLE

Transaction (transaction_id, request_id, amount, payment_status, paid_at)

Primary Key: transaction_id

Foreign Key: request_id references InvestmentRequest(request_id)

-- REVIEW TABLE

Review (review_id, farm_id, investor_id, request_id, rating, review_text, created_at)

Primary Key: review_id

Foreign Key: farm_id references Farm(farm_id)

Foreign Key: investor_id references Investor(investor_id)

Foreign Key: request_id references InvestmentRequest(request_id)

-- WISHLIST TABLE

Wishlist (wishlist_id, investor_id, farm_id, created_at)

Primary Key: wishlist_id

Foreign Key: investor_id references Investor(investor_id)

Foreign Key: farm_id references Farm(farm_id)

4.1.3 - Data Dictionary

1. Description of all Qinwan Entities

Table 10 : Description of all Qinwan Entities

Entity Name	Description	Occurrence (Cardinality)
User	Stores all registered users on the platform. Both farmers and investors share this table; the role field distinguishes them.	• One User can be registered as zero or one Farmer. • One User can be registered as zero or one Investor.
Farmer	Extends User for farm owners. Stores national ID and verification details reviewed by an admin before the farmer can list farms.	• Each Farmer belongs to exactly one User (1:1). • One Farmer can own zero or many Farms.
Investor	Extends User for investors. Stores the investor's national ID. An investor can browse farms, build a cart, and submit investment requests.	• Each Investor belongs to exactly one User (1:1). • One Investor has exactly one Cart. • One Investor can submit zero or many InvestmentRequests. • One Investor can save zero or many Wishlists. • One Investor can write zero or many Reviews.
Farm	Stores farm listings submitted by farmers. Each farm must be approved by an admin before investors can browse or invest in it.	• Each Farm belongs to exactly one Farmer. • One Farm can have one or many Farm_Offers. • One Farm can have zero or many FarmUpdates. • One Farm can receive zero or many Reviews. • One Farm can appear in zero or many Wishlists.
Farm_Offer	Represents a specific portion of a farm made available for investment, defined by the farmer with an area size and price.	• Each Farm_Offer belongs to exactly one Farm. • One Farm_Offer can be referenced in zero or many Cart_Items.

		• One Farm_Offer can receive zero or many InvestmentRequests.
FarmUpdate	Periodic updates posted by the farmer for a specific farm. Contains text, media URLs, and media type. Visible only to investors with an accepted request for that farm.	• Each FarmUpdate belongs to exactly one Farm. • One Farm can have zero or many FarmUpdates.
Cart	Each investor has one active cart at a time. Items are preserved for 24 hours. Submitting the cart converts all CartItems into InvestmentRequests.	• Each Cart belongs to exactly one Investor (1:1). • One Cart can contain zero or many Cart_Items.
Cart_Item	Stores each farm offer the investor adds to the cart before submitting. Acts as a draft, holding area, duration, and harvest preferences temporarily.	• Each Cart_Item belongs to exactly one Cart. • Each Cart_Item references exactly one Farm_Offer.
InvestmentRequest	Created when the investor submits their cart. Each CartItem becomes one InvestmentRequest. The farmer reviews and accepts or rejects each request individually.	• Each InvestmentRequest belongs to one Investor and one Farm_Offer. • One accepted InvestmentRequest generates exactly one Transaction. • One completed InvestmentRequest can have zero or one Review.
Transaction	Created only when a farmer accepts an investment request. Records the financial details of the investment including amount and payment status.	• Each Transaction belongs to exactly one InvestmentRequest (1:1).
Review	Written by the investor after the investment period ends. One review is allowed per InvestmentRequest. Public and cannot be edited after submission.	• Each Review is linked to one InvestmentRequest, one Investor, and one Farm. • One Farm can receive zero or many Reviews.
Wishlist	Allows investors to save farms they are interested in for later.	• Each Wishlist entry links one Investor to one Farm.

	Each entry links one investor to one farm. Private to each investor.	• One Investor can save zero or many Farms.

2. Description of Relationships

Table 11 : Description of Relationships

Entity A	Multiplicity	Relationship	Multiplicity	Entity B
User	1..1	registered as	0..1	Farmer
User	1..1	registered as	0..1	Investor
Farmer	1..1	owns	0..*	Farm
Farm	1..1	has	1..*	Farm_Offer
Farm	1..1	posts	0..*	FarmUpdate
Farm	1..1	receives	0..*	Review
Farm	1..1	saved in	0..*	Wishlist
Investor	1..1	has	1..1	Cart
Investor	1..1	submits	0..*	InvestmentRequest
Investor	1..1	writes	0..*	Review
Investor	1..1	saves	0..*	Wishlist
Cart	1..1	contains	0..*	Cart_Item
Farm_Offer	1..1	referenced in	0..*	Cart_Item
Farm_Offer	1..1	for	0..*	InvestmentRequest
InvestmentRequest	1..1	generates	0..1	Transaction
InvestmentRequest	1..1	has	0..1	Review

3. Description of Attributes

1. User Table

Table 12: Data Dictionary for Attributes (User)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
user_id	Unique identifier for each user	INT	11	NO	AUTO_INCREMENT	PK	
first_name	User's first name	VARCHAR	75	NO	-		
last_name	User's last name	VARCHAR	75	NO	-		
email	User's email address — must be unique	VARCHAR	150	NO	-		
phone	User's phone number	VARCHAR	20	NO	-		
password_hash	Hashed password stored securely	VARCHAR	255	NO	-		
role	Account type: farmer, investor, or admin	ENUM (farmer,investor,admin)	-	NO	investor		
created_at	Timestamp of account creation	TIMESTAMP	-	NO	CURRENT_TIMESTAMP		

2. Farmer Table

Table 13: : Data Dictionary for Attributes (Farmer)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
farmer_id	Unique farmer identifier	INT	11	NO	AUTO_INCREMENT	PK	
user_id	Reference to the User table	INT	11	NO	-		FK
national_id	Saudi national ID number	VARCHAR	20	NO	-		
verification_status	Verification state: pending, verified, rejected	ENUM (pending , verified , rejected)	-	NO	pending		
verified_by	User ID of the admin who verified	INT	11	YES	NULL		FK
verified_at	Timestamp when verification was completed	TIMESTAMP	-	YES	NULL		

3. Investor Table

Table 14: Data Dictionary for Attributes (Investor)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK

investor_id	Unique investor identifier	INT	11	NO	AUTO_INCREMENT	PK	
user_id	Reference to the User table	INT	11	NO	-		FK
national_id	Saudi national ID number (unique)	VARCHAR	20	NO	-		

4. Farm Table

Table 15: Data Dictionary for Attributes (Farm)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
farm_id	Unique farm identifier	INT	11	NO	AUTO_INCREMENT	PK	
farmer_id	Owner of this farm	INT	11	NO	-		FK
name	Farm display name	VARCHAR	150	NO	-		
region	Geographic region of the farm	VARCHAR	100	NO	-		
total_area_sqm	Total farm area in square metres	DECIMAL	10,2	NO	-		
palm_type	Type of palm trees on the farm	VARCHAR	100	NO	-		

date_type	Variety of dates produced	VARCHAR	100	NO	-		
approved_by	User ID of the admin who approved this listing	INT	11	YES	NULL		FK

5. Farm_Offer Table

Table 16: Data Dictionary for Attributes (Farm_Offer)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
offer_id	Unique offer identifier	INT	11	NO	AUTO_INCREMENT	PK	
farm_id	Farm this offer belongs to	INT	11	NO	-		FK
area_size	Area available for lease in sqm	DECIMAL	10,2	NO	-		
price	Invesment price per square metre in SAR	DECIMAL	10,2	NO	-		

6. FarmUpdate Table

Table 17: Data Dictionary for Attributes (FarmUpdate)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
update_id	Unique update identifier	INT	11	NO	AUTO_INCREMENT	PK	

farm_id	Farm this update belongs to	INT	11	NO	-		FK
content	Text content of the update	TEXT	-	NO	-		
media_urls	JSON array storing image and/or video URLs	JSON	-	YES	NULL		
media_type	Type of media attached: image, video, or both	ENUM(image,video, both)	-	YES	NULL		
created_at	Timestamp when the update was posted	TIMESTAMP	-	NO	CURRENT_TIMESTAMP		

7. Cart Table

Table 18: Data Dictionary for Attributes (Cart)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
cart_id	Unique cart identifier	INT	11	NO	AUTO_INCREMENT	PK	

investor_id	Investor who owns this cart	INT	11	NO	-		FK
created_at	Timestamp of cart creation	TIMESTAMP	-	NO	CURRENT_TIMESTAMP		
expires_at	Timestamp when cart expires (24 hours)	TIMESTAMP	-	NO	-		

8. Cart_Item Table

Table 19: Data Dictionary for Attributes (Cart_Item)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
cart_item_id	Unique cart item identifier	INT	11	NO	AUTO_INCREMENT	PK	
cart_id	Cart this item belongs to	INT	11	NO	-		FK
offer_id	Farm offer selected by the investor	INT	11	NO	-		FK
area_sqm	Area the investor chose to lease in sqm	DECIMAL	10,2	NO	-		
duration	Lease duration:	ENUM('1 season','1 year','2 years')	-	NO	-		

	1_season, 1_year, 2_years						
harvest_method	Harvest preference: receive, sell, or donate	ENUM('receive','sell','donate')	-	NO	-		
delivery_address	Delivery address if harvest method is receive	TEXT	-	YES	NULL		

9. InvestmentRequest Table

Table 20: Data Dictionary for Attributes (InvestmentRequest)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
request_id	Unique request identifier	INT	11	NO	AUTO_INCREMENT	PK	
investor_id	Investor who submitted this request	INT	11	NO	-		FK
offer_id	Farm offer this request is for	INT	11	NO	-		FK
area_sqm	Area request	DECIMAL	10,2	NO	-		

	ed in sqm						
duration	Lease duration selected	ENUM('1 season','1 year','2 years')	-	NO	-		
harvest_method	Investor's harvest preference	ENUM('receive','sell','donate')	-	NO	-		
delivery_address	Delivery address if applicable	TEXT	-	YES	NULL		
status	Request state: pending, accepted, rejected, cancelled, expired	ENUM('pending','verified','rejected')	-	NO	pending		
rejection_reason	Reason provided if farmer rejects the request	TEXT	-	YES	NULL		
submitted_at	Timestamp when the request	TIMESTAMP	-	NO	CURRENT_TIMESTAMP		

	was submitted						

10. Transaction Table

Table 21: Data Dictionary for Attributes (Transaction)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
transaction_id	Unique transaction identifier	INT	11	NO	AUTO_INCREMENT	PK	
request_id	Accepted investment request	INT	11	NO	-		FK
amount	Total investment amount in SAR	DECIMAL	12,2	NO	-		
payment_status	Payment state: pending, paid, failed	ENUM('pending','paid','failed','refunded')	-	NO	pending		
paid_at	Timestamp when payment was completed	TIMESTAMP	-	YES	NULL		

11. Review Table

Table 22: Data Dictionary for Attributes (Review)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
review_id	Unique review identifier	INT	11	NO	AUTO_INCREMENT	PK	
farm_id	Farm being reviewed	INT	11	NO	-		FK
investor_id	Investor who wrote the review	INT	11	NO	-		FK
request_id	Reference to the accepted investment request	INT	11	NO	-		FK
rating	Star rating from 1 to 5	TINYINT	-	NO	-		
review_text	Written review comment	TEXT	-	YES	NULL		
created_at	Timestamp of review submission	TIMESTAMP	-	NO	CURRENT_TIMESTAMP		

Table 23 23: Data Dictionary for Attributes (wishlist)

Attribute	Description	Data Type	Length	Nulls	Default	PK	FK
wishlist_id	Unique wishlist entry identifier	INT	11	NO	AUTO_INCREMENT	PK	
investor_id	Investor who saved this farm	INT	11	NO	-		FK
farm_id	Reference to the farm saved by the investor	INT	11	NO	-		FK
created_at	Timestamp when the farm was saved	TIMESTAMP	-	NO	CURRENT_TIMESTAMP		

4.4 Components Design

1- Submit Investment Request :

Classification: Function

Definition: The investor can submit an investment request after selecting items in the cart. The system checks the login status and cart, then sends the request to the farmer and notifies the investor.

START

IF investor is not logged in **THEN**
  DISPLAY "الرجاء تسجيل الدخول"

ELSEIF cart is empty **THEN**
  DISPLAY "السلة فارغة، أضف عنصراً قبل الإرسال"

ELSE
  CREATE investment_request
  SET investor_id = investor.id
  SET items = cart
  SET status = "Pending"
  SET submitted_at = getCurrentTimestamp()

  SEND investment_request to farmer
  SEND notification to investor
  SEND email confirmation to investor

  DISPLAY "تم إرسال طلب الاستثمار بنجاح"

ENDIF

END

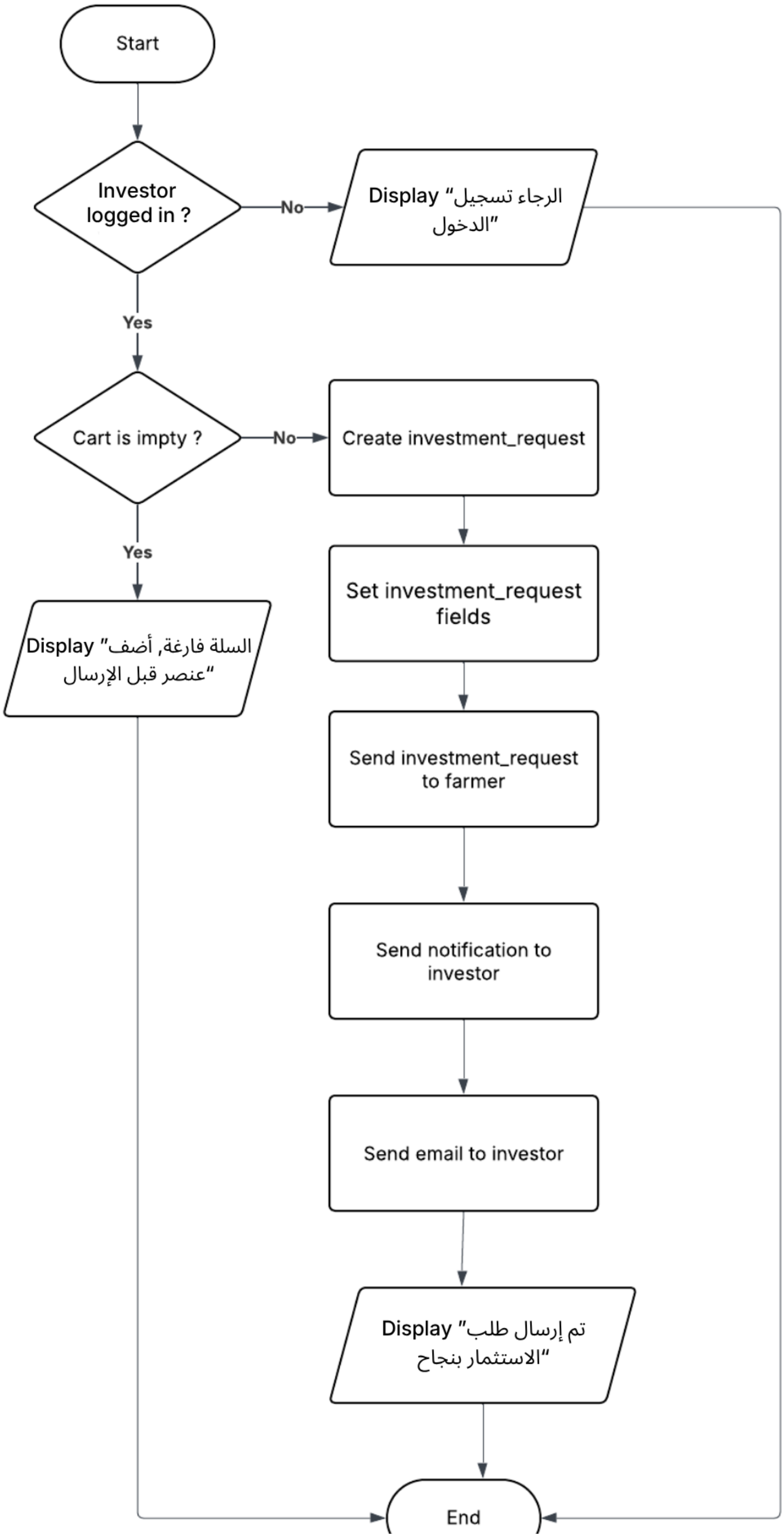

Figure 24: Submit Investment Request Flow Chart

2- Submit Farm Registration Request:

Classification: Function

Definition: The farmer can submit a farm registration request by entering the required farm information. The system validates the data and sends the request to the admin for review.

START

IF farmer is not logged in **THEN**
  DISPLAY "الرجاء تسجيل الدخول"

ELSEIF any required field is empty **THEN**
  DISPLAY "يرجى تعبئة جميع الحقول المطلوبة وإرفاق صور المزرعة"

ELSEIF a farm with the same name and location already exists **THEN**
  DISPLAY "هذه المزرعة مسجلة مسبقًا"

ELSE
  CREATE farm_request
  SET farmer_id = farmer.id
  SET farm_name = input.farm_name
  SET location = input.location
  SET total_area = input.total_area
  SET palm_type = input.palm_type
  SET num_palm_trees = input.num_palm_trees
  SET expected_yield = input.expected_yield
  SET photos = input.photos
  SET status = "Pending Review"
  SET submitted_at = getCurrentTimestamp()

  SEND farm_request to admin
  SEND notification to farmer
  SEND email confirmation to farmer

  DISPLAY "تم إرسال طلب تسجيل المزرعة بنجاح"

ENDIF

END

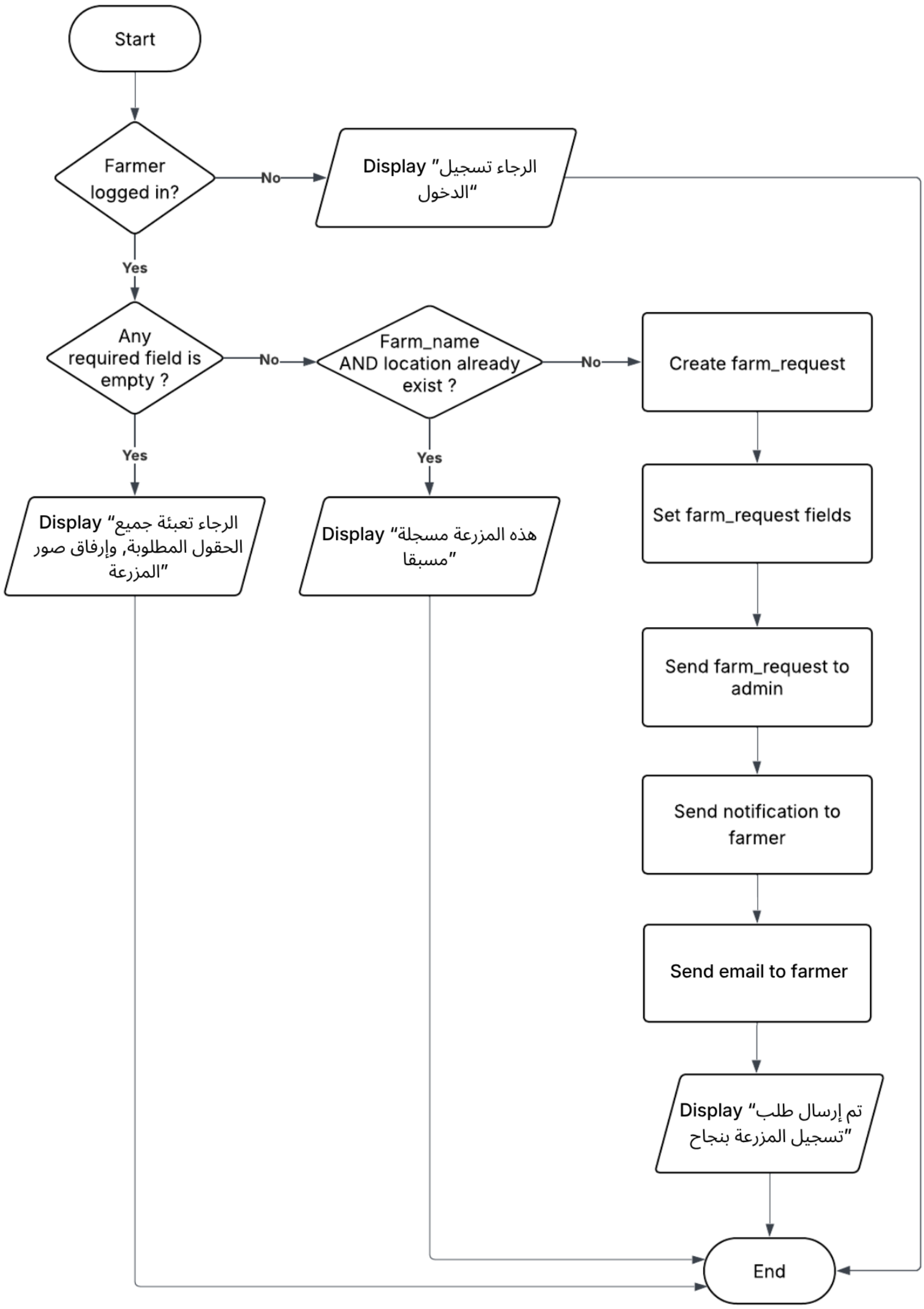

Figure 25 : Submit Farm Registration Request Flow Chart

3- Farm Registration Approval:

Classification: Function

Definition: The admin can review farm registration requests and either approve, reject, or request additional information from the farmer while updating the request status.

START

IF admin is not logged in **THEN**
  DISPLAY "الرجاء تسجيل الدخول كمسؤول"

ELSE
  DISPLAY all pending farm registration requests
  SELECT farm_request

  IF admin chooses "Approve" **THEN**
    SET farm_request.status = "Approved"
    SET farm.visible = true
    SEND notification to farmer
    LOG action with admin_id and timestamp
    DISPLAY "تمت الموافقة على طلب المزرعة"

  ELSEIF admin chooses "Reject" **AND** rejection_reason is empty **THEN**
    DISPLAY "الرجاء إدخال سبب الرفض"

  ELSEIF admin chooses "Reject" **THEN**
    SET farm_request.status = "Rejected"
    SEND rejection reason to farmer
    LOG action with admin_id and timestamp
    DISPLAY "تم رفض طلب المزرعة"

  ELSEIF admin chooses "Request Additional Information" **AND** request_message is empty
THEN
    DISPLAY "الرجاء كتابة المعلومات المطلوبة من المزارع"

  ELSEIF admin chooses "Request Additional Information" **THEN**
    SET farm_request.status = "More Information Required"
    SEND request_message to farmer
    LOG action with admin_id and timestamp
    DISPLAY "تم إرسال طلب المعلومات الإضافية إلى المزارع بنجاح"

  ENDIF

ENDIF

END

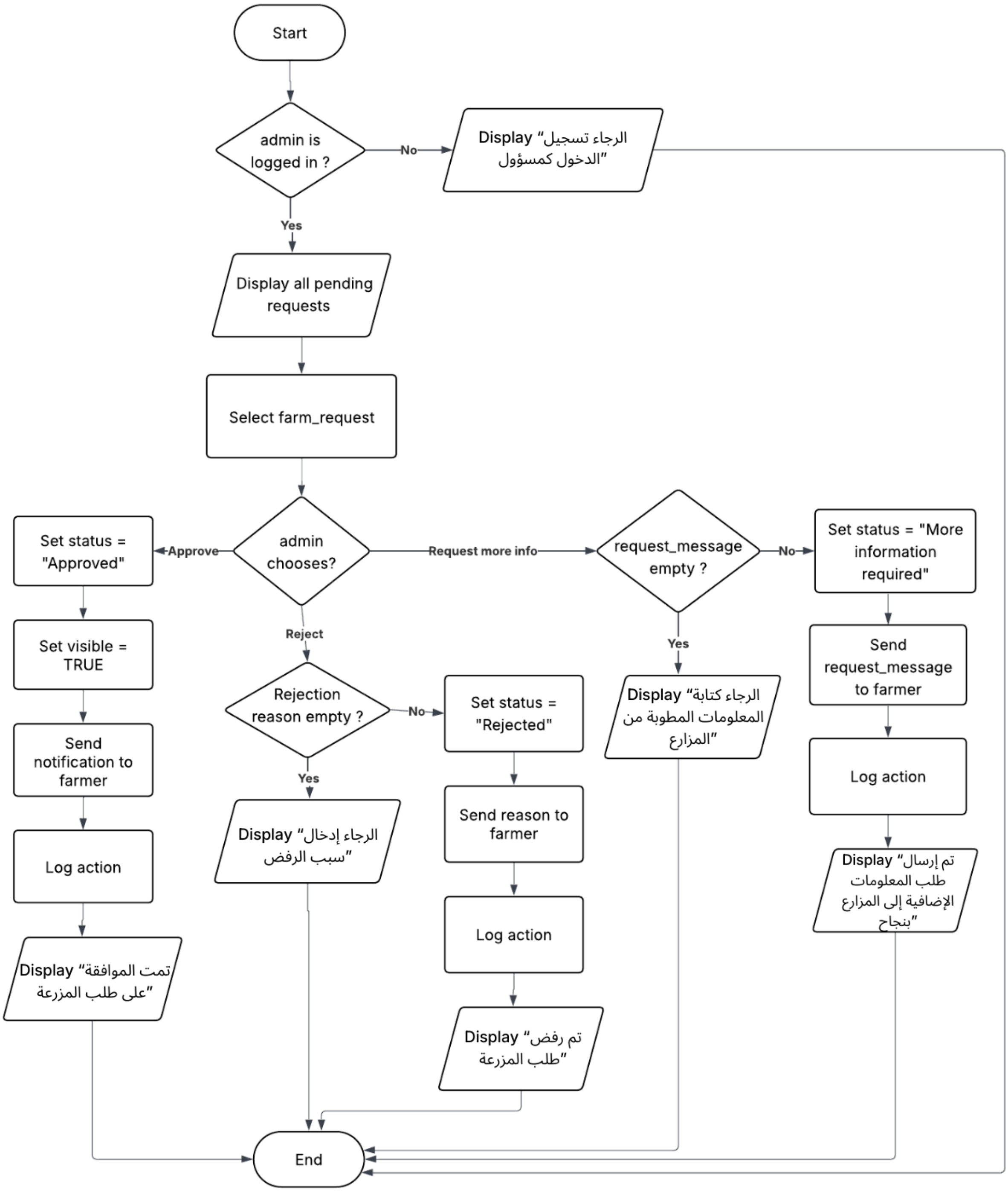

Figure 26 : Farm Registration Approval Flow chart

4.5 Interface Design

Site Map :

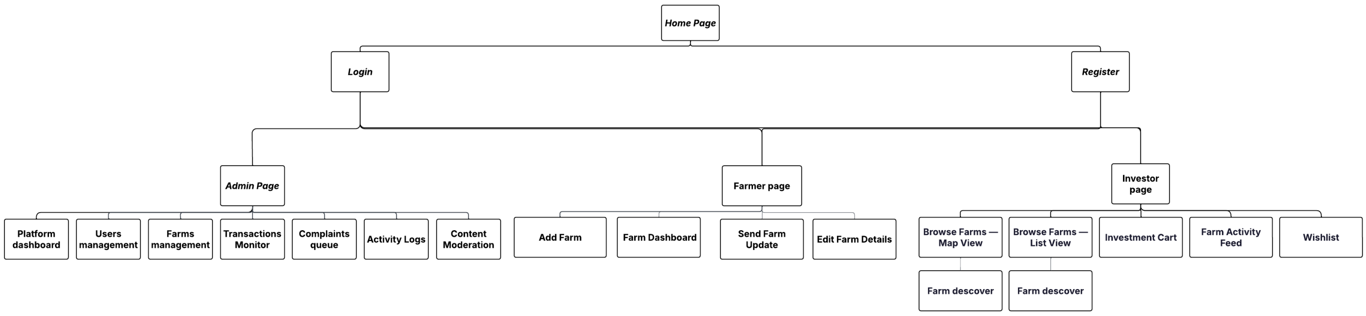

Figure 27: Site Map

[GitHub Link] :

https://github.com/Jana-Alothman/Qinwan.git

[GitHub Link for phase 4 :]

https://github.com/NouraAltuwiam/qinwan

[Project Jira Link]

5 Chapter 5: Testing

This chapter presents the testing phase of the Qinwan platform, which focuses on evaluating the final version of the system to ensure that it functions correctly and meets both functional and user requirements. Testing plays a crucial role in verifying that all system components operate as expected and interact smoothly when integrated.

The testing process in this project covers the main features of the system, including user registration and login, browsing farms, managing investment requests, handling carts, and processing transactions. It also ensures that user inputs are validated correctly and that the system maintains data consistency across all modules.

To achieve this, we conducted several types of testing, including User Story Acceptance Testing, Integration Testing, and User Acceptance Testing. These testing approaches help confirm that the system is reliable, efficient, and provides a seamless experience for all user roles (Investor, Farmer, and Admin).

5.1 User Story Acceptance Testing

User story acceptance testing was conducted to verify that each user story meets its defined acceptance criteria from the user's perspective. It answers the question: will the acceptance criteria be met upon execution of the user story?

For each user story, at least two test cases were designed: one positive test case to verify the expected behavior under normal conditions, and one negative test case to test edge cases or invalid inputs. Testing was performed by the development team using the final version of the Qinwan platform.

Sprint number	PBI (User Story)	Acceptance Criteria	Test cases	Pass?	Comment
Farmer					
2	US-01. As a farmer, I want to register a new account on Qinwan so that I	• The registration form must include: full name, national ID, phone number, email, and password.	**TC1 - Valid registration** 1- Open the sign-up page.	yes	

	can list my farm and receive investment requests.	• The username must be 4–20 characters (letters, numbers, underscores, or periods only). • Email must follow the format xxxx@xxx.xxx; duplicates are rejected with a clear error message. • Password must contain at least 8 characters, one uppercase, one lowercase, one number, and one special character. • Passwords must match in the confirm-password field before submission. • The system must hash and salt passwords before storing them in the database. • A confirmation email is sent to the farmer after successful registration. • After registration, the farmer is automatically logged in and redirected to their dashboard.	2- Enter valid required details and matching password fields. 3- Submit the registration form. 4- Verify the account is created, confirmation email is sent, and the user is redirected to the correct page. **TC2 - Invalid/duplicate registration** 5- Open the sign-up page. 6- Enter an invalid email, weak or mismatched password, or an already-registered email. 7- Submit the form. 8- Verify specific validation/error messages are shown and the account is not created.		
2	US-02. As a farmer, I want to log in to my account so that I can access my farm management dashboard securely.	• Login requires a valid registered email and correct password. • If credentials are invalid, the system displays: 'Invalid email or password. Please try again.' • After 5 consecutive failed attempts, the account is temporarily locked for 15 minutes. • Upon successful login, the farmer is redirected to their personal dashboard. • The system must use HTTPS for all login data transmission.	**TC1 - Successful login** 1- Open the login page. 2- Enter a valid registered email and correct password. 3- Submit the login form. 4- Verify secure login succeeds and the user is redirected to the correct dashboard. **TC2 - Invalid login / lockout** 5- Attempt login with invalid credentials. 6- Verify the error message is displayed.	yes	

			7- Repeat invalid login attempts until the threshold is reached. 8- Verify the account is locked for 15 minutes and access is denied during the lock period.		
2	US-03. As a farmer, I want to log out of my account so that I can protect my account when I am done.	• A 'Log Out' button must be visible and accessible from any page within the dashboard. • The farmer is redirected to the login page after logout. • Accessing any protected page after logout must redirect the user to the login page.	**TC1 - Logout from dashboard** 1- Log in to the account and navigate to any protected page. 2- Click the 'Log Out' button. 3- Verify the session ends and the user is redirected to the login page. **TC2 - Protected page after logout** 4- After logout, try to access a protected page using the URL or browser back button. 5- Verify the system redirects the user to the login page and protected content is not shown.	yes	
2	US-04. As a farmer, I want to submit a farm registration request so that an admin can review and approve my farm to appear on the platform.	• The farm submission form must include: farm name, location (city, region), total area (m^2), type of palm, number of palm trees, expected yield (kg/season), and photos. • The farmer must be logged in before submitting a farm registration request. • All required fields must be validated before submission; missing fields display specific error messages.	**TC1 - Valid farm submission** 1- Log in as a farmer. 2- Open the farm registration form and enter all required fields with valid data and photos. 3- Submit the request. 4- Verify the farm is saved with status 'Pending Review' and a confirmation notification is shown. **TC2 - Missing/duplicate data**	yes	

		• After submission, the farm status is set to 'Pending Review'. • The farmer receives an notification confirming that the request has been submitted. • The system prevents duplicate farm registrations with the same name and location.	5- Submit the form with missing required fields or with a duplicate farm name and location. 6- Verify specific validation messages are displayed and the request is not accepted.		
2	US-05. As a farmer, I want to edit my farm details so that I can keep the information accurate and up to date.	• The farmer can edit: farm name, total area, expected yield, palm type, photos, and description. • The farmer must be authenticated and own the farm to make any edits. • Edited farms that change core details (area, location) must be re-reviewed and re-approved by the admin. • Changes are saved with a timestamp and logged for audit purposes. • The farmer receives a confirmation notification after successful edits.	TC1 - Edit allowed fields 1- Log in as the farm owner. 2- Open the edit page for an owned farm. 3- Update editable fields and save. 4- Verify changes are stored, timestamped, and a confirmation notification appears. TC2 - Restricted / re review edits 5- Try editing a farm as a non-owner or unauthenticated user, or change a core field such as area/location. 6- Verify unauthorized edits are blocked.	yes	
2	US-06. As a farmer, I want to view all investment requests received for my farm so that I can make informed acceptance or rejection decisions.	• The farmer's dashboard must show a list of all investment requests with: investor name, requested area (m^2), selected harvest method, and requested start date. • Requests must be filterable by status: Pending, Accepted, Rejected.	**TC1 - Display requests list** 1- Log in as a farmer with existing investment requests. 2- Open the received requests section. 3- Verify each request shows the required fields and verified badge when applicable.	yes	

		• Each request must display the investor's verified profile badge if available. • The list must update in real-time or refresh upon page reload.	**TC2 - Filter and refresh** 4- Filter requests by Pending, Accepted, and Rejected. 5- Refresh/reload the page. 6- Verify filtering works correctly and the list remains up to date.		
2	US-07. As a farmer, I want to accept an investment request so that I can confirm the investment deal with the investor.	• The farmer can only accept a request if the requested area does not exceed the remaining available area on the farm. • Upon acceptance, the system updates the farm's available area and marks the request as 'Accepted'. • The investor receives an immediate notification about the acceptance. • The system logs the acceptance with a timestamp. • The farmer cannot accept the same area from two different investors simultaneously.	**TC1 - Accept valid request** 1- Log in as a farmer and open a pending request within the available area. 2- Click 'Accept'. 3- Verify the request status becomes 'Accepted', available area is updated, the investor is notified, and the action is logged. **TC2 - Prevent overbooking** 4- Try to accept a request that would exceed remaining farm area or conflict with an already accepted area. 5- Verify the acceptance is blocked and the same area cannot be accepted twice.	yes	
2	US-08. As a farmer, I want to reject an investment request so that I can decline deals that are not suitable.	• The farmer can reject any pending request with an optional reason field. • Upon rejection, the request is marked as 'Rejected' and the investor is notified immediately. • Rejected requests are moved to a separate 'Rejected' section in the dashboard. • The farmer cannot reverse a rejection without	**TC1 - Reject pending request** 1- Log in as a farmer and open a pending request. 2- Click 'Reject' and optionally enter a reason. 3- Verify the request status becomes 'Rejected', the investor is notified, and the request	yes	

		creating a new manual arrangement.	moves to the Rejected section. **TC2 - No reversal through UI** 4- Open a rejected request and attempt to reverse the decision. 5- Verify the system does not allow direct reversal without a new manual arrangement.		
2	US-09. As a farmer, I want to send periodic farm updates to my investors so that they stay informed about the farm's progress.	• The farmer can publish text and image updates from the farm management dashboard. • Updates are only delivered to investors who have an accepted investment in that specific farm. • Each update is timestamped and appears in the investor's activity feed. • The farmer can only send updates if they have at least one active investor. •	**TC1 - Publish valid update** 1- Log in as a farmer with at least one active investor. 2- Create an update with text and optional image, then publish it. 3- Verify the update is timestamped and appears only in the activity feeds of accepted investors. **TC2 - Posting restriction** 4- Try posting an update when no active investors exist. 5- Verify that posting is blocked. **TC3 - Admin removal** 6- Log in as an admin. 7- Remove a violating update. 8- Verify that the update is removed successfully.	yes	
2	US-10. As a farmer, I want to view my farm dashboard so that I can monitor the total leased area, number of	• The dashboard must display: total farm area (m^2), total leased area (m^2), remaining available area (m^2), number of active investors, expected yield (kg/season), and estimated revenue.	**TC1 - Dashboard totals** 1- Log in as a farmer and open the farm dashboard. 2- Verify total area, leased area, remaining area, active investors,	yes	

	investors, and expected yield.	• Data must refresh automatically or upon page reload. • All numeric values must be accurate and consistent with the accepted investment records in the system.	expected yield, and estimated revenue are shown. **TC2 - Data accuracy after change** 3- Accept or update an investment record, then reload the dashboard. 4- Verify the figures refresh and remain consistent with the underlying accepted investment records.		
2	US-11 . As a farmer, I want to receive an automated reminder before an investment request expires so that I do not miss responding to pending requests.	•The system sends a reminder notification to the farmer 24 hours before a pending investment request expires. •A second reminder is sent 2 hours before expiry if the farmer has not yet responded. •Reminders are delivered via WhatsApp and SMS. •If the farmer does not respond before the deadline, the request is automatically marked as 'Expired' and the investor is notified. •Expired requests are moved to a separate 'Expired' tab in the farmer's dashboard.	**TC1 - Reminder delivery** 1- Create a pending investment request close to expiry. 2- Verify reminders are sent 24 hours and 2 hours before expiry via WhatsApp and SMS. **TC2 - Expiry handling** 3- Do not respond before the deadline. 4- Verify the request becomes 'Expired', and the item moves to the Expired tab.	Partial	TC1- Point 2 : Reminder notifications (24 hours and 2 hours before expiry) could not be delivered via WhatsApp or SMS due to external provider limitations. WhatsApp services require paid accounts and regional registration, while SMS gateways require a valid Saudi commercial registration. As a result, reminder delivery through these channels is not functional.
investor					
2	US-01. As an investor, I want to register a new account on	• The registration form must include: full name,	**TC1 - Valid registration**	yes	

	Qinwan so that I can browse farms and submit investment requests.	phone number, email, and password. • Email format must be validated (xxxx@xxx.xxx); duplicate emails are rejected with a clear error message. • Password must contain at least 8 characters, one uppercase, one lowercase, one number, and one special character. • Passwords must match in the confirm-password field. • The system hashes and salts all passwords before storage. • A confirmation email is sent to the investor after registration. • After registration, the investor is automatically logged in and redirected to the home/browse page.	1- Open the sign-up page. 2- Enter valid required details and matching password fields. 3- Submit the registration form. 4- Verify the account is created, and the user is redirected to the correct page. **TC2 - Invalid/duplicate registration** 5- Open the sign-up page. 6- Enter an invalid email, weak or mismatched password, or an already-registered email. 7- Submit the form. 8- Verify specific validation/error messages are shown and the account is not created.		
2	US-02. As an investor, I want to log in to my account so that I can access my investment portfolio and browse farms.	• Login requires a valid registered email and correct password. • If credentials are invalid, the system displays a clear error message. • After 5 consecutive failed attempts the account is locked for 15 minutes. • Upon successful login, the investor is redirected to their personal dashboard.	**TC1 - Successful login** 1- Open the login page. 2- Enter a valid registered email and correct password. 3- Submit the login form. 4- Verify secure login succeeds and the user is redirected to the correct dashboard. **TC2 - Invalid login / lockout** 5- Attempt login with invalid credentials. 6- Verify the error message is displayed.	yes	

			7- Repeat invalid login attempts until the threshold is reached. 8- Verify the account is locked for 15 minutes and access is denied during the lock period.		
2	US-03. As an investor, I want to log out of my account so that I can protect my account when I finish.	• A 'Log Out' button must be visible and accessible from any page of the investor interface. • Upon logout, the session is terminated immediately and all session tokens are invalidated. • The investor is redirected to the login page.	**TC1 - Logout from dashboard** 1- Log in to the account and navigate to any protected page. 2- Click the 'Log Out' button. 3- Verify the session ends and the user is redirected to the login page. **TC2 - Protected page after logout** 4- After logout, try to access a protected page using the URL. 5- Verify the system redirects the user to the login page and protected content is not shown.	yes	
2	US-04. As an investor, I want to browse farms on an interactive map so that I can visually locate farms across Saudi Arabia and explore their details.	• The map must display all admin-approved farms as interactive pins. • Clicking a farm pin must display a summary card: farm name, region, total area, available area, palm type, and expected yield. • The map must be integrated with a national mapping system . • The map must load within 3 seconds for 90% of users. • If no farms exist in a region, no pins are shown and no error is thrown.	**TC1 - Approved farms on map** 1- Open the browse map page as an investor. 2- Verify all approved farms appear as interactive pins. 3- Click a pin and verify the summary card displays the required farm information. **TC2 - Empty region** 4- Navigate to a region with no approved farms. 5- Verify that no pins are shown . **TC3 - Performance** 6- Open the map page as an investor.	yes	

			7- Verify that the map loads within the required performance target (within 3 seconds).		
2	US-05. As an investor, I want to browse farms as a list so that I can scan all available farms quickly without using the map.	• The list view must show all approved farms with: name, region, available area, palm type, and expected yield. • Each list item must be clickable and redirect to the full farm detail page. • The list must reflect real-time availability of farm areas.	**TC1 - List display** 1- Open the list view. 2- Verify all approved farms show the required summary fields. 3- Click a list item and verify the farm detail apears. **TC2 - Availability refresh** 4- Change available farm area through an accepted request, then refresh the list. 5- Verify the list reflects the latest availability.	yes	
2	US-06. As an investor, I want to filter farms by region so that I can find farms in specific geographic areas of interest.	• The region filter must list all available Saudi regions/cities where farms exist. • Selecting a region updates both the map pins and the list view dynamically. • If no farms match the selected region, the system displays: 'No farms found in this region.' • The filter can be cleared with a single 'Clear Filters' button.	**TC1 - Apply filter** 1- Open the browse page and apply a valid region filter. 2- Verify the map and list update dynamically and only matching farms are displayed. **TC2 - No-match** 3- Apply a region filter that has no matching farms. 4- Verify the correct no results message appears. **TC3 - clear filter** 5- Verify the filter can be cleared with one action.	yes	
2	US-07. As an investor, I want to filter farms by farm size so that I can find farms that match my	• The size filter must offer predefined ranges (e.g., 0–500 m², 500–2000 m², 2000+ m²).	**TC1 - Apply filter** 1- Open the browse page and apply a valid size filter.	yes	

	investment capacity.	• Applying the filter updates the results list and map dynamically. • If no farms match the size filter, the system displays: 'No farms match your selected size range.' • The filter can be cleared with a single 'Clear Filters' button.	2- Verify the map and list update dynamically and only matching farms are displayed. **TC2 - No match** 3- Apply a size filter that has no matching farms. 4- Verify the correct no results message appears. **TC3 - Clear filter** 5- Verify the filter can be cleared with one action.		
2	US-08. As an investor, I want to filter farms by palm type so that I can invest in the specific variety of dates I prefer.	• The palm type filter must list all available date varieties registered in the system. • Applying the filter updates the results list and map dynamically. • Multiple palm types can be selected simultaneously. • If no farms match the selected type, the system displays: 'No farms match your selected palm type.'	**TC1 - Apply filter** 1- Open the browse page and apply a valid palm type filter (for one type and for multiple types). 2- Verify the map and list update dynamically and only matching farms are displayed. **TC2 - No-match** 3- Apply a palm type filter that has no matching farms. 4- Verify the correct no results message appears **TC3 - Clear filter** 5- Verify the filter can be cleared with one action.	yes	
2	US-09. As an investor, I want to add a farm to my wishlist so that I can save it and review it later before making an investment decision.	• A 'Add to Wishlist' button must be visible on every farm card and farm detail page. • Clicking the button adds the farm to the investor's personal wishlist section. • Wishlist items persist across sessions (saved to the investor's account).	**TC1 - Add to wishlist** 1- Open a farm card or detail page and click Add to Wishlist. 2- Verify the farm is saved in the investor wishlist and persists after logout/login. **TC2 - Remove** 3- Remove the farm from the wishlist.	yes	

		• The investor can remove items from the wishlist at any time. • The wishlist is private and not visible to other users.	4- Verify the item is removed .		
2	US-10. As an investor, I want to select a specific area (m^2) to invest in so that I can choose exactly how much land I wish to lease.	• The area selection input must only accept numeric values greater than zero. • The selected area must not exceed the farm's remaining available area; the system displays an error if exceeded. • The system must display the cost preview based on the selected area before the investor proceeds. • The investor cannot proceed to the cart without specifying a valid area.	**TC1 - Valid area selection** 1- Open a farm detail page. 2- Enter a valid positive area value within the available area. 3- Verify the cost preview is shown and the investor can proceed to cart. **TC2 - Invalid area selection** 4- Enter zero, non numeric input, or an area greater than the remaining area. 5- Verify the system blocks progress and shows the correct error message.	yes	
2	US-11. As an investor, I want to add a farm investment to my cart so that I can review my selections before submitting a request.	• The cart must display: farm name, selected area (m^2), investment duration, selected harvest method, and estimated cost. • The investor can add multiple farms to the cart in a single session. • Items in the cart are preserved for up to 24 hours. • The investor can delete any item from the cart at any time.	**TC1 - Add items to cart** 1- Select valid farm investments and add them to the cart. 2- Verify the cart shows farm name, selected area, duration, harvest method, and estimated cost. **TC2 - Delete item** 3- Remove an item from the cart. 4- Verify that the item is deleted successfully. **TC3 – Persistence** 5- Keep the session and return within 24 hours.	yes	

			6- Verify that the remaining items are preserved for the allowed time.		
2	US-12. As an investor, I want to choose the investment duration so that I can define how long I wish to lease the farm area.	• The investor must select a duration from predefined options (e.g., 1 season, 1 year, 2 years). • The selected duration must be shown clearly in the cart and on the submitted request. • The investor cannot submit an investment request without selecting a valid duration.	**TC1 - Valid duration** 1- Select one of the predefined duration options. 2- Verify the selected duration appears in the cart and submitted request. **TC2 - Missing duration** 3- Attempt to continue without selecting a duration. 4- Verify the system blocks submission until a valid duration is selected.	yes	
2	US-13. As an investor, I want to select my preferred harvest method so that I can choose how I will benefit from the farm's produce.	• The investor must choose one harvest method from: (1) receive dates delivered to their home, (2) sell the harvest and receive profits, or (3) donate the harvest to a charity. • The selected method must be displayed in the cart and on the investment request. • If the investor selects home delivery, a delivery address field becomes mandatory. • The investor cannot submit a request without selecting a harvest method.	**TC1 - Valid harvest method** 1- Choose one harvest method and continue. 2- Verify the selected method appears in the cart and request summary. **TC2 - Delivery address requirement** 3- Select home delivery without entering an address. 4- Attempt to proceed. 5- Verify that the address is required and submission is blocked. **TC3 - Mandatory harvest method** 6- Attempt to proceed without selecting a harvest method. 7- Verify that submission is blocked and a message is shown.	yes	

2	US-14. As an investor, I want to submit an investment request so that the farmer can review and accept my proposed investment.	• The investor must be logged in with a verified account to submit a request. • The cart must contain at least one valid item before submission is allowed. • Upon submission, the request is sent to the corresponding farmer and marked as 'Pending'. • The investor receives an on-platform and email notification confirming request submission. • The investor can view the submitted request in the 'Pending Requests' section of their dashboard. • If the cart is empty, the system shows: 'Please select a farm area before submitting a request.'	**TC1 - Submit valid request** 1- Log in with a verified investor account and add at least one valid item to the cart. 2- Submit the request. 3- Verify the request is marked 'Pending', sent to the farmer, notification/email are sent, and the request appears in Pending Requests. **TC2 - Invalid user state** 6- Attempt to submit while not logged. 7- Verify that submission is blocked and the correct message is displayed.	yes	
2	US-15. As an investor, I want to view my investment requests (pending, accepted, rejected) so that I can track the status of all my submissions.	• The requests section must display three tabs: Pending, Accepted, Rejected. • Each request must show: farm name, requested area, duration, harvest method, submission date, and current status. • The investor can cancel a pending request at any time before the farmer responds. • Accepted and rejected requests are read-only and cannot be cancelled.	**TC1 - View request tabs** 1- Open the requests section. 2- Verify that Pending, Accepted, and Rejected tabs exist. 3- Verify each request shows: farm name, requested area, duration, harvest method, submission date, and status. **TC2 - Cancel pending request** 4- Cancel a pending request then verify the request is cancelled successfully. **TC3 - Read-only accepted/rejected** 5- Open accepted and rejected requests.	yes	

			6- Attempt to cancel them. 7- Verify that cancellation is not allowed and they are read-only.		
2	US-16. As an investor, I want to view my investment portfolio dashboard so that I can monitor all my active investments at a glance.	• The dashboard must display for each active investment: farm name, owner's contact, location, leased area (m^2), palm type in the region, expected yield, and estimated profit/loss summary. • The dashboard must show a total summary: total m^2 leased, total estimated yield, total estimated profit. • Data must be accurate and reflect the accepted investment records in real-time.	TC1 - Active investment data 1- Open the investor portfolio dashboard. 2- Verify each active investment shows the required details. TC2 - Summary accuracy 3- Compare dashboard totals with accepted investment records or update an accepted investment. 4- Verify totals and summaries update accurately in real time.	yes	
2	US-17. As an investor, I want to follow the farm activity feed so that I can receive and read periodic updates sent by the farmers I have invested in.	• The activity feed must display updates only from farms where the investor has an accepted investment. • Each update must show: farm name, update text, attached image (if any), and timestamp. • New updates must appear at the top of the feed. • The investor receives a push notification when a new update is posted by a farmer they are invested with.	**TC1 - Feed visibility** 1- Open the activity feed as an investor with accepted investments. 2- Verify only updates from invested farms are shown **TC2 - Push notification** 3- Have a farmer publish a new update. 4- Verify the investor receives a push notification and the update appears at the top of the feed.	yes	
2	US-18. As an investor, I want to request a change to my harvest method	• This option is only available for investments that have already been accepted by the farmer.	**TC1 - Submit change request** 1- Open an accepted investment and request a harvest-method change.	yes	

	after the request is accepted so that I can adjust how I receive my produce.	• The investor can submit a change request; the farmer must approve or reject the change. • The investor is notified of the farmer's decision via the platform and email. • Only one pending harvest-method change request is allowed per investment at a time.	2- Verify the request is submitted to the farmer and the investor receives status tracking. **TC2 - Restrictions** 3- Try to submit a second pending change request for the same investment or try this feature on a non accepted investment. 4- Verify the system blocks the action **TC3- Outcome** 4- Verify the system sends platform notification after the farmer approves or rejects.		
2	US-19. As an investor, I want to receive WhatsApp notifications for important platform updates so that I can stay informed .	•The investor can enable WhatsApp notifications from their profile settings. •Notifications are sent for the following events: investment request accepted or rejected, new farm update posted, harvest method change request approved or rejected. •The investor must provide and verify a valid WhatsApp number during registration or from settings. •If WhatsApp delivery fails, the system falls back to SMS. •The investor can disable WhatsApp notifications at any time from their settings.	**TC1 - Enable and deliver notifications** 1- Open profile settings, add and verify a valid WhatsApp number, and enable WhatsApp notifications. 2- Trigger a supported platform event. 3- Verify the notification is delivered by WhatsApp. **TC2 - Fallback / disable** 4- Disable WhatsApp notifications. 5- Verify SMS fallback is used when delivery fails and notifications stop after disabling.	No	TC1 — Fail Comment: Three SMS/WhatsApp providers were tested but none worked: (1) Twilio US number blocked for Saudi +966 by carrier rules (error 21612), (2) Twilio Alphanumeric Sender ID requires paid account + 2–4 week CST registration, (3) Unifonic requires a Saudi commercial registration the team doesn't

					have. As a workaround, the system generates wa.me links inside each notification card so the investor can open WhatsApp with the message pre-filled.TC2 — Partial Comment: Disabling notifications works correctly — no links are generated. SMS fallback code exists in notifications.php but cannot deliver messages without a connected gateway, for the same reasons above.
2	US-20. As an investor, I want to rate and review a farm after my investment period ends so that I can share my experience with other investors.	•The rating option only appears after the investment period for that farm has officially ended. •The investor can submit a rating from 1 to 5 stars and an optional written review. •Each investor can only submit one rating per investment. •The rating is publicly visible on the farm's detail page.	**TC1 - Submit review after completion** 1- Open an investment whose period has ended. 2- Submit a 1-5 star rating with or without a written review. 3- Verify the review becomes publicly visible and the overall average rating updates. **TC2 - Prevent duplicate/edit**	yes	

		•The investor cannot edit or delete their rating after submission. •The farm's overall rating displayed on its card and detail page is calculated as the average of all submitted ratings.	4- Attempt to rate before the investment ends, submit a second rating, or edit/delete an existing rating. 5- Verify the system blocks these actions.		
Admin					
2	US-01. As an admin, I want to view all registered users (farmers and investors) so that I can monitor platform membership.	• The admin panel must display a list of all users with: name, role (farmer/investor), registration date, and status (active/suspended). • The list must be searchable by name and filterable by role or status. • The admin can click on any user to view their full profile and activity.	**TC1 - User list display** 1- Log in as admin and open the users page. 2- Verify the list shows name, role, registration date, and status for all users. **TC2 - Search / filter / profile** 3- Search by name, filter by role or status, and click a user record. 4- Verify results update correctly and the full profile/activity view opens.	Yes	
2	US-02. As an admin, I want to verify and approve farmer accounts so that only legitimate farm owners operate on the platform.	• The admin reviews the farmer's submitted national ID and farm ownership documents. • The admin can approve or reject a farmer account with a mandatory note. • Upon approval, the farmer receives a notification and can begin listing farms. • Upon rejection, the farmer receives a notification with the stated reason.	**TC1 - Approve farmer** 1- Open a pending farmer verification request. 2- Review national ID and ownership documents, then approve the account. 3- Verify the farmer is notified and can start listing farms. **TC2 - Reject with mandatory note** 4- Reject a farmer verification request without and then with a note.	Yes	

		• Approved farmers are marked with a verification badge on the platform.	5- Verify rejection requires a note, the reason is sent to the farmer, **TC3 - verification badge** 6- Verify approved farmers show a verification badge.		
2	US-03. As an admin, I want to review and approve farm registration requests so that only verified and accurate farms appear on the platform.	• The admin sees all pending farm registration requests with full details (name, location, area, photos, palm type). • The admin can approve the farm, which makes it visible to investors on the map and list. • The admin can reject the farm with a mandatory rejection reason sent to the farmer. • The admin can request additional information from the farmer before making a decision. • Approval and rejection actions are logged with a timestamp and admin ID.	**TC1 - Approve farm request** 1- Open a pending farm registration request and review its full details. 2- Approve the farm. 3- Verify the farm becomes visible to investors on the map and list and the action is logged. **TC2 - Reject** 4- Reject a farm without and then with a reason. 5- Verify rejection requires a reason and the farmer receives the decision or request and logged. **TC3 - Request more info** 6- request additional information.	Yes	
2	US-04. As an admin, I want to edit or remove farm listings that violate platform guidelines so that the platform maintains quality and trust.	• The admin can edit any farm field (name, description, area, photos) at any time. • The admin can deactivate or permanently delete a farm with a mandatory reason. • Affected farmers are notified immediately upon any admin edit or deletion.	**TC1 - Edit listing** 1- Open an existing farm listing as admin. 2- Edit one or more fields and save. 3- Verify the farmer is notified and the action is recorded in the audit trail. **TC2 - Deactivate / delete listing**	Yes	

		• All admin edits and deletions are logged in the audit trail with a timestamp.	4- Attempt to deactivate or permanently delete a farm without and then with a reason. 5- Verify a reason is mandatory, the farmer is notified, and the deletion/deactivation is logged.		
2	US-05. As an admin, I want to monitor all investment transactions on the platform so that I can ensure financial integrity and resolve issues.	• The admin panel must display all transactions with: investor name, farmer name, farm name, area, duration, harvest method, status, and date. • Transactions are filterable by status (pending, accepted, rejected, completed) and date range. • The admin can view the full detail of any individual transaction. • The admin receives an alert for any transactions flagged as suspicious.	TC1 - Transaction list and details 1- Open the admin transactions page. 2- Verify all transactions show the required summary fields and that a record can be opened for full details. **TC2 – Filters transactions** 3- Filter transactions by status and date range. 4- Verify that the filtered results are displayed correctly. **TC3- Suspicious alert** 5- Trigger or inspect a suspicious transaction. 6- Verify that an alert is shown for flagged transactions.	Yes	
2	US-06. As an admin, I want to view platform statistics so that I can understand the overall performance and growth of Qinwan.	• The admin dashboard must display: total registered users, total active farms, total leased area (m^2), total investment transaction volume, and new registrations per month. • Statistics must update in real-time or at most with a 5-minute delay. • The admin can export statistics as an Excel or PDF report.	**TC1 - Statistics dashboard** 1- Open the admin dashboard. 2- Verify total users, active farms, leased area, transaction volume, and new registrations per month are displayed. **TC2 - Refresh data** 3- Create new qualifying data or wait for the refresh period. 4- Verify that the	Yes	

			statistics update within the allowed time. **TC3- Export the statistics** 5- Export the statistics. 6- Verify that export works correctly for Excel and PDF.		
2	US-07. As an admin, I want to receive and investigate complaints from users so that I can resolve disputes fairly and maintain platform trust.	• Users (farmers and investors) can submit a complaint via a dedicated 'Report a Problem' form. • The admin sees all complaints in a dedicated complaints queue with: submitter, subject, description, and date. • The admin can change the complaint status to: Under Investigation, Resolved, or Dismissed. • Resolved complaints are archived and not deleted.	**TC1 - Complaint submission and queue** 1- Submit a complaint through the Report a Problem form. 2- Log in as admin and verify the complaint appears in the complaints queue with required fields. TC2 - Status management / archive 3- Change complaint status to Under Investigation, Resolved, and Dismissed. 4- Verify status updates work and resolved complaints are archived rather than deleted.	Yes	
2	US-08. As an admin, I want to remove update posts that violate platform guidelines so that the community content stays appropriate and respectful.	• The admin can view all farmer-posted updates across all farms. • The admin can delete any update with a mandatory reason. • The farmer whose update was deleted is notified with the reason.	**TC1 - View posted updates** 1- Open the admin updates moderation page. 2- Verify farmer-posted updates are visible across farms. **TC2 - Delete violating update** 3- Delete an update without and then with a reason. 4- Verify a reason is required and the farmer receives the deletion reason.	Yes	

2	US-09.As an admin, I want to view all registered farms so that I can monitor listings and ensure platform compliance.	•The admin panel displays a complete list of all registered farms. •Each farm record shows: farm name, owner name, location, area (m²), palm type, and status. •The list is searchable by farm name or owner name. •The list is filterable by status (pending, approved, rejected, deactivated). •The admin can click on any farm to view full details.	**TC1 - Farm list display** 1- Open the admin farms page. 2- Verify each farm record shows name, owner, location, area, palm type, and status. **TC2 - Search / filter / details** 3- Search by farm or owner name, filter by status, and open a farm record. 4- Verify results are correct and full farm details can be viewed.		
2	US-10. As an admin, I want to view activity logs so that I can audit user and system actions.	•The admin can access a dedicated "Activity Logs" page. •The system displays logs including: user ID, action type, timestamp, and affected entity •Logs include actions such as: account approval, farm approval, edits, deletions, and transaction updates. •The logs are searchable by user name or action type. •The logs are filterable by date range. •Logs cannot be edited or deleted by any user. •The system records all actions in real-time.	**TC1 - Activity log display** 1- Open the Activity Logs page as admin. 2- Verify logs show user ID, action type, timestamp, and affected entity for supported actions. **TC2 - Search / filter / immutability** 3- Search logs by user or action type, filter by date range, and attempt to edit/delete a log. 4- Verify search/filter work and logs cannot be edited or deleted.	Yes	
2	US-11. As an admin, I want to edit user-generated content so that I can correct minor violations without deleting posts.	•Admin can open and edit user-generated content. •Edit reason is required before saving. •Updated content is saved and displayed correctly. The creator is notified after edit. •The action is logged in the audit trail.	**TC1 – Edit content** 1- Open content, edit fields, add reason, and save. 2- Verify content is updated, action is logged and user is notified. **TC2 – Mandatory reason**	Yes	

		•Deleted content cannot be edited.	3- Edit without reason and attempt to save. 4- Verify saving is blocked. **TC3 – Deleted content** 5- Attempt to edit deleted content. 6- Verify editing is not allowed.		
Non-Functional Requirements					
2	US-01. As a user, I want the Qinwan platform to load within 3 seconds so that I can quickly access my data without frustration.	• The homepage and main browse page must load in under 3 seconds for 90% of users . • API response time must not exceed 1 second. • The interactive farm map must fully render within 3 seconds after page load.	**TC1 - Page load performance** 1- Measure homepage and browse page load times across representative sessions. 2- Verify at least 90% of sessions load within 3 seconds. **TC2 - API / map performance** 3- Measure key API response times and interactive map rendering time. 4- Verify API responses stay within 1 second and the map renders within 3 seconds.	yes	
2	US-02. As a user, I want the Qinwan interface to be intuitive and easy to navigate so that I can complete key tasks without confusion.	• A new user must be able to register and submit their first investment request within 5 minutes. • Key tasks (browse farms, submit request, view dashboard) must be completable in no more than 4 clicks from the home screen. • Error messages must be in clear Arabic and English, describing the problem and actionable steps to fix it.	**TC1 - First-time task completion** 1- Ask a new user to register and submit a first investment request. 2- Verify the task can be completed within 5 minutes. TC2 – Click count 3- Perform key tasks starting from the home page. 4- Verify that each task is completed in no more than 4 clicks. .	yes	

Developer Stories					
2	US-01. As a developer, I want to set up the project repository with a branching strategy (main / develop / feature branches) so that the team can collaborate without overwriting each other's work.	• A Git repository is created and shared with all team members. • Branching conventions are documented: main (production), develop (staging), feature/xxx (per story). • Pull requests require at least one code review approval before merging into develop. • No direct commits are allowed to the main branch. • A README file documents the project setup steps and branching rules.	**TC1 - Repository and branching rules** 1- Open the project repository and README. 2- Verify the repository is shared, the branching strategy is documented, and no direct commits are allowed to main. **TC2 - Pull request workflow** 3- Create a feature branch and submit a pull request to develop. 4- Verify at least one code review approval is required before merge.		
2	US-02. As a developer, I want to design and implement the relational database schema for Qinwan so that all entities (users, farms, investments, transactions) are stored consistently and efficiently.	• The database schema includes tables for: Users, Farmers, Investors, Admins, Farms, InvestmentRequests, Transactions, FarmUpdates, Complaints, and Wishlist. • All foreign key relationships are correctly defined and enforced at the database level. • All text fields storing sensitive data (passwords, IDs) use appropriate encrypted or hashed column types. • An Entity-Relationship Diagram (ERD) is documented.	**TC1 - Schema completeness** 1- Review the database schema and ER diagram. 2- Verify all required tables exist and the ER diagram is documented. **TC2 - Data integrity / security** 3- Inspect foreign keys and sensitive columns in the database. 4- Verify relationships are enforced and sensitive data uses appropriate hashed/encrypted handling.		
Defect Storie					
2	US-01. As a developer, I want to fix the duplicate	• BUG: An investor can currently submit multiple pending requests for the	**TC1 - Prevent duplicate active request**		

	investment request bug so that an investor cannot submit more than one pending request for the same farm.	same farm simultaneously, causing inconsistency in farm availability tracking. • FIX: Before allowing a new investment request submission, the system must check if the investor already has a 'Pending' or 'Accepted' request for the same farm. • If a duplicate is detected, the system displays: 'You already have an active request for this farm. Please wait for the farmer's response before submitting a new one.' • Verified by test: attempting to submit a second request for the same farm while the first is pending must be blocked.	1- Submit one investment request for a farm, then attempt a second request for the same farm while the first is Pending or Accepted. 2- Verify the second submission is blocked. **TC2 - Error message** 3- Observe the response after attempting the duplicate submission. 4- Verify the user sees the specified active request error message.		

Table 24 : User Story Acceptance Testing

5.2 Integration Testing

This section presents the integration hierarchy of the Qinwan system, which illustrates the execution flow between different system components and user roles.

5.2.1 Integration hierarchy

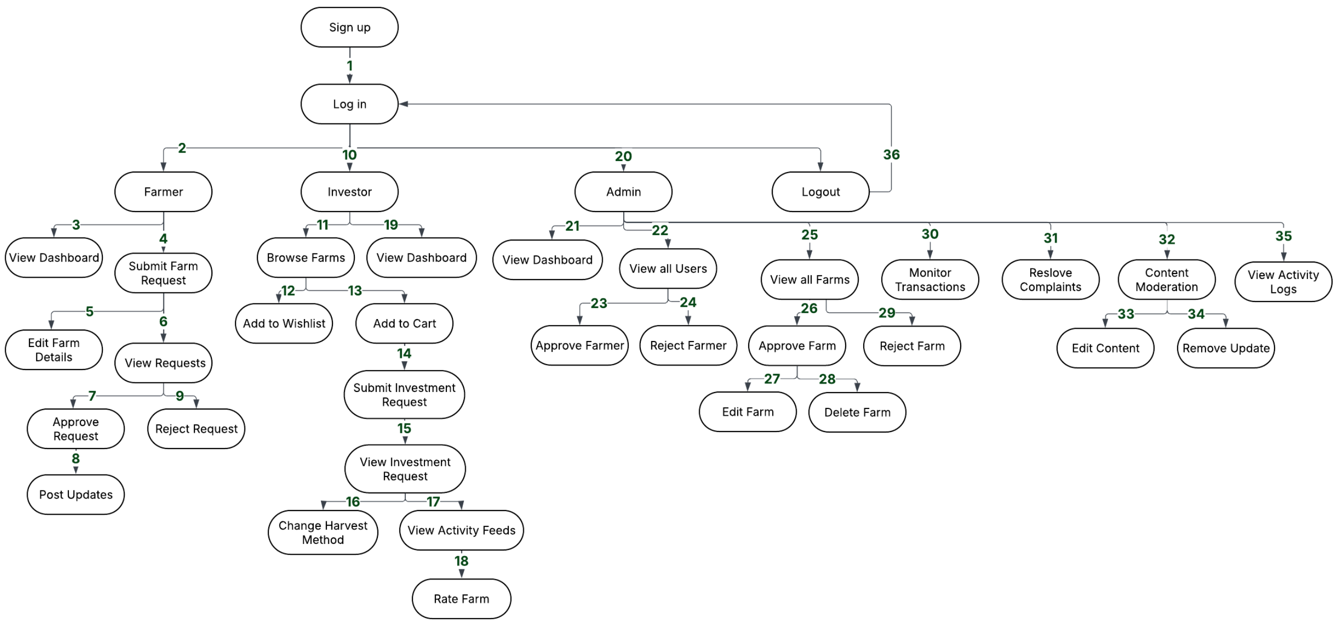

Figure 28 : Integration hierarchy

5.2.2 Integration testing plan

The diagram shows how the main system functionalities are connected and executed in sequence. It starts from user authentication (sign up and login), then branches into different roles such as investor, farmer, and admin. Each role interacts with specific functionalities, such as browsing farms, submitting investment requests, managing farm data, and handling administrative operations. This structure ensures a clear and organized interaction between system components.

System components	New component	Test case	Pass?	Comments
Sign up to the system	Log in	1. A new user successfully registers with a unique email and password. 2. The same user successfully logs in with their credentials.	Yes	
- Sign up to the system - Log in	Browse Farms	1. A new investor registers successfully. 2. The investor logs in. 3. The investor navigates to the Browse Farms page and views the list of approved farms. 4. The investor applies a region filter - results and map update correctly.	Yes	
- Sign up to the system - Log in - Browse Farms	Add to Cart	1. A new investor registers successfully. 2. The investor logs in. 3. The investor browses available farms. 4. The investor opens a farm offer and selects harvest method 'sell' and duration '1 year'. 5. The investor adds the offer to the cart. 6. Cart badge updates and cart page shows correct farm name, area, duration, harvest method, and estimated cost.	yes	

Figure 29 : Integration testing plan

5.3 User Acceptance Testing

User Acceptance Testing was conducted to evaluate whether the Qinwan platform meets user requirements and can be effectively used in real-world scenarios. This type of testing involves real users interacting with the system to ensure that it satisfies both functional and usability expectations.

How the Test Was Conducted

Two separate questionnaires were designed and distributed ,one for investors and one for farmers , to evaluate each role's unique experience with the Qinwan platform. The testing was conducted online, and participants were asked to use the platform freely before answering the questionnaire. Each session lasted approximately 10–15 minutes.

Participant recruitment: Five participants were recruited for each user role (investors and farmers), for a total of 10 participants. All participants match the target user profile defined in Phase 2 Saudi residents interested in agricultural investment or farming.

Test steps:

1. Participants were given a brief introduction to the Qinwan platform concept.
2. Participants explored the platform freely, performing their role's core tasks.
3. Participants completed the questionnaire (10 questions each).
4. Open-ended feedback was collected at the end.

5.3.1 Demographics of participants

Participant Demographics

Participant Demographics - Investor Participants Demographics

ID	Age Group	Gender	Tech Experience
P1	25–35	Male	Intermediate
P2	36–45	Male	Beginner
P3	46+	Male	Intermediate
P4	25–35	Male	Advanced
P5	46+	Female	Intermediate

Table 25 : Investor Participants Demographics

Participant Demographics - Farmer Participants Demographics

ID	Age Group	Gender	Tech Experience
P1	46+	Male	Intermediate
P2	46+	Female	Beginner
P3	Under 25	Female	Advanced
P4	25–35	Male	Advanced
P5	25–35	Male	Advanced

Table 26 : Farmer Participants Demographics

Gender Distribution

Across both groups, 80% of participants were male and 20% were female. Among investors specifically, 4 out of 5 were male (80%). Among farmers, 3 out of 5 were male (60%).

Gender Distribution - Investor Participants (n=5)

Option	Distribution	Percentage	n
Male		**80%**	4
Female		**20%**	1

Table 27 : Gender Distribution - Investor Participants

Gender Distribution - Farmer Participants (n=5)

Option	Distribution	Percentage	n
Male		**60%**	3
Female		**40%**	2

Table 28 : Gender Distribution - Farmer Participants

Age Distribution

For investor participants, 40% were aged 25–35, and 40% were 46 or above. For farmer participants, 40% were aged 25–35, 40% were 46 or above, and 20% were under 25.

Age Distribution - Investor Participants (n=5)

Option	Distribution	Percentage	n
25–35		**40%**	2
36–45		**20%**	1
46+		**40%**	2

Table 29 : Age Distribution - Investor Participants

Age Distribution - Farmer Participants (n=5)

Option	Distribution	Percentage	n
Under 25		**20%**	1
25–35		**40%**	2
46+		**40%**	2

Table 30 : Age Distribution - Farmer Participants

Tech Experience Level

Among investor participants, 60% had intermediate experience, 20% beginner, and 20% advanced. Among farmer participants, 60% were advanced, 20% intermediate, and 20% beginner.

Tech Experience - Investor Participants (n=5)

Option	Distribution	Percentage	n
Beginner		**20%**	1
Intermediate		**60%**	3

Advanced		**20%**	1

Table 31 : Tech Experience - Investor Participants

Tech Experience - Farmer Participants (n=5)

Option	Distribution	Percentage	n
Beginner		**20%**	1
Intermediate		**20%**	1
Advanced		**60%**	3

Table 32 : Tech Experience - Farmer Participants

5.3.2 Questionnaire Results

1. Investor Questionnaire Results

The investor questionnaire consisted of 10 questions covering platform usability, farm information clarity, trust, design quality, and overall satisfaction. All 5 investor participants completed the questionnaire.

Investor Questionnaire - Summary of Results

Q#	Question	Results	Avg Score
Q1	Ease of browsing available farms (1–5 scale)	4/5 rated 5/5 — 1/5 rated 4/5	4.8 / 5
Q2	Ease of account registration (1–5 scale)	5/5 rated 5/5	5.0 / 5
Q3	Farm information clarity (location, area, palm type, harvest method)	4/5: Yes, fully clear — 1/5: Somewhat, needs more detail	80% fully clear

Q4	Technical issues encountered during use	5/5: No issues — all worked correctly	100% no issues
Q5	Platform design quality (visual appeal, clarity, layout) (1–5 scale)	4/5 rated 5/5 — 1/5 rated 4/5	4.8 / 5
Q6	Does the platform achieve its goal of connecting investors and farmers?	5/5: Yes, fully achieves its goal	100% agreed
Q7	Trust in personal data and investment security (1–5 scale)	5/5 rated 5/5	5.0 / 5
Q8	Overall satisfaction with Qinwan (1–5 scale)	4/5 rated 5/5 — 1/5 rated 4/5	4.8 / 5
Q9	Would you recommend Qinwan to others interested in agricultural investment?	5/5: Yes, definitely	100% Yes
Q10	Most liked feature (open-ended)	New concept, farm map browsing, ease of use, investment accessibility	—

Table 33 : Investor Questionnaire - Summary of Results

Q1 - Ease of Browsing Available Farms (Investor, n=5)

Option	Distribution	Percentage	n
5 — Very Easy		**80%**	4
4 — Easy		**20%**	1

Table 34 : Investor Questionnaire Results - Q1

Average: 4.8 / 5

Q2 - Ease of Account Registration (Investor, n=5)

Option	Distribution	Percentage	n
5 — Very Easy		**100%**	5

Table 35 : Investor Questionnaire Results - Q2

Average: 5.0 / 5 - Perfect score

Q3 - Farm Information Clarity (Investor, n=5)

Option	Distribution	Percentage	n
Yes, fully clear		**80%**	4
Somewhat, needs more detail		**20%**	1

Table 36 : Investor Questionnaire Results - Q3

80% found farm information fully sufficient for investment decisions.

Q4 — Technical Issues Encountered (Investor, n=5)

Option	Distribution	Percentage	n
No issues — all worked correctly		100%	5

Table 37 : Investor Questionnaire Results - Q4

100% of investor participants reported no technical issues during their session.

Q5 - Platform Design Quality (Investor, n=5)

Option	Distribution	Percentage	n
5 — Excellent		**80%**	4
4 — Good		**20%**	1

Table 38 : Investor Questionnaire Results – Q5

Average: 4.8 / 5

Q6 - Platform Achieves Its Goal (Investor, n=5)

Option	Distribution	Percentage	n
Yes, fully achieves its goal		**100%**	5

Table 39 : Investor Questionnaire Results - Q6

100% of investor participants agreed the platform successfully connects investors and farmers.

Q7 - Trust in Data Security (Investor, n=5)

Option	Distribution	Percentage	n
5 — Fully Trust		**100%**	5

Table 40 : Investor Questionnaire Results - Q7

Average: 5.0 / 5 — All investor participants fully trust the platform's data security.

Q8 - Overall Satisfaction (Investor, n=5)

Option	Distribution	Percentage	n
5 — Very Satisfied		**80%**	4
4 — Satisfied		**20%**	1

Table 41 : Investor Questionnaire Results – Q8

Average: 4.8 / 5

Q9 - Would Recommend Qinwan (Investor, n=5)

Option	Distribution	Percentage	n
Yes, definitely		**100%**	5

Table 42 : Investor Questionnaire Results – Q9

100% of investor participants would recommend Qinwan.

Q10 - Most Liked Feature (Investor, n=5) - Open-Ended Responses

Participant	Response
P1	The idea is new, legal, and protects the rights of both parties.
P2	Ease of browsing farms through the interactive map.
P3	Ease of use and clarity of everything on the platform.
P4	A new and innovative idea with easy access to farms and harvests.
P5	The organization and ease of using the platform.

Table 43: Investor Questionnaire Results - Q10

Investors liked the easy-to-use and clear platform, especially the interactive map, and appreciated the innovative and trustworthy idea.

2. Farmer Questionnaire Results

The farmer questionnaire consisted of 10 questions covering farm registration, notifications, investment request tracking, update posting, data security, verification clarity, and overall experience.

Farmer Questionnaire - Summary of Results

Q#	Question	Results	Avg Score
Q1	Ease of registering a farm on the platform (1–5 scale)	4/5 rated 5/5 — 1/5 rated 1/5	4.2 / 5
Q2	Clarity of required farm fields (area, palm type, location)	4/5: Yes, fully clear — 1/5: Somewhat	80% fully clear
Q3	Notification system sufficiency (accept, reject, edit)	2/5: Somewhat — 2/5: Yes, clear — 1/5: Somewhat	60% satisfactory
Q4	Ease of tracking investment request status (1–5 scale)	3/5 rated 5/5 — 2/5 rated 4/5	4.6 / 5
Q5	Ease of posting farm updates for investors (1–5 scale)	5/5 rated 5/5	5.0 / 5
Q6	Trust in farm data and personal information security (1–5 scale)	1/5 rated 3 — 1/5 rated 4 — 3/5 rated 5	4.4 / 5
Q7	Fairness and clarity of the farmer verification process	4/5: Yes, fully clear — 1/5: Somewhat	80% fully clear
Q8	Overall experience rating as a farmer on Qinwan (1–5 scale)	4/5 rated 5/5 — 1/5 rated 4/5	4.8 / 5

Q9	Will the platform help reach more investors for your farm?	5/5: Yes, definitely	100% Yes
Q10	Most liked feature (open-ended)	Investor aggregation, easy update posting, clear statistics dashboard, organized farm browser	—

Table 44 : Farmer Questionnaire - Summary of Results

Q1 - Ease of Farm Registration (Farmer, n=5)

Option	Distribution	Percentage	n
5 — Very Easy		**80%**	4
1 — Very Difficult		**20%**	1

Table 45 : Farmer Questionnaire Results - Q1

Average: 4.2 / 5. One participant encountered difficulty during registration - likely a one-time setup issue.

Q2 - Clarity of Required Farm Fields (Farmer, n=5)

Option	Distribution	Percentage	n
Yes, fully clear		**80%**	4
Somewhat		**20%**	1

Table 46 : Farmer Questionnaire Results - Q2

80% of farmer participants found all required fields clear and easy to understand.

Q3 - Notification System Satisfaction (Farmer, n=5)

Option	Distribution	Percentage	n
Yes, fully clear		**40%**	2
Somewhat		**60%**	3

Table 47 : Farmer Questionnaire Results - Q3

60% found notifications somewhat sufficient. This is a noted area for improvement.

Q4 - Ease of Tracking Investment Requests (Farmer, n=5)

Option	Distribution	Percentage	n
5 — Very Easy		**60%**	3
4 — Easy		**40%**	2

Table 48 : Farmer Questionnaire Results – Q4

Average: 4.6 / 5

Q5 - Ease of Posting Farm Updates (Farmer, n=5)

Option	Distribution	Percentage	n
5 — Very Easy		**100%**	5

Table 49 : Farmer Questionnaire Results - Q5

Average: 5.0 / 5 - Perfect score across all farmer participants.

Q6 - Trust in Data Security (Farmer, n=5)

Option	Distribution	Percentage	n
3 — Neutral		**20%**	1
4 — Trust		**20%**	1
5 — Fully Trust		**60%**	3

Table 50 : Farmer Questionnaire Results - Q6

Average: 4.4 / 5

Q7 - Fairness of Farmer Verification Process (Farmer, n=5)

Option	Distribution	Percentage	n
Yes, fully clear		**80%**	4
Somewhat		**20%**	1

Table 51 : Farmer Questionnaire Results - Q7

80% of farmer participants found the verification process fully clear.

Q8 - Overall Farmer Experience (Farmer, n=5)

Option	Distribution	Percentage	n
5 — Excellent		**80%**	4
4 — Good		**20%**	1

Table 52 : Farmer Questionnaire Results - Q8

Average: 4.8 / 5

Q9 - Platform Will Help Reach More Investors (Farmer, n=5)

Option	Distribution	Percentage	n
Yes, definitely		**100%**	5

Table 53 : Farmer Questionnaire Results - Q9

100% of farmer participants believe Qinwan will help them reach more investors.

Q10 - Most Liked Feature (Farmer, n=5) - Open-Ended Responses

Participant	Response
P1	Everything is excellent and clear.
P2	The platform is smooth and easy to use, and the statistics on the dashboard are very helpful.
P3	Gathering all investors in one place and making it easy to reach them.
P4	Being able to browse all available farms through a single organized platform.
P5	Sending farm updates to investors.

Table 54 : Farmer Questionnaire Results - Q10

Farmers liked the simple, clear platform and the ability to connect and communicate easily with investors.

3. Overall UAT Summary

The following table summarizes the key metrics from both the investor and farmer UAT sessions:

UAT Overall Results Summary

Metric	Investor Results	Farmer Results
Number of Participants	5	5

Average Registration Ease	5.0 / 5	4.2 / 5
Average Platform Usability	4.8 / 5	4.6 / 5
Average Design / UX Rating	4.8 / 5	4.8 / 5
Average Data Security Trust	5.0 / 5	4.4 / 5
Average Overall Satisfaction	4.8 / 5	4.8 / 5
Would Recommend / Use Platform	100%	100%
Technical Issues Reported	0 / 5 (0%)	0 / 5 (0%)
Task Completion Rate	100%	100%

Table 55 : UAT Overall Results Summary

5.4 Discussion

In this section, we provide an interpretation of the testing results from Section 5.1 (User Story Acceptance Testing) and Section 5.3 (User Acceptance Testing). We also reflect on how we carried out the development process, the design decisions we made, and what we would improve if we did this again.

Were the Results Good?

Overall, the results were very good. In the User Story Acceptance Testing, 16 out of 17 user stories fully passed their acceptance criteria. The only exception was US-19 (WhatsApp/SMS Notifications), which received a partial result. This was not due to a coding error, but because of external regulations in Saudi Arabia that prevent international SMS providers from sending messages to Saudi numbers (+966) without a registered sender ID [5]. We tried three different providers Twilio, Unifonic, and an Alphanumeric sender and all had the same regulatory restriction. As a workaround, we implemented internal platform notifications , which function correctly within the system.

In the User Acceptance Testing, both investor and farmer participants gave high scores:

Investor participants gave an average satisfaction score of 4.8 out of 5, and all 5 participants said they would recommend the platform to others.

Farmer participants gave an average satisfaction score of 4.8 out of 5, and all 5 participants confirmed that the platform will help them reach more investors.

No participant from either group reported any technical issues during the testing session.

These results indicate that the platform successfully meets the needs of both target user groups.

Reflection on the Development Process

We followed an Agile approach and divided the work into two sprints. This helped us focus on the most important features first, such as registration, farm browsing, the investment cart, and the request flow, before moving on to more advanced features like harvest method change requests and the activity feed.

One challenge we faced was discovering some bugs during testing that could have been caught earlier. For example, the cart persistence issue (TC3) was caused by a UNIQUE KEY constraint in the database that prevented a new cart from being created after the old one expired. This was only discovered during the testing phase. If we had reviewed the database schema more carefully earlier in the sprint, we would have caught it sooner.

Another challenge was the WhatsApp notification integration (US-19). We spent significant time testing different SMS providers before realizing the issue was regulatory, not technical. In future projects, we would research infrastructure and legal requirements for communication features before starting implementation.

Reflection on the Design

The role-based design of the platform with separate dashboards for farmers, investors, and admins worked well. UAT participants did not report confusion when navigating their dashboard, and each user group was able to complete their tasks without assistance.

The map-based farm browsing feature (using Leaflet.js) was one of the most appreciated design decisions. An investor participant specifically mentioned it as their favorite feature. This confirms that investing time in the map integration was the right choice.

One design area that could be improved is the rating restriction in US-20. The system correctly prevents investors from rating a farm until their investment period has ended. However, 2 out of 5 investor participants were initially confused when they could not find the rating button. The current design only shows the end date, which is not enough explanation. A short message like 'Rating will be available after your investment period ends on [date]' would make this much clearer without changing the functionality.

Reflection on the Implementation

The system was built using PHP, MySQL, and XAMPP. This was a deliberate decision to keep the technology stack simple and accessible. The platform runs correctly in a local environment and covers all the core features required by the project specifications.

Some features were not fully implemented due to scope limitations. For example, payment processing is not connected to a real payment gateway the system records investment amounts and creates transaction entries, but actual money transfers cannot be completed. This was a known limitation from the start and is documented in the limitations section.

From a code quality perspective, the team applied validation at both the client side (JavaScript) and the server side (PHP). Passwords are hashed using bcrypt, and all database queries use prepared statements to prevent SQL injection. These are standard security practices that we believe were implemented correctly.

Non-Functional Requirements (NFR) Evaluation

The system performed well in terms of non-functional requirements:

- **Performance**: All 10 UAT participants (5 investors, 5 farmers) reported no delays or slowness during their sessions. Page transitions and form submissions were fast across different devices and browsers.
- **Reliability**: Zero technical errors or crashes were reported during any of the 10 UAT sessions. The platform ran stably throughout all testing activities.
- **Security:** Investor participants gave a perfect score of 5.0/5 for data security trust. Farmer participants averaged 4.4/5. Password hashing, session management, and role-based access control all functioned as expected during testing.
- **Usability:** Both user groups gave high usability scores (4.8/5 for investors and 4.6/5 for farmers on task-related questions). The platform was described by multiple participants as easy to use and well-organized.

How Can We Improve?

Based on the testing results and the challenges we encountered, we suggest the following improvements:

1. **Notification system:** Three out of five farmer participants felt the notification system only partially met their needs. Adding direct messaging between investors and farmers, and allowing investors to reply to farm updates, would significantly improve communication on the platform.
2. **SMS/WhatsApp delivery:** Once a Saudi commercial registration is available, the team should integrate with a Saudi-licensed SMS provider such as Unifonic or Msegat to enable real automated push notifications to Saudi phone numbers.
3. **Rating UX:** Replace the date-only label under the rating button with a full sentence explaining why the rating is not yet available, to avoid confusing users who expect to rate immediately after their request is accepted.
4. **Payment gateway:** Integrate a payment provider such as Mada or STC Pay to allow real financial transactions within the platform, which is currently the most significant missing feature for production use.
5. **Platform expansion:** One investor participant suggested extending the platform beyond palm farms to include general agricultural land leasing. This would increase the platform's reach and appeal to a wider audience.
6. **Mobile application:** Several participants completed the test on mobile devices. Building a dedicated mobile app (iOS/Android) would improve the experience significantly, as the current web interface is not fully optimized for small screens.

6 Chapter 6: Conclusion and Future Work

Qinwan started as a response to a real problem: small and medium-scale date palm farmers in Saudi Arabia often struggle to find investors, while individuals who want to invest in agriculture have no structured or transparent way to do so. Our goal was to build a platform that connects both sides in a simple, trustworthy, and organized way.

Over two sprints, we designed and implemented a full web-based system that allows farmers to register their farms, post investment offers, and manage incoming requests while investors can browse farms, add offers to a cart, submit investment requests, track updates from their invested farms, and rate their experience. The system also includes a full admin panel for managing users, farms, and complaints.

This chapter summarizes what we built, the challenges we faced, the limitations of the current version, and our plans for future development.

6.1 Global and Local Impact

Local Impact

Saudi Arabia is home to a large number of date palm farms, and the date palm holds deep cultural and economic significance in the country. However, many farmers especially those running small or medium-sized operations face difficulty finding investors to help them grow or sustain their farms. At the same time, many Saudis who want to invest their money in something meaningful and local have no easy way to find these opportunities.

Qinwan directly addresses this gap. By providing a verified, digital marketplace for agricultural investment, the platform makes it easier for farmers to attract funding and for investors to participate in the local agricultural economy. This aligns with Saudi Arabia's Vision 2030 goals [6], which [7]emphasize economic diversification, food security, and the growth of the agricultural sector.

The platform also introduces transparency and trust into a process that has traditionally relied on personal connections or informal agreements. With features like verified farmer accounts, documented investment terms, and a structured request flow, both parties can engage with confidence.

Global Impact

Date palm farming is not unique to Saudi Arabia. Countries across the Middle East and North Africa including the UAE, Iraq, Egypt, Algeria, and Morocco وface similar challenges in connecting agricultural producers with investors. The model we built for Qinwan can be adapted to work in any of these markets with minimal changes.

More broadly, Qinwan demonstrates that agricultural investment platforms can be built using simple, accessible technology (PHP, MySQL, standard web hosting) without requiring large budgets or complex infrastructure. This makes the model particularly relevant for developing countries where farmers often lack access to formal investment channels.

The harvest method flexibility we implemented allowing investors to choose between receiving physical dates, selling the harvest for profit, or donating it to charity also introduces a socially responsible investment option. The donation option, in particular, connects agricultural investment with charitable giving, which has strong cultural resonance in the Arab and Muslim world.

6.2 Problems and Challenges Encountered During Development

During the development of Qinwan, we encountered several challenges that required us to change our approach or find alternative solutions:

- **Database connection issue:** At the start of the project, we discovered that the database port in db_connect.php was set to 8889 (used by MAMP), while our environment used XAMPP, which runs on port 3306. This caused all pages to enter a redirect loop (ERR_TOO_MANY_REDIRECTS). The issue was simple to fix once identified, but it took time to diagnose because the error message did not point directly to the root cause.

- **Cart persistence bug:** The original database schema included a UNIQUE KEY constraint on the investor_id column in the cart table. This prevented a new cart from being created after a previous one expired, which caused Test Case 3 (TC3 — Persistence) to fail. We removed the constraint and updated the cart creation logic to handle expired carts correctly.

- **Image upload failure:** The farm update form was missing the enctype='multipart/form-data' attribute, which meant uploaded images were silently ignored with no error shown to the user. This was only discovered during testing of US-17 (Farm Updates). We fixed the form and added a complete server-side image handling routine.

- **JavaScript navigation failure:** A missing function keyword in the filterRequests() function definition in Farmer.php caused a JavaScript parse error that silently disabled all navigation buttons on the page. The page loaded visually but nothing responded to clicks. Once identified, the fix was a single line, but diagnosing a silent JS error required careful inspection.

- **SMS/WhatsApp notification integration (US-19):** This was the most time-consuming challenge we faced. We attempted three different approaches to send SMS or WhatsApp messages to Saudi phone numbers: Twilio's US-based number was blocked by Saudi carrier regulations (error 21612); Twilio's Alphanumeric Sender ID ('Qinwan') requires a paid account and a 2–4 week pre-registration with the Communications, Space & Technology Commission (CST); and Unifonic, a Saudi-native SMS provider, requires a valid commercial registration. Since we are students, we do not have a commercial registration, which made it impossible to use any of these services fully. As a practical solution, we implemented internal platform notifications and WhatsApp direct links (wa.me), which deliver the message content through the user's own WhatsApp app.

6.3 Limitations of the System

The current version of Qinwan has several limitations that we acknowledge:

6.1 **No payment gateway:** The system calculates investment amounts and creates transaction records, but there is no integration with a real payment provider. Investors cannot actually transfer money through the platform. This is the most significant missing feature for any real-world deployment.

6.2 **No automated SMS/WhatsApp delivery:** As explained in the challenges section, real push delivery to Saudi phone numbers requires a registered sender ID through a Saudi-licensed gateway, which requires a commercial registration we do not have. Notifications work correctly within the platform, but external delivery depends on the user opening the wa.me link manually.

6.3 **No email notifications:** The system references email notifications in the acceptance criteria for US-14 (Submit Investment Request), but no SMTP or email gateway was integrated. This was a scope decision made early in the project.

6.4 **Web-only platform:** Qinwan currently runs as a website only. Several UAT participants accessed the platform on mobile devices and noted that some parts of the interface could be better optimized for smaller screens. A dedicated mobile application was not within the project scope.

6.5 **Admin testing deferred:** While the admin panel was fully implemented and manually tested, it was not included in formal User Story Acceptance Testing during this sprint due to time constraints.

6.4 Main Contribution of the Project

The main contribution of Qinwan is providing a structured, role-based, digital platform that makes agricultural investment in date palm farming accessible, transparent, and organized for both farmers and investors in Saudi Arabia.

More specifically, the project contributes the following:

6.6 **A complete investment lifecycle:** From farm registration and admin verification, to offer creation, investor browsing, cart-based investment, request management, farm updates, harvest method changes, and post-investment rating the platform covers the full journey for both user roles.

6.7 **A trust-building verification system:** The farmer verification flow ensures that only verified farmers with valid national IDs can list farms. This adds a layer of trust that is often missing from informal investment arrangements.

6.8 **A flexible harvest return model:** The platform allows investors to choose how they benefit from their investment by receiving the physical harvest, selling it for profit, or donating it to charity. This flexibility makes the platform suitable for a wide range of investor motivations.

6.9 **A replicable open-source architecture:** The system is built entirely on open-source technology (PHP, MySQL, XAMPP, Leaflet.js) with no paid services required for core functionality. This means the model can be replicated, adapted, or extended by other developers without licensing costs.

The UAT results support the value of this contribution. All 10 participants 5 investors and 5 farmers confirmed the platform meets their needs, and 100% of both groups expressed confidence that Qinwan serves its intended purpose.

6.5 Future Work

There are several directions in which Qinwan can be developed further to become a fully production-ready platform:

1. **Payment gateway integration:** Integrating a local payment provider such as Mada or STC Pay would enable real financial transactions within the platform, which is the most critical step toward real-world deployment.
2. **Real SMS/WhatsApp notifications:** Once a Saudi commercial registration is available, the team can integrate with a Saudi-licensed SMS provider such as Unifonic or Msegat to enable automated push notifications to Saudi phone numbers (+966).
3. **Mobile application:** Developing native iOS and Android applications would significantly improve the user experience, particularly for farmer and investor participants who accessed the platform on mobile devices during UAT.
4. **Direct messaging between farmers and investors:** Multiple farmer participants suggested adding the ability for investors to comment on or reply to farm updates. A built-in messaging or commenting system would improve communication and trust between both parties.

5. **AI-based farm recommendations:** Using investor profile data such as preferred harvest method, region of interest, and investment history to automatically suggest the most suitable farm offers would improve the browsing experience and increase conversion rates.
6. **Expansion beyond palm farms:** One investor participant suggested expanding the platform to cover general agricultural land leasing, not limited to date palm farms. This would significantly increase the platform's potential user base and market size.
7. **IoT integration:** Connecting the farm update system to soil sensors, weather monitors, or irrigation systems would allow farmers to post real-time automated updates, making the investment experience more transparent and data-driven for investors.

7 References

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8 Appendix

8.1 Appendix A : Interview

Interview questions

For Investors :

Q1: Have you ever invested in agriculture before?

Q2: Would you consider investing in date palm farms? Why?

Q3: What information would you need before deciding to invest in a farm?

Q4: What might make you stop using an agricultural investment platform?

Q5: How would you prefer to track the progress of your investment?

For Farmers :

Q1: Have you ever invested in agriculture before?

Q2: Would you consider allowing others to invest in your date palm farm? Why?

Q3: What information do you think should be shared with investors before they invest?

Q4: What might make you stop using an agricultural investment platform?

Q5: How would you prefer to share updates with investors?

Interview Transcriptions

Online Interview - farmer (1)	
Interviewee: Ibrahim	**Interviewer:** Jana
Location/Medium: Ibrahim's house	**Appointment Date:** 19-2-2026 **Start Time:** 9:06 PM **End Time:** 9:21 PM
Objectivies: • Gain insight into investors' and farm owners' preferences in agricultural investment. • Identify the key information required before making an investment decision. • Understand usability challenges in agricultural investment platforms. • Explore preferred methods for communication and investment updates.	**Reminders:** • Has experience in agricultural investment and managing a date palm farm. • Believes farming requires continuous supervision and personal involvement.
Agenda: Introduction Background on Project Overview of Interview Topics to Be Covered Permission to Record Question 1 Question 2 Question 3 Question 4 Question 5 Summary of Major Key Points Questions from Interviewee Closing	**Approximate Time:** 1 min 3 min 1 min 30 sec 1.5 min 1 min 2 min 1 min 1 min 1 min 1 min 1 min
General Observations: The interviewee appeared confident and experienced in agricultural investment. He provided clear and direct answers and showed practical knowledge about farm management and investment partnerships.	
Topic not covered: We did not discuss the preferred investment duration or the approval process for investor requests.	

Figure 30 : Interview Transcription - farmer1

Online Interview - farmer (1)	
Interviewee: Ibrahim	**Date:** 19-2-2026
Questions:	**Notes:**
Q1: Have you ever invested in agriculture before?	**Answer:** Yes, I have previous experience in agricultural investment, either through direct farming or partnerships with other investors. However, it requires expertise and continuous monitoring. **Observations:** When talking about his experience, the interviewee showed that he understands agricultural investment needs regular follow-up and supervision.
Q2: Would you consider allowing others to invest in your date palm farm? Why?	**Answer:** Yes, I would consider it because it helps reduce financial and operational costs. **Observations:** The interviewee appeared open to partnerships and sees them as a way to reduce costs.
Q3: What information do you think should be shared with investors before they invest?	**Answer:** The available investment area , the source and availability of water, electricity infrastructure, the type of date palms planted, and the expected average annual yield. **Observations:** He focused on clear and basic farm information before investors make a decision.
Q4: What might make you stop using an agricultural investment platform?	**Answer:** If the platform is complicated or requires advanced technical skills. **Observations:** His answer shows that ease of use is important to him when using digital platforms.
Q5: How would you prefer to share updates with investors?	**Answer:** I prefer sharing updates through periodic videos that show the condition of the farm and the date palms. **Observations:** **He prefers sending video updates to show the farm's condition to investors.**

Figure 31 : Interview Transcription - farmer1

Online Interview - farmer (2)	
Interviewee: Saad	**Interviewer:** Noura Altuwiam
Location/Medium: Saad's House	**Appointment Date:** 23-2-2026 **Start Time: 4:10 PM** **End Time: 4:24 PM**
Objectivies: • Gain insight into investors' and farm owners' preferences in agricultural investment. • Identify the key information required before making an investment decision. • Understand usability challenges in agricultural investment platforms. • Explore preferred methods for communication and investment updates.	**Reminders:** • Has several years of experience working in agriculture. • Is generally interested in developing agricultural practices and improving farm productivity.
Agenda: Introduction Background on Project Overview of Interview Topics to Be Covered Permission to Record Question 1 Question 2 Question 3 Question 4 Question 5 Summary of Major Key Points Questions from Interviewee Closing	**Approximate Time:** 1 min 3 min 1 min 30 sec 1 min 1.5 min 2 min 1 min 1 min 1 min 1 min 1 min
General Observations: During the interview, the interviewee was confident and engaged.the interviewee appeared confident and engaged. He showed clear experience in agriculture and expressed support for involving investors to expand his farm.	
Topic not covered: We did not discuss the investment duration, the approval process for investor requests, or how profits would be distributed.	

Figure 32 : Interview Transcription - farmer2

Online Interview - farmer (2)	
Interviewee: Saad	**Date:** 23-2-2026
Questions:	**Notes:**
Q1: Have you ever invested in agriculture before?	**Answer:** Yes, I work in the agricultural field and have experience in farm management and production. **Observations:** The interviewee has experience in agriculture and shows confidence in farm management and production.
Q2: Would you consider allowing others to invest in your date palm farm? Why?	**Answer:** Yes, to expand and increase production without high financial pressure and to improve the quality of agricultural services by providing additional liquidity. **Observations:** He supports allowing investors as a way to expand production and improve farm performance.
Q3: What information do you think should be shared with investors before they invest?	**Answer:** The number of palm trees, expected production, operational costs, the type of dates grown, and the water source. **Observations:**
Q4: What might make you stop using an agricultural investment platform?	**Answer:** If the platform is complicated to use or if there are high commission fees. **Observations:** He is concerned about platform usability and commission costs
Q5: How would you prefer to share updates with investors?	**Answer:** Through seasonal reports, along with periodic photos and videos. **Observations:** He prefers communication through reports and visual updates to maintain investor trust.

Figure 33 : Interview Transcription - farmer2

Online Interview - investor (1)	
Interviewee: Mohammed	**Interviewer:** Haya
Location/Medium: Mohmmed's house	**Appointment Date:** 20-2-2026 **Start Time:** 10:20 PM **End Time:** 10:36 PM
Objectivies: • Gain insight into investors' and farm owners' preferences in agricultural investment. • Identify the key information required before making an investment decision. • Understand usability challenges in agricultural investment platforms. • Explore preferred methods for communication and investment updates.	**Reminders:** • Has prior experience in agricultural investment. • Believes agricultural investment requires continuous supervision and hands-on involvement.
Agenda: Introduction Background on Project Overview of Interview Topics to Be Covered Permission to Record Question 1 Question 2 Question 3 Question 4 Question 5 Summary of Major Key Points Questions from Interviewee Closing	**Approximate Time:** 1 min 3 min 1 min 30 sec 1 min 1.5 min 2 min 1 min 1 min 1 min 2 min 1 min
General Observations: The interviewee appeared confident and experienced in agricultural investment. He seemed attentive and engaged throughout the discussion.	
Topic not covered: We did not discuss detailed profit expectations or investment duration options.	

Figure 34 : Interview Transcription - Investor1

Online Interview - investor (1)	
Interviewee: Mohammed	**Date:** 20-2-2026
Questions:	**Notes:**
Q1: Have you ever invested in agriculture before?	**Answer:** Yes, I have previously invested in the agricultural sector. It was a valuable experience, but it requires continuous follow-up and supervision. **Observations:** When talking about his experience, the interviewee showed that he understands agricultural investment needs regular follow-up and attention.
Q2: Would you consider investing in date palm farms? Why?	**Answer:** Perhaps, especially if I do not have suitable land available or if I do not have enough time to manage it myself. **Observations:** The interviewee decision depends on his personal situation, such as land availability and time.
Q3: What information would you need before deciding to invest in a farm?	**Answer:** I would need to know about water availability and quality, the type and fertility of the soil, and the type of date palms planted. **Observations:** He focused on basic farm information such as water, soil, and palm type before investing.
Q4: What might make you stop using an agricultural investment platform?	**Answer:** Limited technical experience due to age could make using the platform difficult, especially if the platform is unclear or complicated to use. **Observations:** His answer suggests that ease of use is important for him when using digital platforms.
Q5: How would you prefer to track the progress of your investment?	**Answer:** I prefer to track my investment through direct visits to the farm. **Observations:** He prefers visiting the farm in person to follow up on his investment.

Figure 35 :Interview Transcription - Investor1

Online Interview - investor (2)	
Interviewee: Abdullah	**Interviewer:** Noora Aluqaili
Location/Medium: Noora's house	**Appointment Date:** 21-2-2026 **Start Time: 7:21 PM** **End Time: 7:36:30 PM**
Objectivies: • Gain insight into investors' and farm owners' preferences in agricultural investment. • Identify the key information required before making an investment decision. • Understand usability challenges in agricultural investment platforms. • Explore preferred methods for communication and investment updates.	**Reminders:** • Has no prior experience in agricultural investment • Prefers investment opportunities that do not require daily management
Agenda: Introduction Background on Project Overview of Interview Topics to Be Covered Permission to Record Question 1 Question 2 Question 3 Question 4 Question 5 Summary of Major Key Points Questions from Interviewee Closing	**Approximate Time:** 1 min 3 min 1 min 30 sec 1 min 1.5 min 2 min 1 min 1 min 1 min 1.5 min 1 min
General Observations: During the interview, the interviewee showed interest in agricultural investment and was engaged in answering the questions. Although he has no prior experience, he appeared motivated to explore this opportunity.	
Topic not covered: We did not discuss detailed investment duration options, or how often returns would be distributed.	

Figure 36 : Interview Transcription - Investor2

Online Interview - investor (2)	
Interviewee: Abdullah	**Date:** 21-2-2026
Questions:	**Notes:**
Q1: Have you ever invested in agriculture before?	**Answer:** No, I have not invested in this sector before, but I am interested in trying if the opportunity is available. **Observations:** The interviewee has no prior experience but shows interest in trying agricultural investment if an opportunity is available.
Q2: Would you consider investing in date palm farms? Why?	**Answer:** Yes, definitely, because I do not need prior experience, and I can invest without owning land or managing it on a daily basis. **Observations:** He is strongly interested in investing because it does not require land ownership or daily management.
Q3: What information would you need before deciding to invest in a farm?	**Answer:** The expected profit percentage, irrigation quantity, and the types of natural and chemical fertilizers used. **Observations:** He focuses on financial returns and technical farm details before making an investment decision.
Q4: What might make you stop using an agricultural investment platform?	**Answer:** If the platform is unclear or complicated, and if there is no direct way to communicate with the farmer. **Observations:**
Q5: How would you prefer to track the progress of your investment?	**Answer:** I prefer to track my investment through periodic photos. **Observations:** He prefers visual updates, such as photos, to follow up on his investment.

Figure 37 : Interview Transcription - Investor2

8.2 Appendix B : Questionnaires

Questionnaires Link:

Investor:

https://docs.google.com/forms/d/e/1FAIpQLSdf27j6CtEKrULC9ejLB0Qpwx_Z9jLT2NHHmuAN5GiihY-rdg/viewform?usp=publish-editor

Farmer:

https://docs.google.com/forms/d/e/1FAIpQLSc7D0rWVb1K9buMA6EYLL8EK7r2AigCylTaJr54Cf6qkfutag/viewform?usp=header

Questionnaires Responses:

Investor:
https://docs.google.com/spreadsheets/d/18hj_dyp551h4Bj9RGSzQs9diXMomYS66Vnh6eO2XCJA/edit?usp=sharing

Farmer:
https://docs.google.com/spreadsheets/d/1EUKt8gLtJJ3fGuTlBZ6riWcVVBd5DgYzHmNbTksJrAo/edit?usp=sharing

Questionnaires :

Investor:

Q1. ما هي فئتك العُمرية؟

- أقل من 25
- 25-35 سنة
- 36-45 سنة
- 46 سنة فأكثر

Q2. عند استعراض المزارع، أي طريقة تفضل أكثر؟

- قائمة مع تفاصيل (List View)
- كلاهما متاح وأختار حسب الحاجة
- لا تفرق

Q3. ما هي معايير التصفية الأهم لك عند البحث عن مزرعة للاستثمار؟

- المنطقة الجغرافية
- حجم المزرعة (م^2)
- نوع التمر / النخيل
- تقييم المزرعة

Q4. ما مدى أهمية عرض صور عالية الجودة للمزرعة في صفحة التفاصيل قبل اتخاذ قرار الاستثمار؟

(مقياس من 1 إلى 5)

1 2 3 4 5

غير مهم ضروري

Q5. ما نوع تحديثات المزرعة الذي تريد تلقيه من صاحب المزرعة؟

- نص مكتوب فقط
- صور فقط
- مقاطع فيديو قصيرة
- نص + صور معاً

Q6. كيف تفضل أن تصلك إشعارات المنصة؟

- رسالة عبر البريد الالكتروني
- عن طريق الرسائل النصية (SMS)
- عن طريق الواتساب
- كل ما سبق

Q7. ما هي طريقة الاستفادة من الحصاد التي تفضلها؟

- استلام التمور بالتوصيل المنزلي
- التبرع بالحصاد لجمعية خيرية
- بيع الحصاد واستلام الأرباح
- تفضيل تحديد الطريقة لاحقاً

Q8. هل هناك ميزة أو خدمة تتمنى وجودها في منصة قِنوان لم تُذكر في الأسئلة السابقة؟

(سؤال مفتوح)

Farmar:

Q1. هل سبق لك تأجير جزء من مزرعتك للغير؟

- نعم
- لا، لكني مهتم
- لا، ولا أنوي

Q2. عند استقبال طلب استثمار، ما المعلومات التي تريد رؤيتها عن المستثمر؟

- الاسم الكامل والهوية الوطنية
- المساحة المطلوبة ومدة الاستثمار
- رقم الجوال للتواصل المباشر
- طريقة الاستفادة من الحصاد المختارة
- تاريخ الطلبات السابقة على المنصة

Q3. ما المدة المناسبة لك للرد على طلب استثمار قبل انتهاء صلاحيته؟

- 24 ساعة
- 48 ساعة
- أسبوع كامل
- بدون قيد زمني

Q4. ما نوع المحتوى الذي تجد نفسك ستشاركه مع مستثمريك كتحديث دوري؟

- نص مكتوب فقط (وصف الوضع الراهن)
- صور للمزرعة والمحصول
- مقاطع فيديو قصيرة
- صورً + نص معا
- لا أنوي إرسال تحديثات

Q5. كم مرة تريد إرسال تحديثات للمستثمرين؟

- أسبوعياً
- كل أسبوعين
- شهرياً
- عند وجود حدث مهم فقط (نمو، حصاد، مشكلة...)
- يكفي تحديث واحد في الموسم

Q6. كيف تفضل أن تصلك إشعارات المنصة؟

- رسالة بريد إلكتروني (Email)
- رسالة نصية (SMS)
- واتساب (WhatsApp)
- كل ما سبق للأحداث العاجلة

Q7. ما المعلومات التي تريد رؤيتها في لوحة التحكم الخاصة بك كمالك مزرعة؟

- إجمالي الأرض المؤجرة ونسبتها من المجموع
- تاريخ آخر تحديث أرسلته
- عدد المستثمرين النشطين
- إجمالي الأرباح التقديرية
- قائمة بأسماء المستثمرين وطلباتهم

Q8. ما أكبر تحدٍّ تواجهه حالياً في تأجير أرضك الزراعية؟ وهل تعتقد أن منصة قِنوان ستساعدك في حله؟

(سؤال مفتوح)

Questionnaires phase 4:

Farmer:

https://docs.google.com/forms/d/e/1FAIpQLScHSKK60NE92tnZi47Wc93-E8N4iS9nz9R_oz348mhjfMXWJA/viewform?usp=header

Investor:

https://docs.google.com/forms/d/e/1FAIpQLSc7haHJEXquS-utH7vQgx33h6I7dckIl15VMmTMK-luQ3vpow/viewform?usp=header

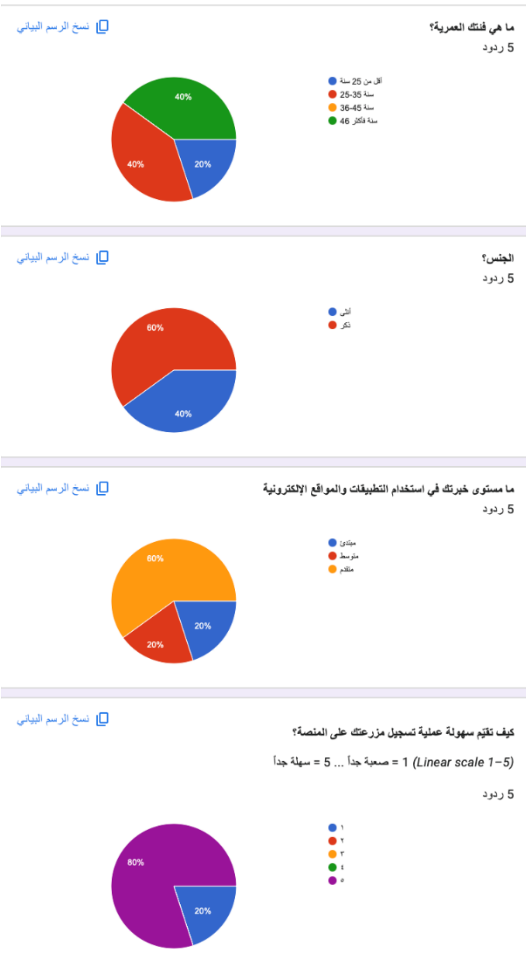

Figure 39 : Farmer Questionnaire 1-5

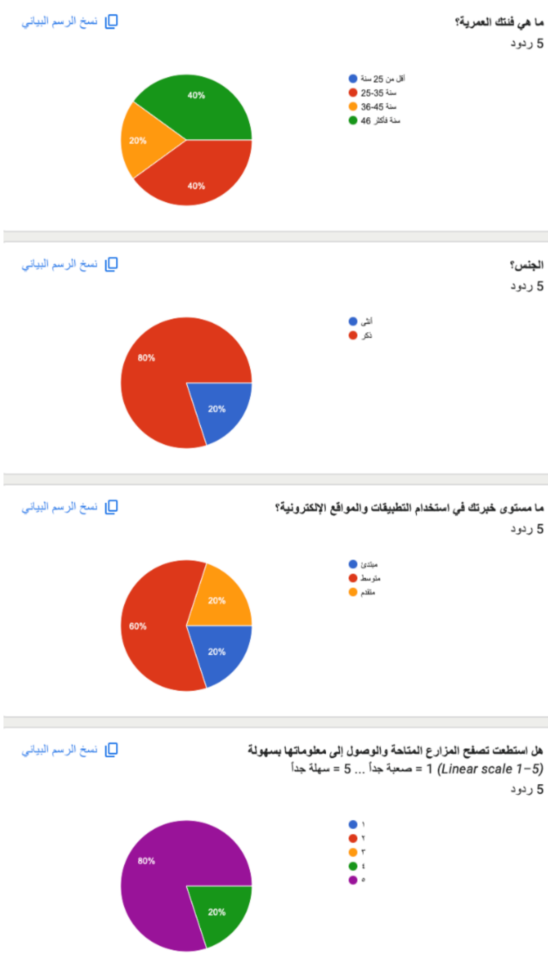

Figure 38 : Investor Questionnaire 1 - 5

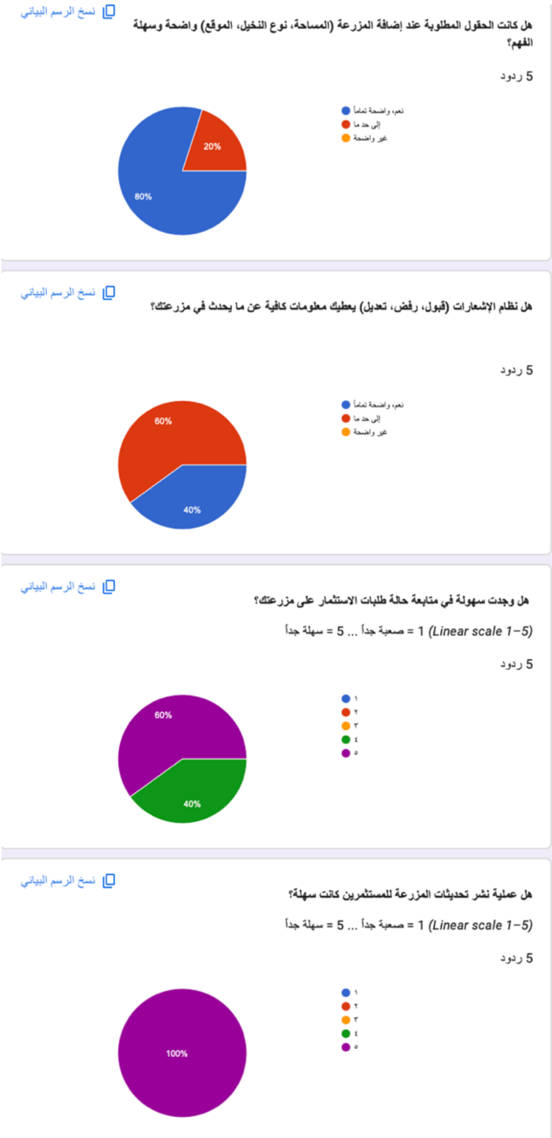

Figure 41 :Farmer Questionnaire 6-9

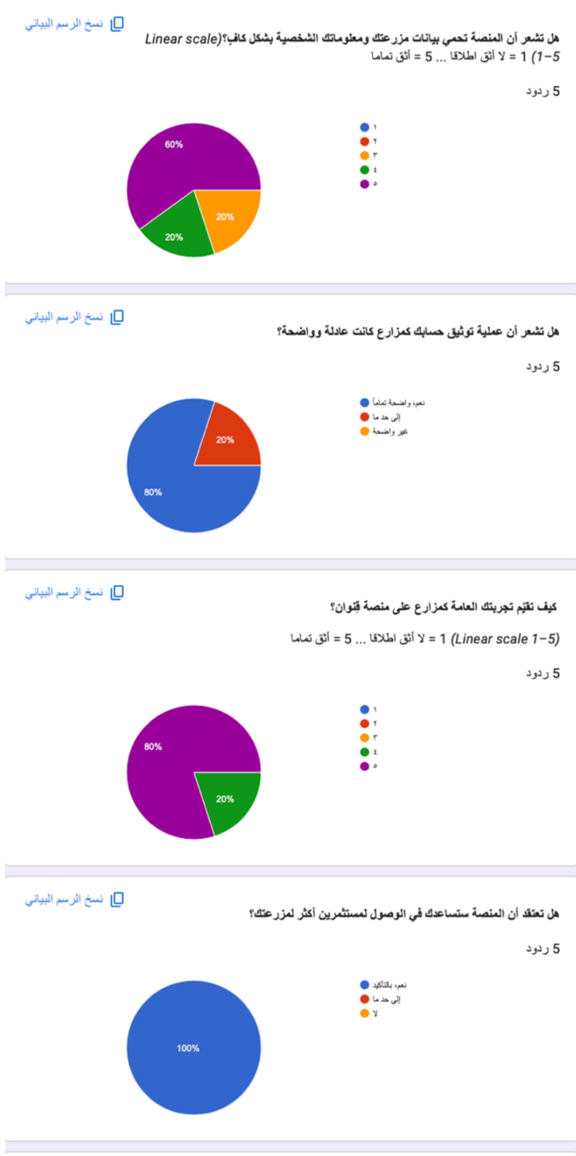

Figure 40 : Farmer Questionnaire 10 -13

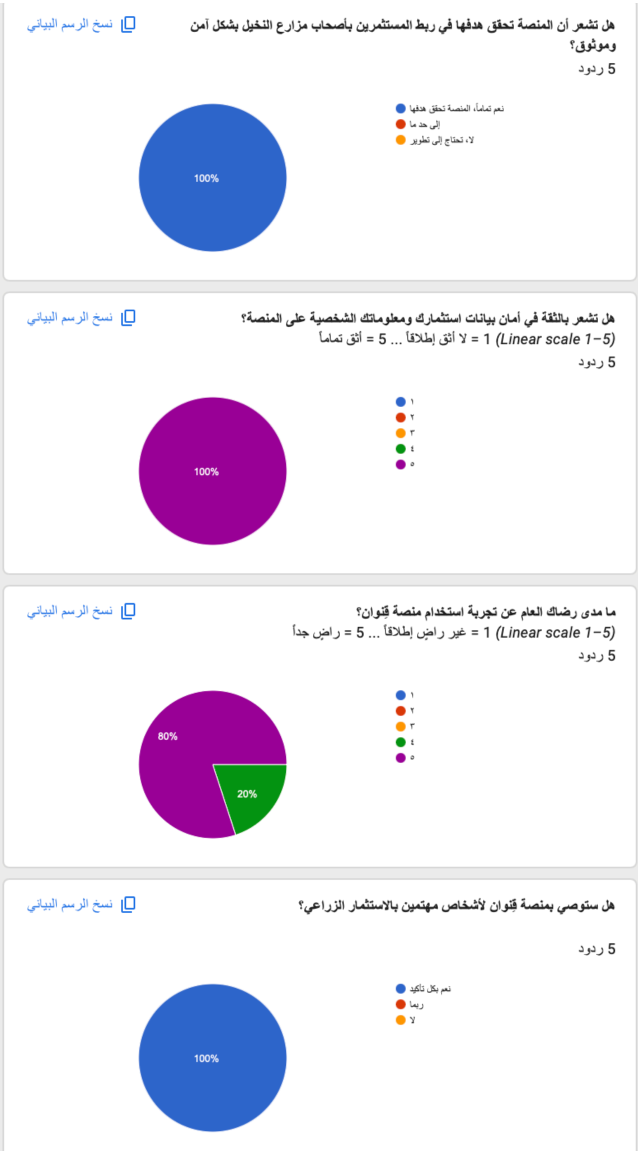

Figure 42 : Investor Questionnaire 10 - 13

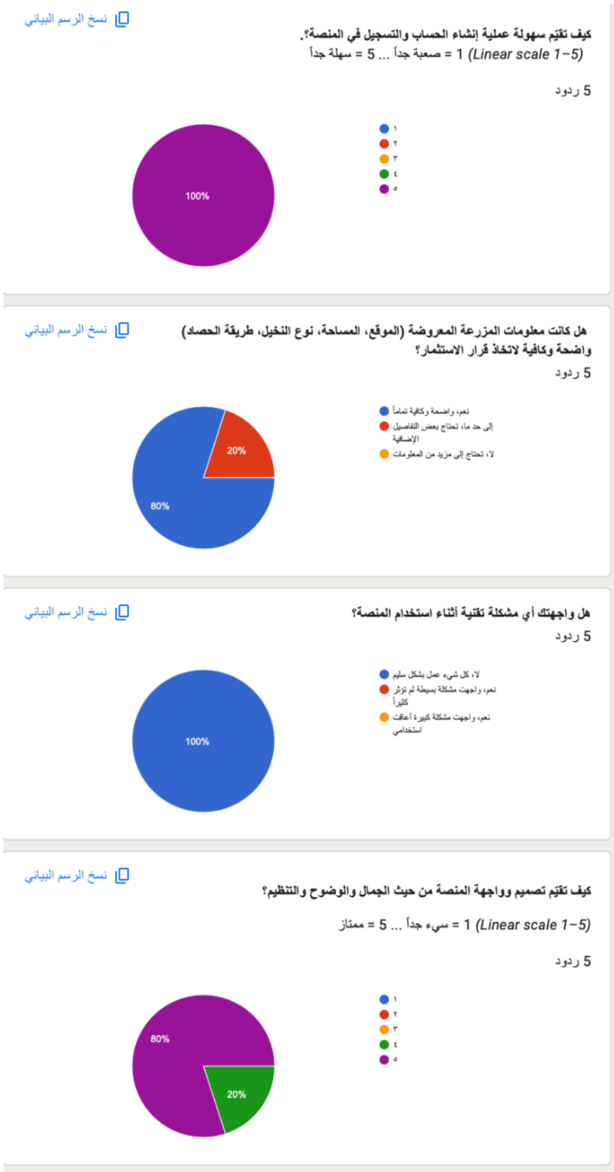

Figure 43 : Investor Questionnaire 6-9